



Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY

Parul Institute of Engineering Technology

Cyber Security Laboratory

(303105342)

6th SEMESTER (3RD YEAR)

COMPUTER SCIENCE & ENGINEERING DEPARTMENT

LAB MANUAL



CERTIFICATE

This is to certify that Mr. **KUNAM MANI SAI CHARAN REDDY** with enrollment No **2303031250067** has successfully completed his/her laboratory experiments in the **Cyber Security Laboratory** (303105342) from the department of COMPUTER SCIENCE & ENGINEERING during the academic year 2025-2026.

Date of Submission:

Staff In charge:

Head of Department:

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PRACTICAL NO: 1

Aim: Implementation to gather information from any PC's connected to the LAN using who.is, port scanners, network scanning, Angry IP scanners etc.

Objective : To know how to gather information about the networks by using different n/w reconnaissance tools.

Requirements : Laptop, who.is, nmap, angry ip scanner.

1. Who.is

- **Theory**

whois search for an object in a WHOIS database. WHOIS is a query and response protocol that is widely used for querying databases that store the registered users of an Internet resource, such as a domain name or an IP address block but is also used for a wider range of other information. Most modern versions of whois try to guess the right server to ask for the specified object. If no guess can be made, whois will connect to whois.networksolutions.com for NIC handles or whois.arin.net for IPv4 addresses and network names.

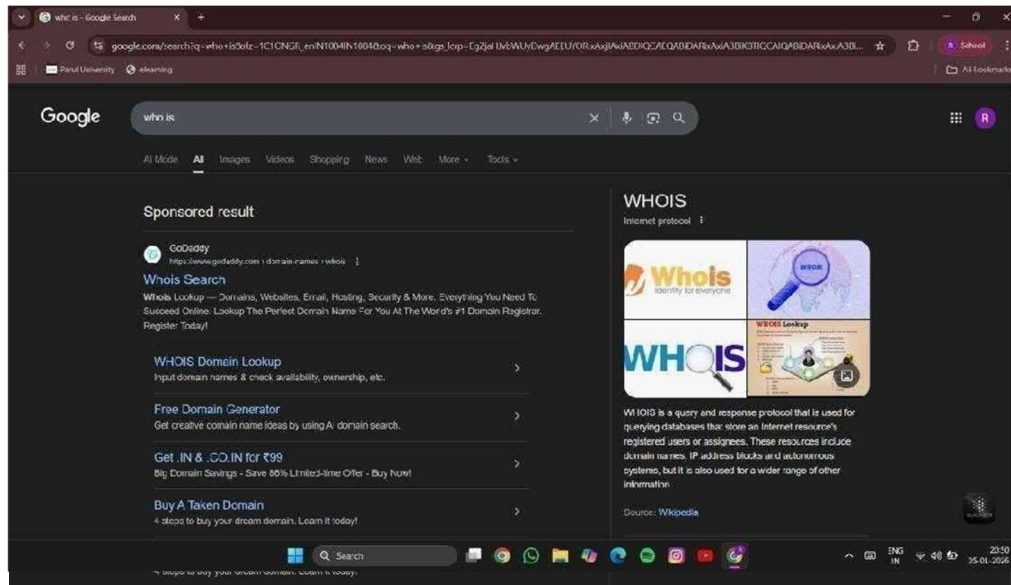
EXAMPLES:

- Obtaining the domain WHOIS record by IP Querying
- WHOIS record by IP querying
- Querying WHOIS in google search engine

To use the WHO.IS lookup tool, just enter the domain name whose information you'd like to view into the search field on the WHOIS main page. You can retrieve key data about a domain in this way, including availability, **domain owner lookup**, and creation and expiration details. If you own multiple domains of your own, it can be helpful to download exportable lists from the tool to analyze large amounts of domains data.

- Implementation

Step-1: search who.is in web browser



Step-2: open the who.is tool

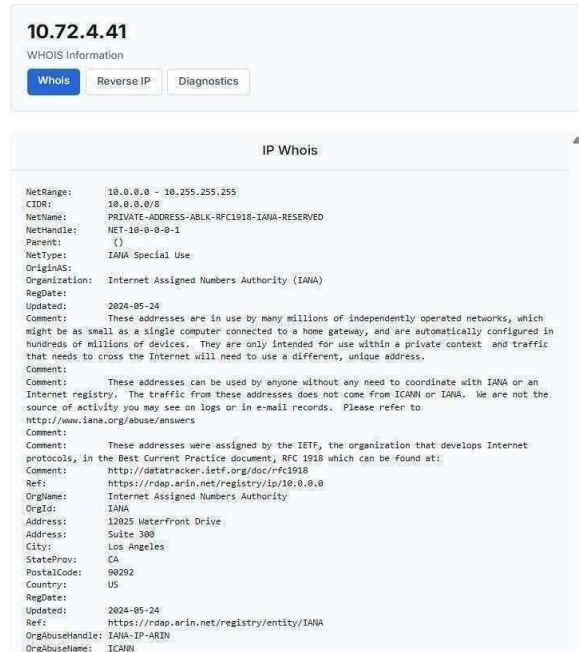


Step-3: Enter the Domain name or IP Address of which devices' information you want to gather



- **Observation**

Get the information about searched domain name or IP Address



2. Port Scanner

Nmap is convenient during penetration testing of networked systems. Nmap provides the network details, and also helps to determine the security flaws present in the system. Nmap is **platform-independent** and runs on popular **operating systems** such as **Linux**, **Windows** and **MAC**

Nmap is a useful tool for network scanning and auditing purposes.

- It can search for hosts connected to the Network.
- It can search for free **ports** on the target host.
- It detects all services running on the host with the help of **operating system**.
- It also detects any **flaws** or **potential vulnerabilities** in networked systems.

It is effortless to work with the Nmap. With the release of a new graphical user interface called **GenMap User**, it performs many tasks such as saving and comparing scan results, scanning the results in a database, and visualize the network system topology graphically, etc.

- **Advantage of Nmap:**

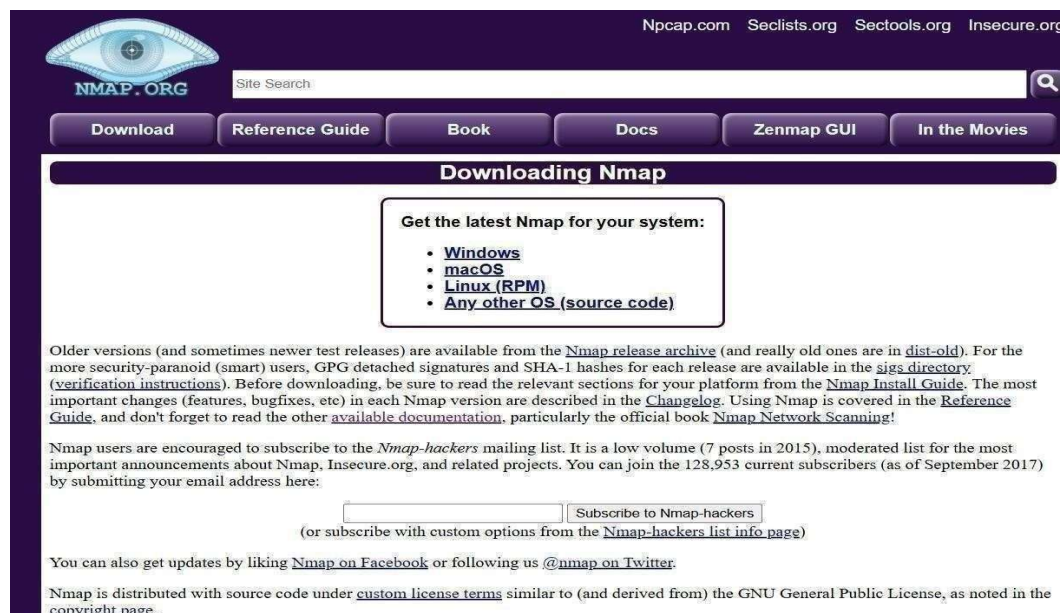
Nmap has a lot of advantages that make it different from open-source and **free** to use.

Some other advantages are listed below.

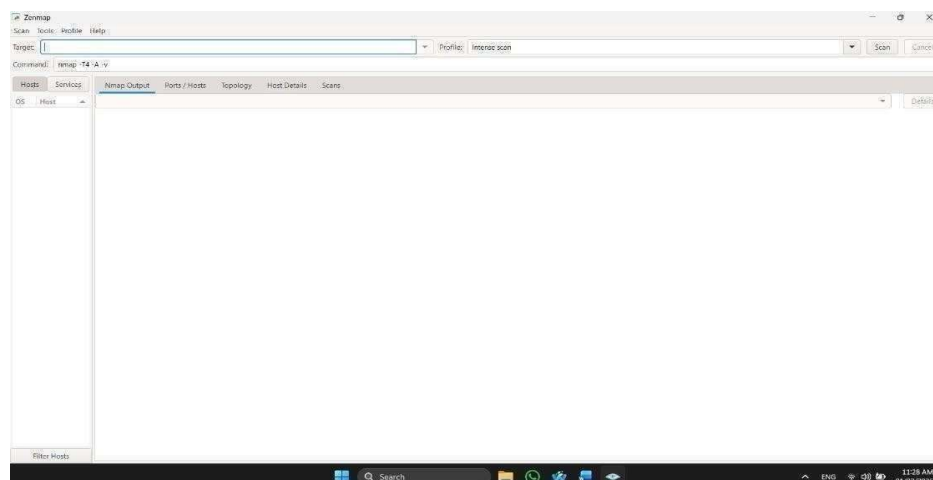
- It is used for auditing network systems as it can detect new servers.
- It will search for **subdomain** and Domain Name System.
- With the help of Nmap Scripting Engine (**NSE**), interaction can be made with the target host.
- It determines the nature of the service in the host and performs whether the host is a mail service or a web server

• Implementation

Step-1: Install Nmap in Windows



Step-2: Open Nmap



Step-3: Open command prompt and give command ipconfig to see local network

```

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

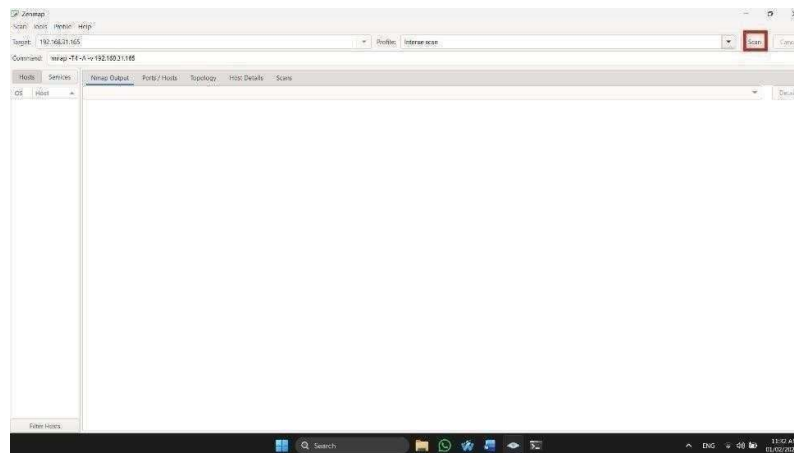
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::6a59:4d60:311f:14be%14
    IPv4 Address. . . . . : 10.72.4.41
    Subnet Mask . . . . . : 255.255.224.0
    Default Gateway . . . . . : 10.72.0.1

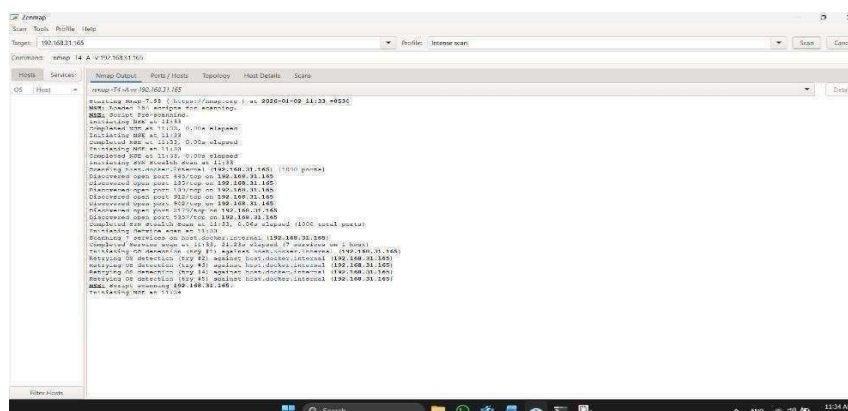
Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    
```

Step 4: paste ip address in target select scan type and press scan button



Nmap Output:



Step-5: Open Command prompt Run as administrator
To see the open ports run:

```
C:\Windows\System32>nmap solvetic.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-02 19:01 +0530
Nmap scan report for solvetic.com (178.33.118.246)
Host is up (0.22s latency).
rDNS record for 178.33.118.246: mail.solvetic.com
Not shown: 984 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
25/tcp    filtered smtp
53/tcp    open  domain
80/tcp    open  http
110/tcp   open  pop3
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
143/tcp   open  imap
443/tcp   open  https
445/tcp   filtered microsoft-ds
465/tcp   open  smtps
587/tcp   open  submission
993/tcp   open  imaps
995/tcp   open  pop3s
2875/tcp  filtered dxmessagebase2

Nmap done: 1 IP address (1 host up) scanned in 92.08 seconds
```

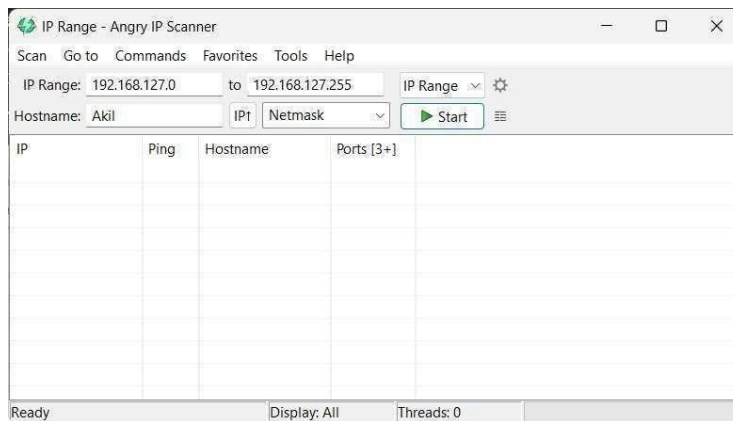
3. Angry IP Scanner

- **Implementation**

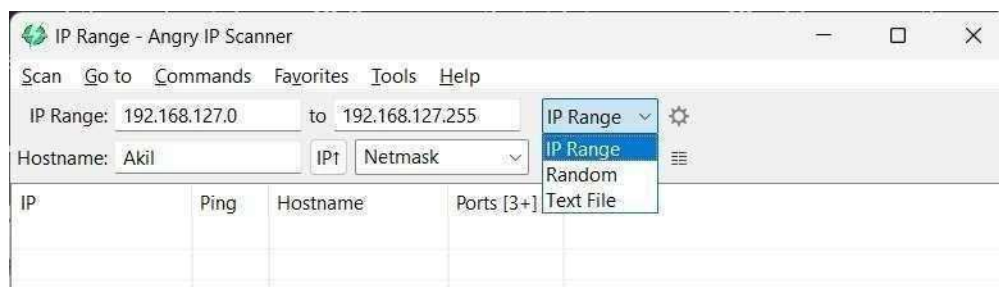
Install Angry IP Scanner



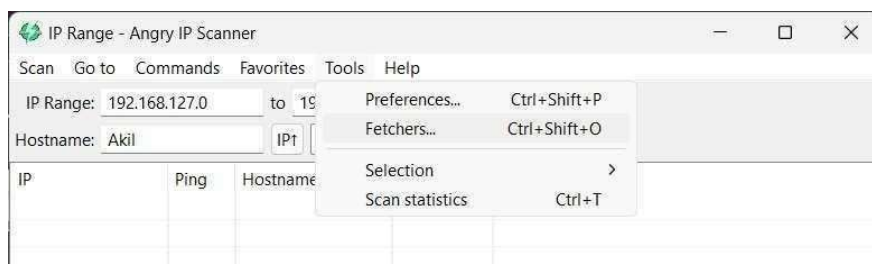
Once installed, open the application by searching for it in the Start Menu. As you can see, the home screen of the application is pretty simple and straightforward. By default, Angry IP scanner will enter your local IP address range and your computer name as the hostname.



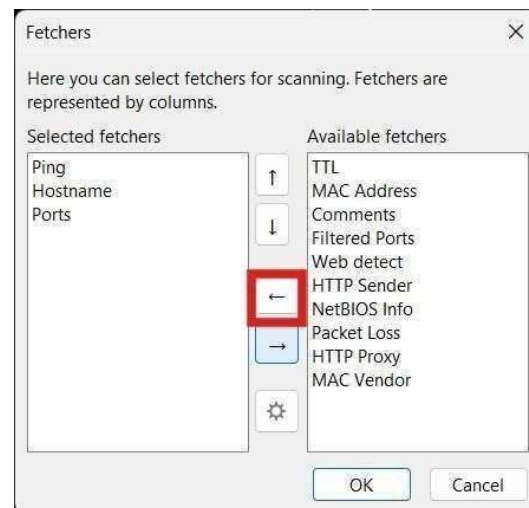
The good thing about Angry IP Scanner is that it lets you scan IP addresses in three different ways. They are, the range you specified, a random IP address or a list of IP addresses from a text file. You can easily select the scan mode from the drop-down menu next to the IP address field.



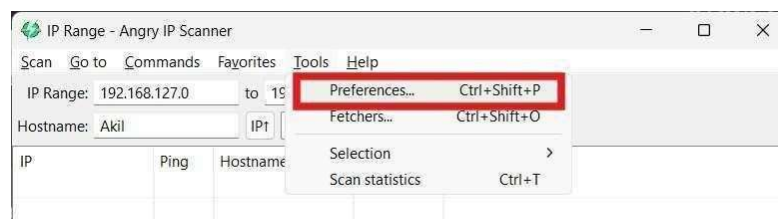
As you can see from the above image, the Angry IP Scanner will only include default fetchers like Ping, Hostname, and Ports. However, you can add more fetchers to get and see more information about an IP address. To do that, select “Tools > Fetchers.”



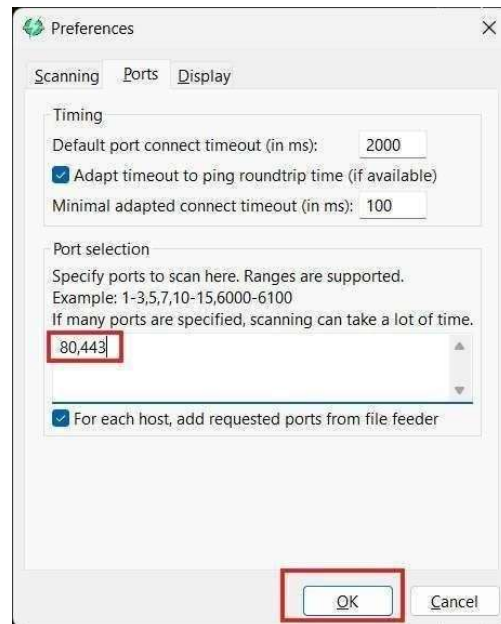
In this window, you will see all the current fetchers on the left pane and all the available fetchers in the right pane. To add a fetcher, select the fetcher on the right pane and then click on the button that looks like “Less than” sign. In my case, I’ve added new fetchers like MAC address, NetBIOS info, filtered ports, and Web detectors.



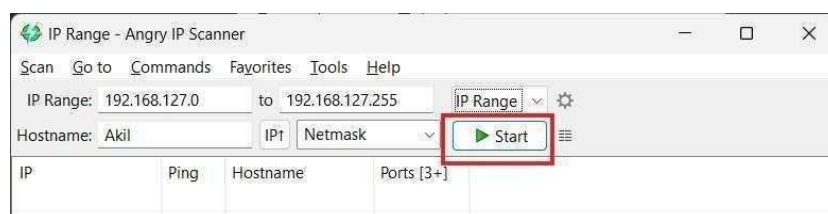
Moreover, Angry IP Scanner will only tell whether the ports are open or not. It will not list the individual ports that are open. So, if you want to do a port scan, then you need to configure the application. To do that, simply navigate to “Tools” and then select the option “Preferences.”



Here, navigate to the “Ports” tab and enter the ports you want to scan in the “Ports Selection” field. As you can see, I’m scanning for port 80 and port 443. If you want to scan a range of ports, then enter the port range like “1-1000.” After you have entered the port number, click OK to save the changes.



Once you are done configuring the Angry IP Scanner, you can continue to scan. To start off, set the scan mode to “IP Range,” enter the IP address range in the “IP address” fields and then click on the button “Start.” For instance, I’ve entered an IP range that is known to have live devices connected to it.



Depending on the number of addresses in the range, it may take some time to complete. Once completed, the application will show you a summary of the scan. The summary includes the number of hosts that are alive and the number of hosts that have open ports. Just click on the button “Close” to continue.



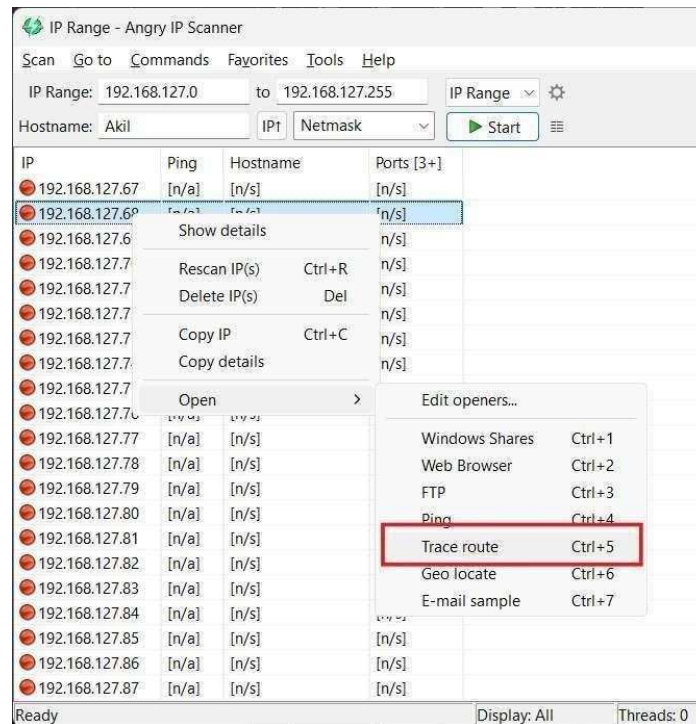
Once you close the summary window, you will see the list of all the IP addresses. You can also see additional details in different “fetcher” columns. In case you are wondering, here’s what the coloured dots next to each IP address mean.

Red: The IP address is inactive, dead or there is no device connected to this IP address.

Blue: The IP address is either active or busy and not responding to the requests sent by Angry IP Scanner. This usually will be your own IP Address.

Green: The IP address is active, and the device connected to it is responding to the requests made by Angry IP Scanner. There may also be open ports.

Apart from copying the details of an IP address, you can also perform a range of different activities on the entries. You can open an IP address in the web browser, do an FTP, trace routing, etc. For instance, if you want to traceroute an IP address, simply right-click on the target IP address. After that, select the option Open and click on Traceroute.



Once you are done scanning an IP address or the IP address range, you can save the scan results. To do that, select the option Scan from the menu bar.

PRACTICAL NO:2

Aim: Experiments with open source firewall/proxy packages like iptables, squid etc.

- **IP Tables:**

- Simply put, iptables is a firewall program for Linux. It will monitor traffic from and to your server using tables. These tables contain sets of rules, called **chains**, that will filter incoming and outgoing data packets.
- When a packet matches a rule, it is given a target, which can be another chain or one of these special values:
- **ACCEPT** – will allow the packet to pass through.
- **DROP** – will not let the packet pass through.
- **RETURN** – stops the packet from traversing through a chain and tell it to go back to the previous chain.

- **Iptables chains:**

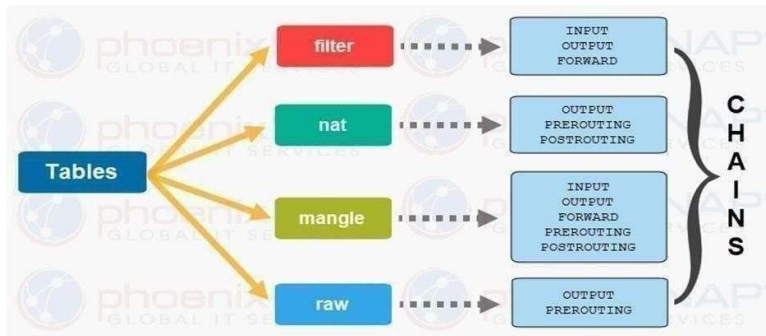
- In this iptables practical, we are going to work with one of the default tables, called filter. It consists of three chains:
- **INPUT** – controls incoming packets to the server.
- **FORWARD** – filters incoming packets that will be forwarded somewhere else.
- **OUTPUT** – filter packets that are going out from your server.

- **How to Install and Use Iptables Linux Firewall**

- Installing of Iptables
- sudo apt-get update
- sudo apt-get install iptables
- Check the status of your current iptables configuration by running.
- sudo iptables -L

```
(kali@kali)-[~]  
$ sudo iptables -L  
[sudo] password for kali:  
Chain INPUT (policy ACCEPT)  
target    prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target    prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target    prot opt source                destination
```

- Types of tables in iptables



- Commands:

- Iptables -A INPUT -s your ip -j DROP
- Iptables -A INPUT -s your ip -j ACCEPT
- ➔ to stop https websites
- ➔ to stop http websites
- Iptables -A OUTPUT -p tcp -o eth0 -s your ipaddress --dport 80 -j DROP

- Follow the instructions to create a firewall rule that prevents the communication between two systems via ping.

Step-1 : CHECK YOUR IPADDRESS IN BOTH OPERATING SYSTEM(**KALI LINUX AND WINDOWS**) BY FOLLWING COMMANDS IN THE BELOW PICTURES.

```
(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.236.128 netmask 255.255.255.0 broadcast 192.168.236.255
    inet6 fe80::d96a:4096:b6e2:2076 prefixlen 64 scopeid 0<20<link>
    ether 00:0c:29:4d:b2:ec txqueuelen 1000 (Ethernet)
    RX packets 27 bytes 12184 (11.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55 bytes 12436 (12.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Step-2: Attempt to ping both devices to verify whether they are responsive or not.

```
(kali㉿kali)-[~]
$ ping 192.168.31.165
PING 192.168.31.165 (192.168.31.165) 56(84) bytes of data:
64 bytes from 192.168.31.165: icmp_seq=1 ttl=128 time=0.879 ms
64 bytes from 192.168.31.165: icmp_seq=2 ttl=128 time=0.984 ms
64 bytes from 192.168.31.165: icmp_seq=3 ttl=128 time=0.695 ms
64 bytes from 192.168.31.165: icmp_seq=4 ttl=128 time=1.20 ms
64 bytes from 192.168.31.165: icmp_seq=5 ttl=128 time=1.01 ms
64 bytes from 192.168.31.165: icmp_seq=6 ttl=128 time=1.09 ms
64 bytes from 192.168.31.165: icmp_seq=7 ttl=128 time=1.12 ms
64 bytes from 192.168.31.165: icmp_seq=8 ttl=128 time=0.676 ms
64 bytes from 192.168.31.165: icmp_seq=9 ttl=128 time=1.05 ms
64 bytes from 192.168.31.165: icmp_seq=10 ttl=128 time=1.14 ms
64 bytes from 192.168.31.165: icmp_seq=11 ttl=128 time=1.02 ms
64 bytes from 192.168.31.165: icmp_seq=12 ttl=128 time=0.891 ms
64 bytes from 192.168.31.165: icmp_seq=13 ttl=128 time=1.17 ms
64 bytes from 192.168.31.165: icmp_seq=14 ttl=128 time=0.692 ms
64 bytes from 192.168.31.165: icmp_seq=15 ttl=128 time=0.838 ms

PS C:\Users\khatr> ping 192.168.236.128

Pinging 192.168.236.128 with 32 bytes of data:
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.236.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\khatr> |
```

Step-3 : Please create an iptables rule to stop communication between both devices, as shown in the picture.

```
(kali㉿kali)-[~]  
$ sudo iptables -A INPUT -s 192.168.31.165 -j DROP
```

Step-4 : Now try to ping on both devices its working or not.

```
(kali㉿kali)-[~]  
$ ping 192.168.31.165  
PING 192.168.31.165 (192.168.31.165) 56(84) bytes of data.  
^C  
— 192.168.31.165 ping statistics —  
19 packets transmitted, 0 received, 100% packet loss, time 18439ms
```

C:\Users\Rakshith>ping 192.168.185.141

Pinging 192.168.185.141 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.185.141:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Step-5 : Again if you want resume the communication then you have to write another rule to “**ACCEPT**” the communication both device by iptable rule. Before you accept the rule We have to remove the previous rule which we have given the “**DROP**” rule.

```
(kali㉿kali)-[~]  
$ sudo iptables -D INPUT -s 192.168.31.165 -j DROP  
  
(kali㉿kali)-[~]  
$
```

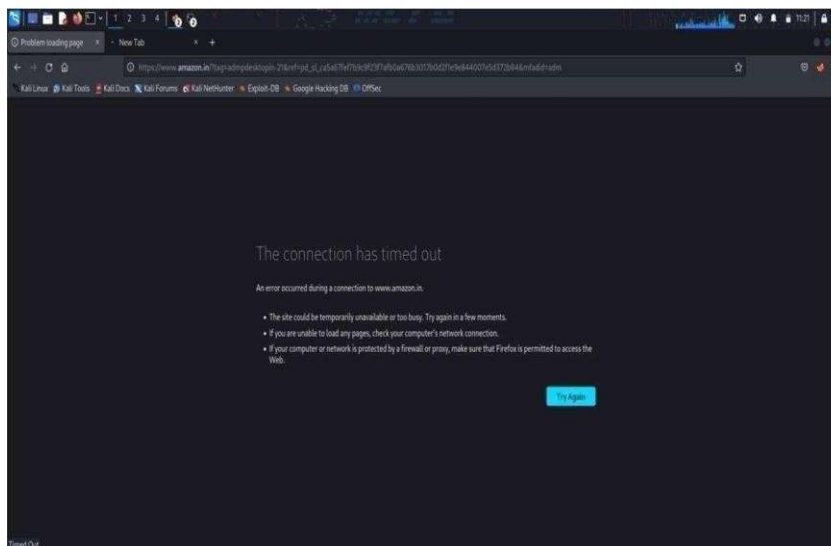
```
(kali㉿kali)-[~]  
$ ping 192.168.31.165  
PING 192.168.31.165 (192.168.31.165) 56(84) bytes of data:  
64 bytes from 192.168.31.165: icmp_seq=1 ttl=128 time=1.26 ms  
64 bytes from 192.168.31.165: icmp_seq=2 ttl=128 time=0.962 ms  
64 bytes from 192.168.31.165: icmp_seq=3 ttl=128 time=1.26 ms  
64 bytes from 192.168.31.165: icmp_seq=4 ttl=128 time=1.31 ms  
64 bytes from 192.168.31.165: icmp_seq=5 ttl=128 time=1.59 ms  
^C  
— 192.168.31.165 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4007ms  
rtt min/avg/max/mdev = 0.962/1.277/1.591/0.199 ms  
  
(kali㉿kali)-[~]  
$
```


Step-6: To remove the rule from iptable as see in picture.

```
(kali㉿kali)-[~]  
$ sudo iptables -D INPUT 1  
[sudo] password for kali:  
iptables: No chain/target/match by that name.
```

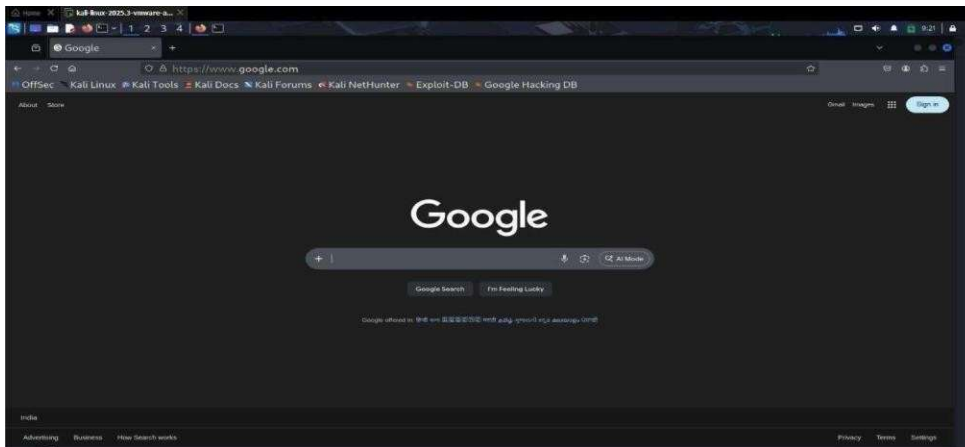
Step-7: To stop https request in our browser by ip address.

```
(kali㉿kali)-[~]  
$ sudo iptables -A OUTPUT -p tcp -o eth0 -s 192.168.236.128 --dport 443 -j DROP  
(kali㉿kali)-[~]  
$
```



Step-8 : To resume the https request in browser by accept rule

```
(kali㉿kali)-[~]  
$ sudo iptables -A OUTPUT -p tcp -o eth0 -s 192.168.236.128 --dport 443 -j ACCEPT  
(kali㉿kali)-[~]  
$
```



PRACTICAL NO:3

Aim : Implementation of Steganography.

Scope : Encompasses the detection of concealed messages and data within various media files, emphasizing the application of steganographic methods for security and forensic purposes.

Requirements : The requirements are Hardware and Devices, Steg software, S-Tools, sample image, Documentation Templates etc.

- **Theory:**

Steganography

- The root “steganos” is Greek for “hidden” or “covered,” and the root “graph” is Greek for “to write.” Steganography is the practice of hiding a secret message inside of (or even on top of) something that is not secret.
- Examples of steganography involve embedding a secret piece of text inside of a picture. Or hiding a secret message or script inside of a Word or Excel document. The purpose of steganography is to conceal and deceive. It is a form of covert communication and can involve the use of any medium to hide messages.
- It’s not a form of cryptography, because it doesn’t involve scrambling data or using a key. Instead, it is a form of data hiding and can be executed in clever ways.

Types of Steganography?

1. Text steganography

Text steganography conceals a secret message inside a piece of text. The simplest version of text steganography might use the first letter in each sentence to form the hidden message. Other text steganography techniques might include adding meaningful typos or encoding information through punctuation.

2. Image steganography

In image steganography, secret information is encoded within a digital image. This technique relies on the fact that small changes in image color or noise are very difficult to detect with the human eye. For example, one image can be concealed within another by using the least significant bits of each pixel in the image to represent the hidden image instead.

3. Video steganography

Video steganography is a more sophisticated version of image steganography that can encode entire videos. Because digital videos are represented as a sequence of consecutive images, each video frame can encode a separate image, hiding a coherent video in plain sight.

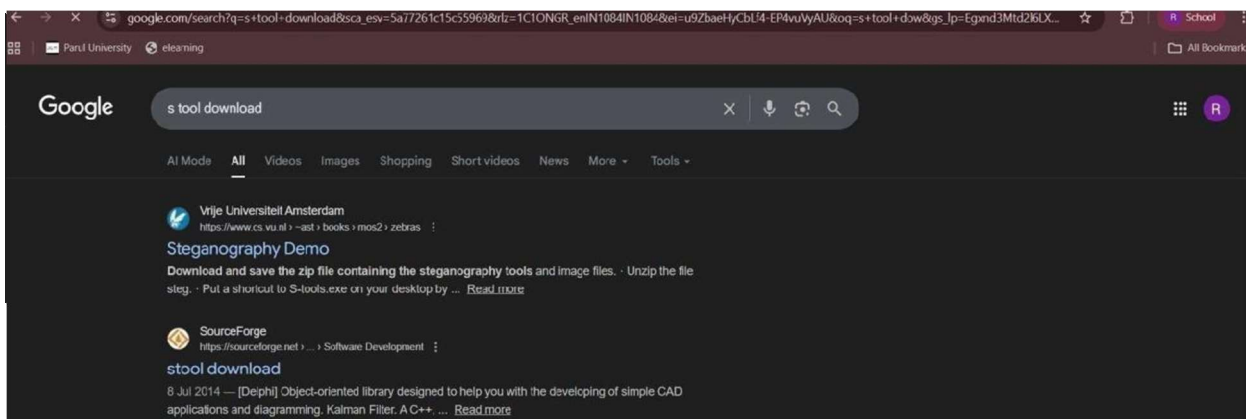
4. Audio steganography

Audio files, like images and videos, can be used to conceal information. One simple form of audio steganography is “backmasking,” in which secret messages are played backwards on a track (requiring the listener to play the entire track backwards). More sophisticated techniques might involve the least significant bits of each byte in the audio file, similar to image steganography.

5. Network steganography

Last but not least, network steganography is a clever digital steganography technique that hides information inside network traffic. For example, data can be concealed within the TCP/IP headers or payloads of network packets. The sender can even impart information based on the time between sending different packets.

- **Download S tools**



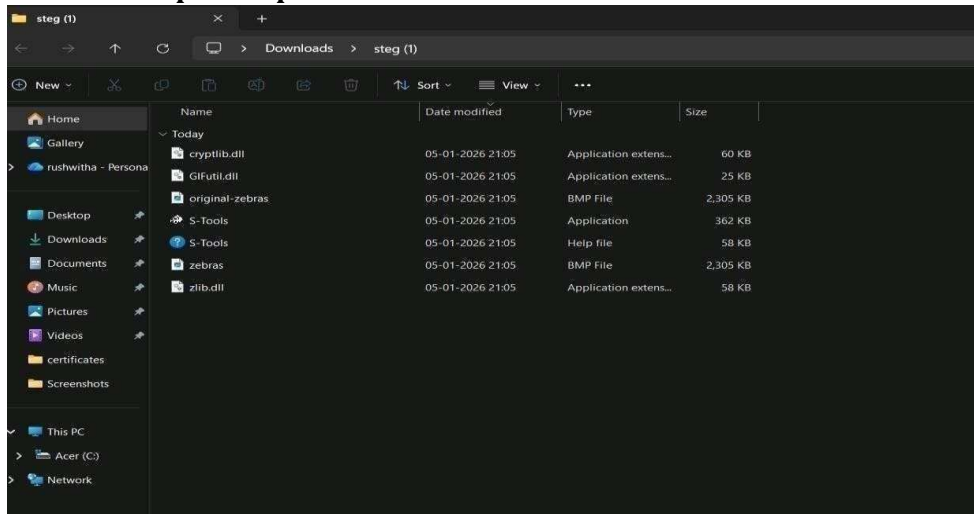
Steganography Demo

To try this demo you need a Windows 98/Me/NT/2000/XP/Vista/Win 7 computer. Please follow the instructions below.

1. Download and save the zip file containing the [steganography tools and image files](#).
2. Unzip the file *steg.zip* into an empty directory. If you do not have the *unzip* program already, you can get one [here](#).
3. Put a shortcut to *S-tools.exe* on your desktop by dragging it from Windows Explorer.
4. Drag the *zebras.bmp* file to your desktop. Do not make a shortcut. The file itself must be moved there.
5. Start *S-tools.exe* by double clicking on the icon on the desktop. A window will appear.
6. Drag the *zebras.bmp* file to the S-tools window.
7. Right click on the zebras pictures and select Reveal from the menu.
8. Fill in the 3-character pass phrase 'abc' (without the quotes) in two places. Leave IDEA as the encryption algorithm. Click on OK.
9. Wait until the Revealed Archive dialog box appears. This may take a minute or two.
10. Right click on any item and select Save As to save the file. Repeat for the other ones. These are the hidden files.
11. The file *original-zebras.bmp* is the file before the steganography was done, in case you wish to compare the 'before' and 'after' images.

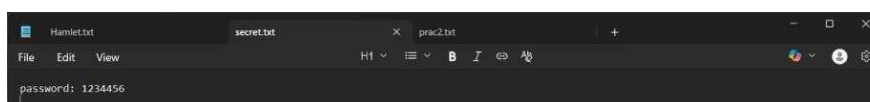
Note: The image at the left has been compressed for quick downloading of this page. It does not contain any secret messages so do not download it unless you really like zebras or [my photography](#).

Extract the zip file in particular folder



• Examination:

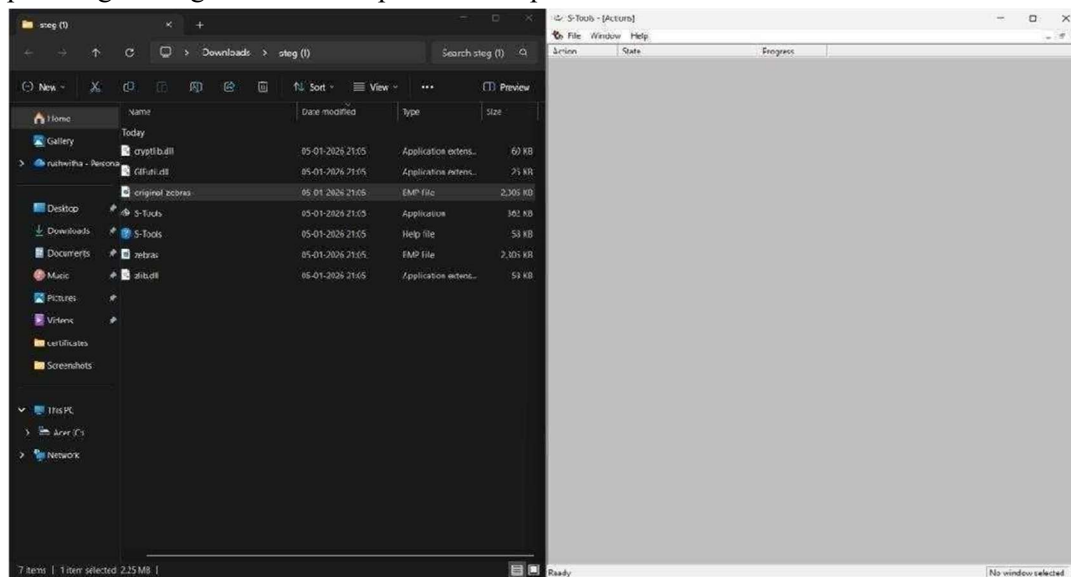
Step-1: Write a secret message in .txt format (Ex: evidence a.txt).



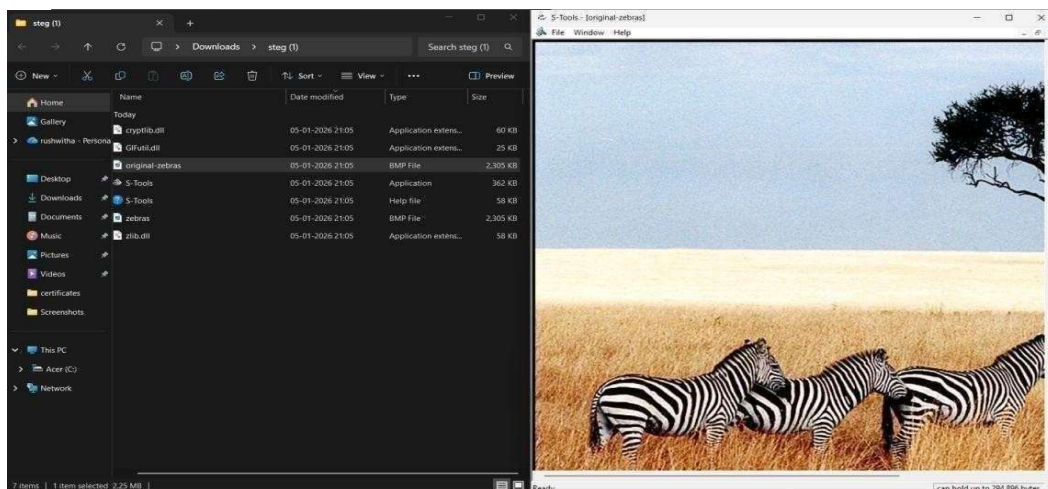
Step 2: Open the S tools



Step 3: Drag the original-zebras.bmp file and drop it into S tools.



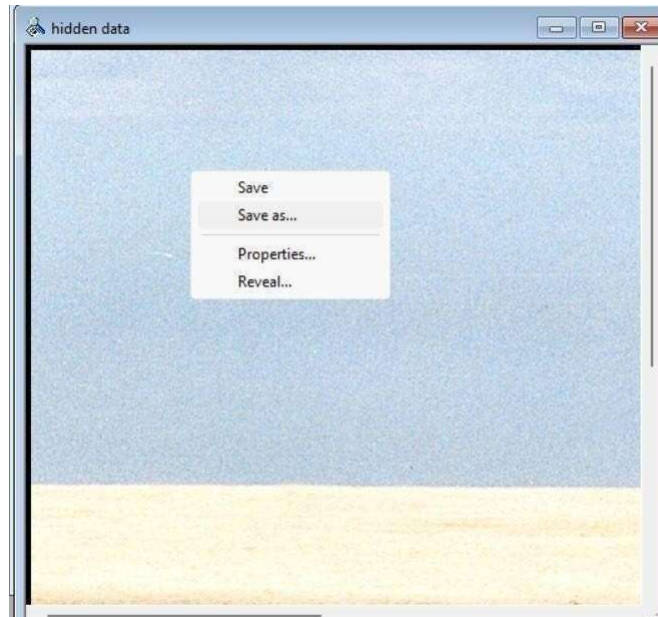
Step 4: Using S-Tools, drag and drop the secret message file on top of image file.



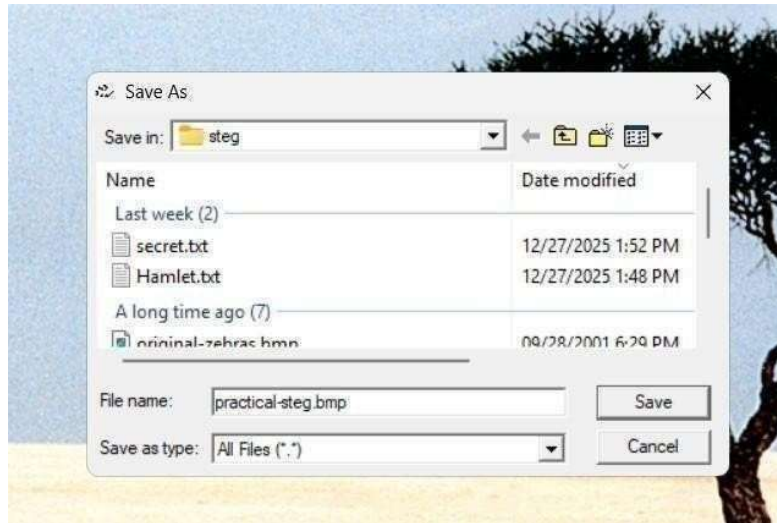
Step 5: Now to encrypt the stego file. Create a new password for the encryption and click on the “OK” button.



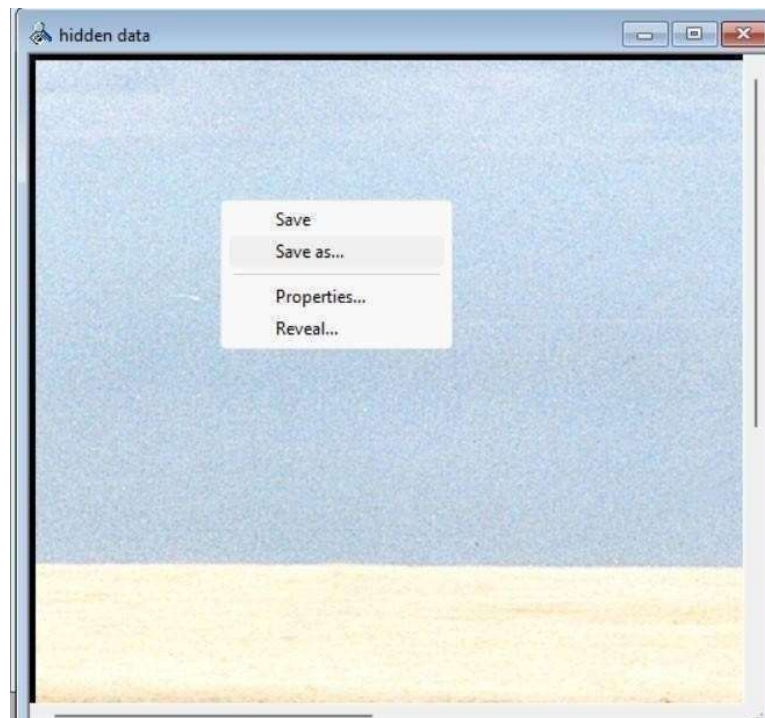
Step 6: generate a new stego image. To save the stego file, right-click on the image and select the “Save as...” option.



Step 7: Specify the destination to save the stego image and click on the “Save” button in order to save the file use .BMP extension.



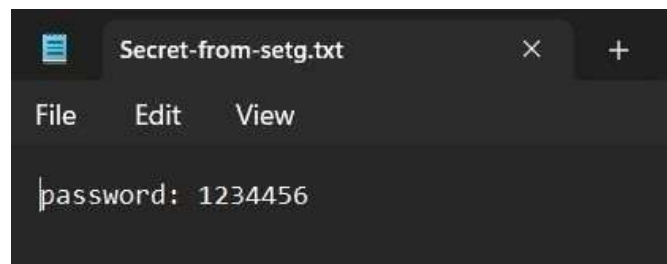
Step 8: Now, to extract the concealed information from the stego image, run the S-Tools and then drag and drop suspected file into it. Right-click on the image and select the “Reveal” option from the top-down menu.



Step 9: “Revealed Archive” window displaying the secret message file name.



Step 10 : Now, Right-click on the file name and then select the “Save as...” option to save it in a location. Next check the extracted file.



PRACTICAL NO: 4

Aim: Implementation of MITM- attack using Wireshark / network sniffers.

Tools: Wireshark or Ettercap

We'll need a client machine as well whose network traffic we will spoof and sniff to get cleartext submission of passwords from certain vulnerable websites.

- **Ettercap: Ettercap is GUI based tool built into Kali so need to download and install anything, so let's get started doing a MiTM attack with Ettercap.**

Start Ettercap

Let's view the help file for ettercap by typing; kali > ettercap -h



```
root@kali:~# ettercap -h
ettercap 0.8.0 copyright 2001-2013 Ettercap Development Team

Usage: ettercap [OPTIONS] [TARGET1] [TARGET2]

TARGET is in the format MAC/IP/PORTs (see the man for further detail)

Sniffing and Attack options:
-M, --mitm <METHOD:ARGS>    perform a mitm attack
-o, --only-mitm              don't sniff, only perform the mitm attack
-b, --broadcast              sniff packets destined to broadcast
-B, --bridge <IFACE>         use bridged sniff (needs 2 ifaces)
-p, --nopromisc              do not put the iface in promisc mode
-S, --noSSLmitm              do not forge SSL certificates
-u, --unoffensive            do not forward packets
-r, --read <file>            read data from pcapfile <file>
-f, --pcapfilter <string>    set the pcap filter <string>
-R, --reversed               use reversed TARGET matching
-t, --proto <proto>          sniff only this proto (default is all)
--certificate <file>         certificate file to use for SSL MiTM
--private-key <file>         private key file to use for SSL MiTM

User Interface Type:
-T, --text                   use text only GUI
-q, --quiet                  do not display packet contents
-s, --script <CMD>           issue these commands to the GUI
-C, --curses                 use curses GUI
-D, --daemon                 daemonize ettercap (no GUI)
-G, --gtk                    use GTK+ GUI
```

As you can see, ettercap has a significant help file for running it from a command line, but the only thing we need from here is the switch to run it in graphical mode. In the bottom line of the screenshot (not the bottom line of the actual help file as I have truncated it in the interest of space), you can see the -G switch. This after the command ettercap will launch the ettercap GUI.

Start Ettercap GUI

kali > ettercap -G

When we do, the ettercap GUI will start as seen below.



The first step in launching our MiTM attack is to start sniffing. Go to pulldown menu that says "Sniff" and click on "Sniffing at startup".

When we do that, it asking us what interface we want to use and defaults to eth0.

Then click on TICK mark



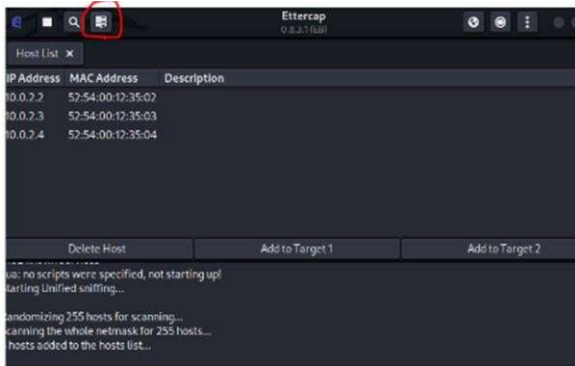
When we click "OK", ettercap launches it sniffing and loads its plugins.

Our next step to find the hosts on the network. Click on the "Hosts" tab and you will see a menu that includes "Scan for Hosts". Click on it and ettercap will begin scanning the network for hosts.



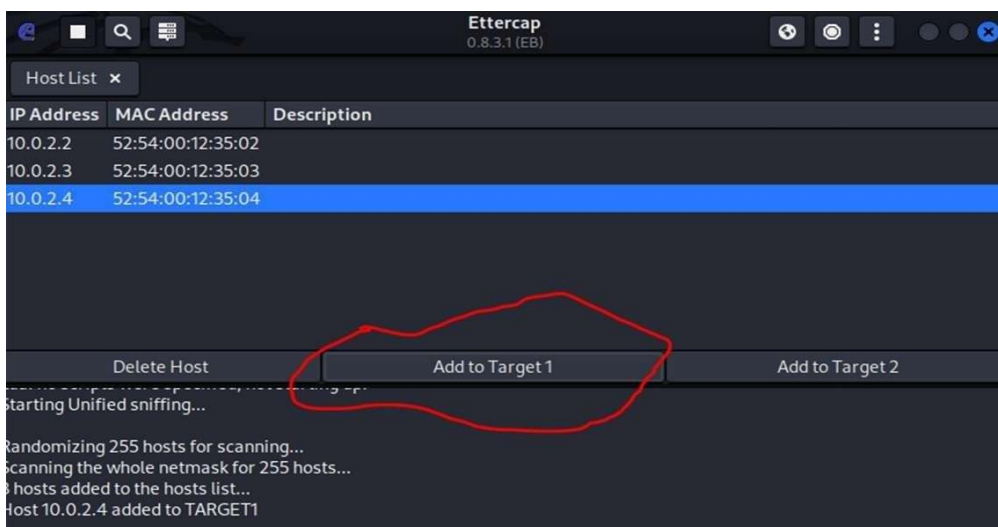
Now, using that same "Hosts" tab, click on "Hosts List". This will display all the hosts that ettercap has discovered on your network as seen in the screenshot below.

En no :2303031250067

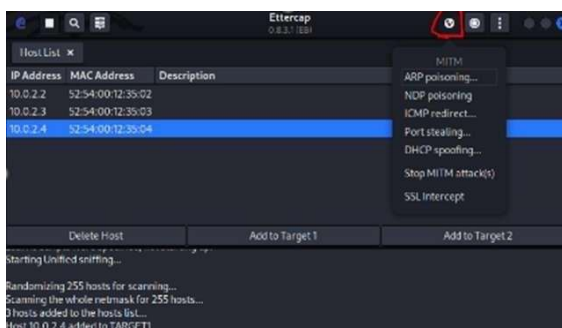


Now, select one of the hosts that will be the target of this attack in the window by clicking on it and then click on "Add to Target 1" at the bottom of the window. When you do so, ettercap will add that host as the first target in our MiTM attack as seen in the screenshot below.

Next, select the second host in this attack and then click "Add to Target 2".

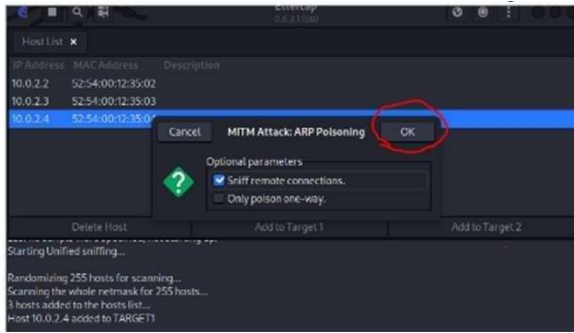


Finally, go to the menu above and click on MITM tab and the drop down menu will have a selection called "ARP Poisoning" as seen in the screenshot below.



Select it and it will open a pop window like below. Select "Sniff remote connections". When we press OK, ettercap will begin ARP poisoning and you will see ettercap respond in its main windows with the message below.

Now, we have successfully placed ourselves between the two targets systems and all their traffic must flow through us. This is where the fun begins as we can now delete, manipulate, impersonate and view



Host 192.168.1.116 added to TARGET1

Host 192.168.1.103 added to TARGET2

ARP poisoning victims:

GROUP 1 : 192.168.1.116 08:00:27:46:69:8D

GROUP 2 : 192.168.1.103 70:1A:04:F4:B9:D0

Now, open any browser on target IP address system and open any HTTP websites of LOGIN PAGE type your id and password then it will show output in ettercap output panel.

• Wireshark

A basic setup is complete and victim network traffic will now pass through the attacker machine. To listen to these packets, we will use Wireshark.

- Open up a new terminal and type wireshark. Go to the interface which is capturing all the data flow (here eth0) and start the capture.
- Filter out packets according to what you are looking for. For the purpose of this demo, the user is logging in to a vulnerable website DVWA which uses HTTP instead of the secure version HTTPS. Filter protocol as http and search for required data.

Disclaimer: This tutorial is purely intended for educational purposes and should not be misused.

- Right click on the packet and follow TCP stream to open up the data contained within. We can clearly obtain the login credentials of the user, that is the username and password.

MITM is one of the classic hacks and on a LAN connection, ARP spoofing is much preferred. Today there have been various measures to prevent such an attack by use of HTTPS, use of VPN and, strong WEP/WAP encryption on access points.

PRACTICAL NO: 5

Aim : Implementation of Windows security using firewalls and other tools.

Tools : Implementation of Windows security using firewalls and other tools.

Description : Windows Firewall is a security feature that helps to protect your device by filtering network traffic that enters and exits your device. This traffic can be filtered based on several criteria, including source and destination IP address, IP protocol, or source and destination port number. Windows Firewall can be configured to block or allow network traffic based on the services and applications that are installed on your device. This allows you to restrict network traffic to only those applications and services that are explicitly allowed to communicate on the network.

The default behavior of Windows Firewall is to:

- block all incoming traffic, unless solicited or matching a rule
- allow all outgoing traffic, unless matching a rule

• Firewall Rules

Firewall rules identify allowed or blocked network traffic, and the conditions for this to happen. The rules offer an extensive selection of conditions to identify traffic, including:

- Application, service or program name
- Source and destination IP addresses
- Can make use dynamic values, like default gateway, DHCP servers, DNS servers and local subnets
- Protocol name or type. For transport layer protocols, TCP and UDP, you can specify ports or port ranges. For custom protocols, you can use a number between 0 and 255 representing the IP protocol
- Interface type
- ICMP/ICMPv6 traffic type and code

• Implementation

How can you blocked the program?

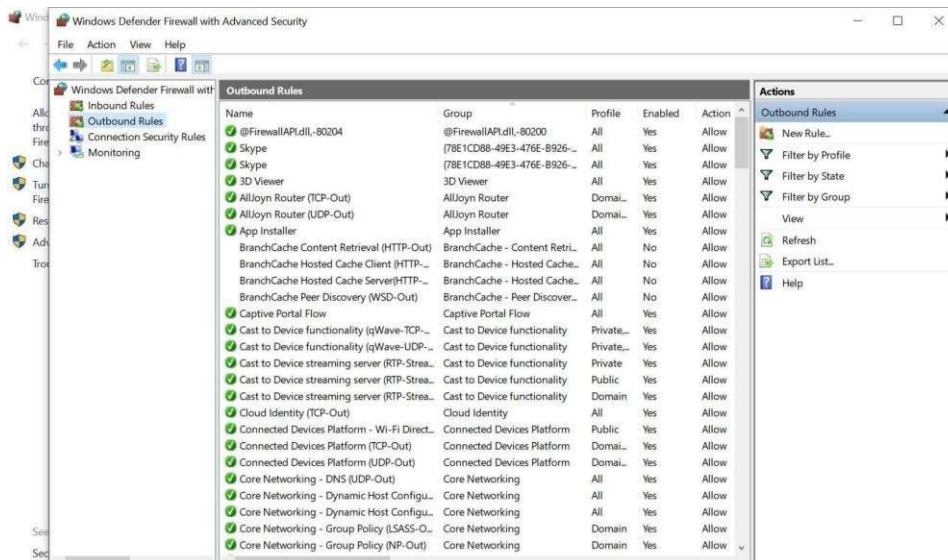
Step-1 : Open Windows defender Firewall



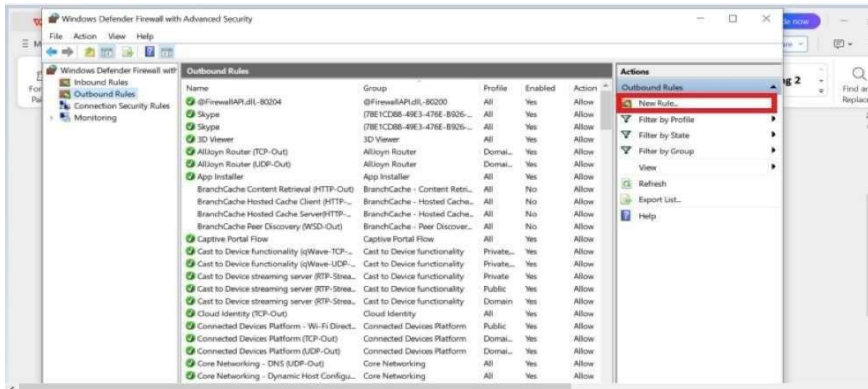
Step-2 : Select Advance Settings



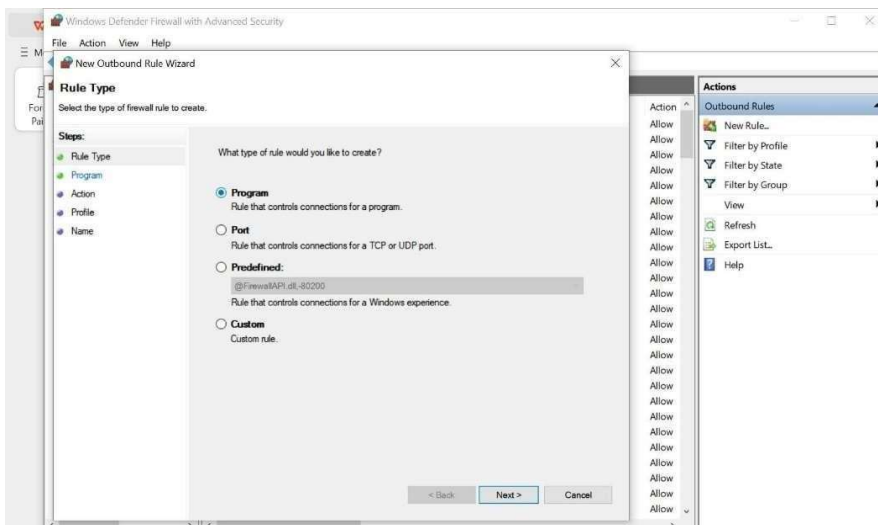
Step-3 : Select outbound Rules



Step-4 : Select New Rule



Step-5 : Select Program and Click on a next Button



PRACTICAL NO: 6

Aim : Implementation to identify web vulnerabilities, using OWASP project

Tools : OWASP ZAP Proxy (windows and kali)

Description :

- Introduction to ZAP

Zed Attack Proxy (ZAP) is a free, open-source penetration testing tool being maintained under the umbrella of the Open Web Application Security Project (OWASP). ZAP is designed specifically for testing web applications and is both flexible and extensible. At its core, ZAP is what is known as a “man-in-the-middle proxy.” It stands between the tester’s browser and the web application so that it can intercept and inspect messages sent between browser and web application, modify the contents if needed, and then forward those packets on to the destination. It can be used as a stand-alone application, and as a daemon process.



If there is another network proxy already in use, as in many corporate environments, ZAP can be configured to connect to that proxy.



ZAP provides functionality for a range of skill levels – from developers, to testers new to security testing, to security testing specialists. ZAP has versions for each major OS and Docker, so you are not tied to a single OS. Additional functionality is freely available from a variety of add-ons in the ZAP Marketplace, accessible from within the ZAP client. Because ZAP is open-source, the source code can be examined to see exactly how the functionality is implemented. Anyone can volunteer to work on ZAP, fix bugs, add features, create pull requests to pull fixes into the project, and author add-ons to support situations.

- Install and Configure ZAP

ZAP has installers for Windows, Linux, and Mac OS/X. There are also Docker images available on the download site listed below.

Install ZAP The first thing to do is install ZAP on the system you intend to perform pen testing on. Download the

appropriate installer from ZAP's download location at <https://www.zaproxy.org/download/> and execute the installer.

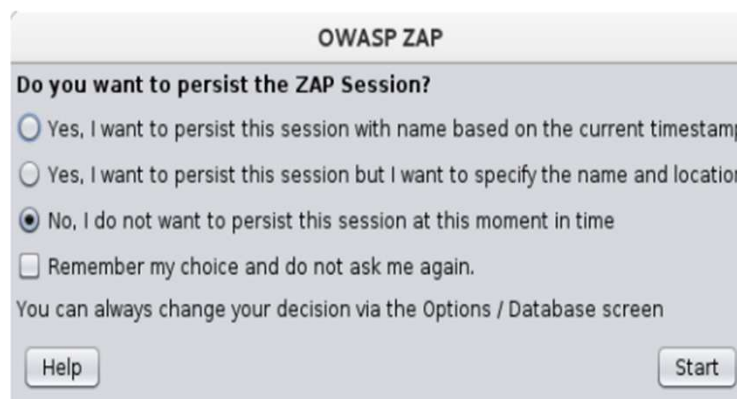
Note that ZAP requires Java 8+ in order to run. The Mac OS/X installer includes an appropriate version of Java but you must install Java 8+ separately for Windows, Linux, and Cross-Platform versions. The Docker versions do not require you to install Java.

Once the installation is complete, launch ZAP and read the license terms. Click Agree if you accept the terms, and ZAP will finish installing, then ZAP will automatically start.

Persisting a Session When you first start ZAP, you will be asked if you want to persist the ZAP session. By default, ZAP sessions are always recorded to disk in a HSQLDB database with a default name and location.

If you do not persist the session, those files are deleted when you exit ZAP.

If you choose to persist a session, the session information will be saved in the local database so you can access it later, and you will be able to provide custom names and locations for saving the files.



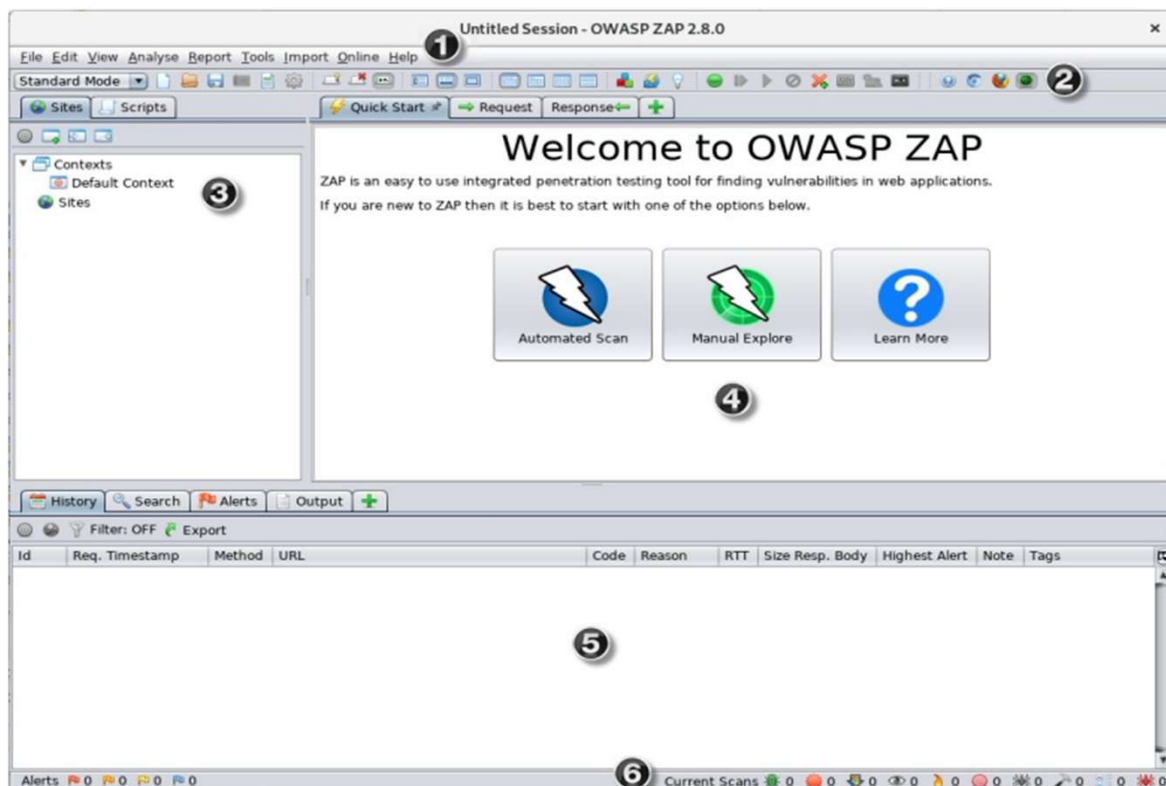
For now, select No, I do not want to persist this session at this moment in time, then click Start. The ZAP sessions will not be persisted for now.

- Zap Desktop UI

The ZAP Desktop UI is composed of the following elements:

1. **Menu Bar** – Provides access to many of the automated and manual tools.
2. **Toolbar** – Includes buttons which provide easy access to most commonly used features.

3. **Tree Window** – Displays the Sites tree and the Scripts tree.
4. **Workspace Window** – Displays requests, responses, and scripts and allows you to edit them.
5. **Information Window** – Displays details of the automated and manual tools.
6. **Footer** – Displays a summary of the alerts found and the status of the main automated tools.



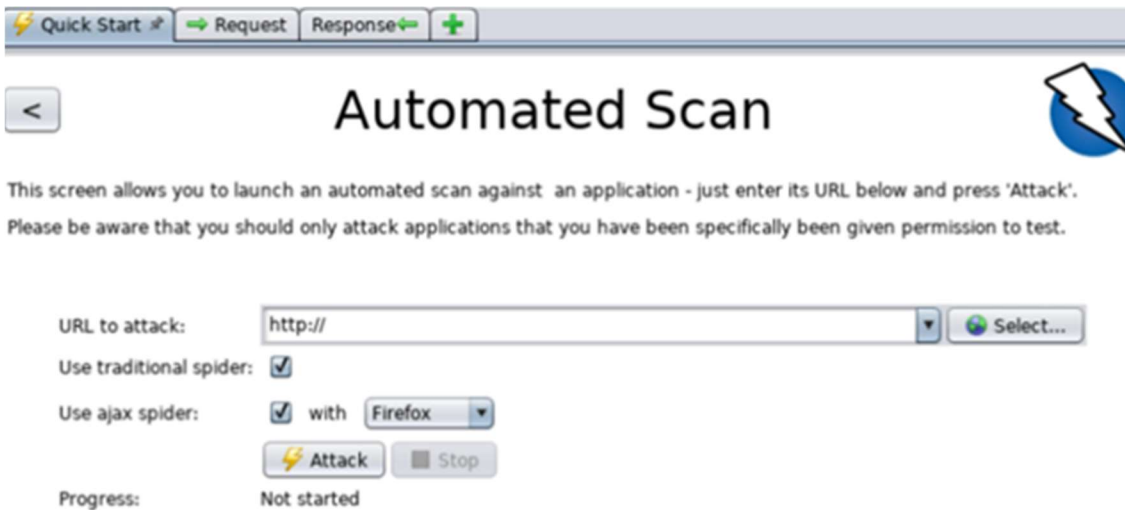
- Running an Automated Scan

The easiest way to start using ZAP is via the Quick Start tab. Quick Start is a ZAP add-on that is included automatically when you installed ZAP.

To run a Quick Start Automated Scan :

1. Start ZAP and click the Quick Start tab of the Workspace Window.
2. Click the large Automated Scan button.
3. In the URL to attack text box, enter the full URL of the web application you want to attack.

4. Click the Attack button.



○ ZAP will proceed to crawl the web application with its spider and passively scan each page it finds. Then ZAP will use the active scanner to attack all of the discovered pages, functionality, and

parameters.

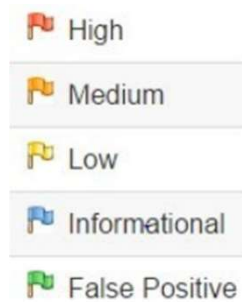
- ZAP provides 2 spiders for crawling web applications, you can use either or both of them from this screen.
- The traditional ZAP spider which discovers links by examining the HTML in responses from the web application. This spider is fast, but it is not always effective when exploring an AJAX web application that generates links using JavaScript.
- For AJAX applications, ZAP's AJAX spider is likely to be more effective. This spider explores the web application by invoking browsers which then follow the links that have been generated. The AJAX spider is slower than the traditional spider and requires additional configuration for use in a "headless" environment.
- ZAP will passively scan all of the requests and responses proxied through it. so far ZAP has only carried out passive scans of your web application. Passive scanning does not change responses in any way and is considered safe. Scanning is also performed in a background thread to not slow down exploration.
- Passive scanning is good at finding some vulnerabilities and as a way to get a feel for the basic security state of a web application and locate where more investigation may be warranted. Active scanning, however, attempts to find other vulnerabilities by using known attacks against the selected targets.
- Active scanning is a real attack on those targets and can put the targets at risk, so do not use active scanning against targets you do not have permission to test.

- **See Explored page**

To examine a tree view of the explored pages, click the Sites tab in the Tree Window. You can expand the nodes to see the individual URLs accessed.

- **View Alerts and Alert Details**

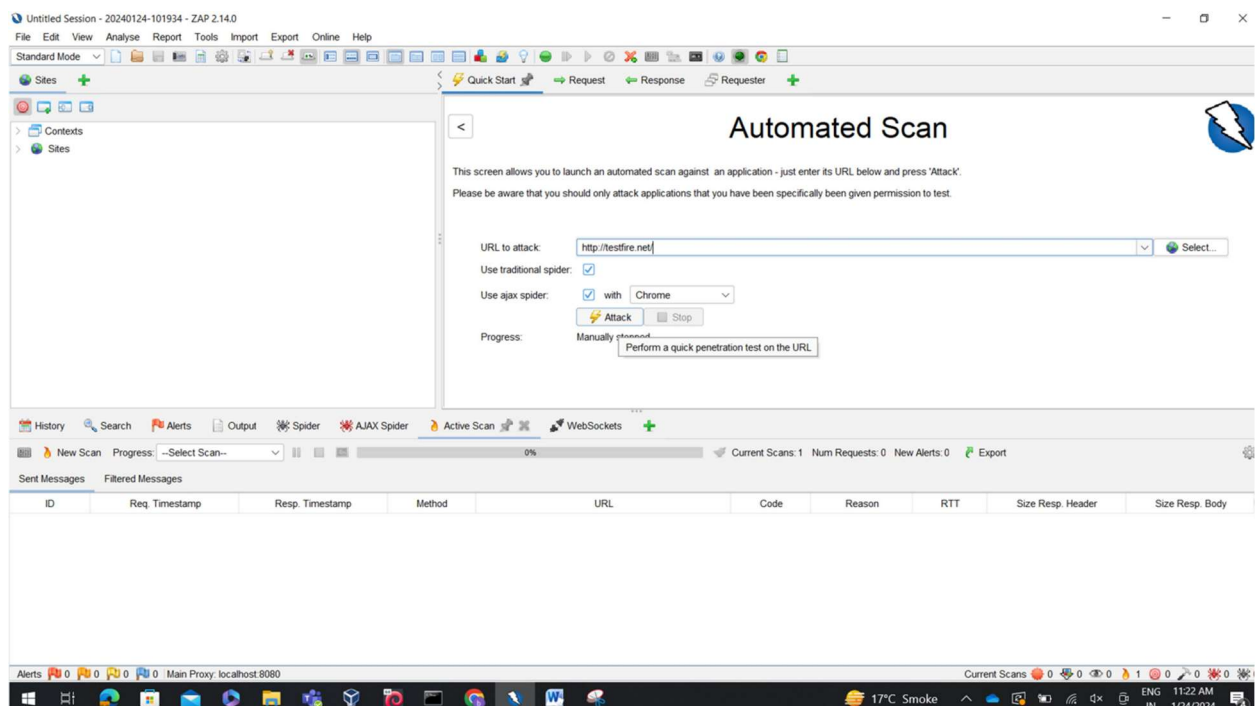
The left-hand side of the Footer contains a count of the Alerts found during your test, broken Out into risk categories. These risk categories are:



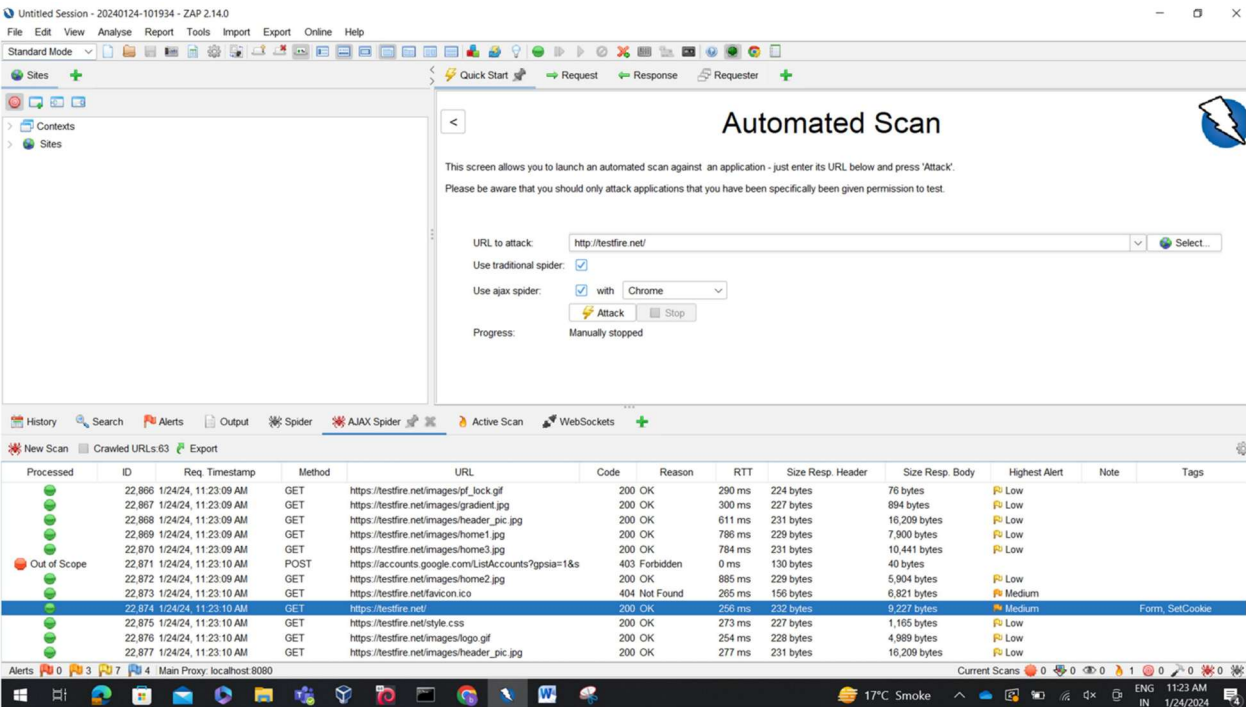
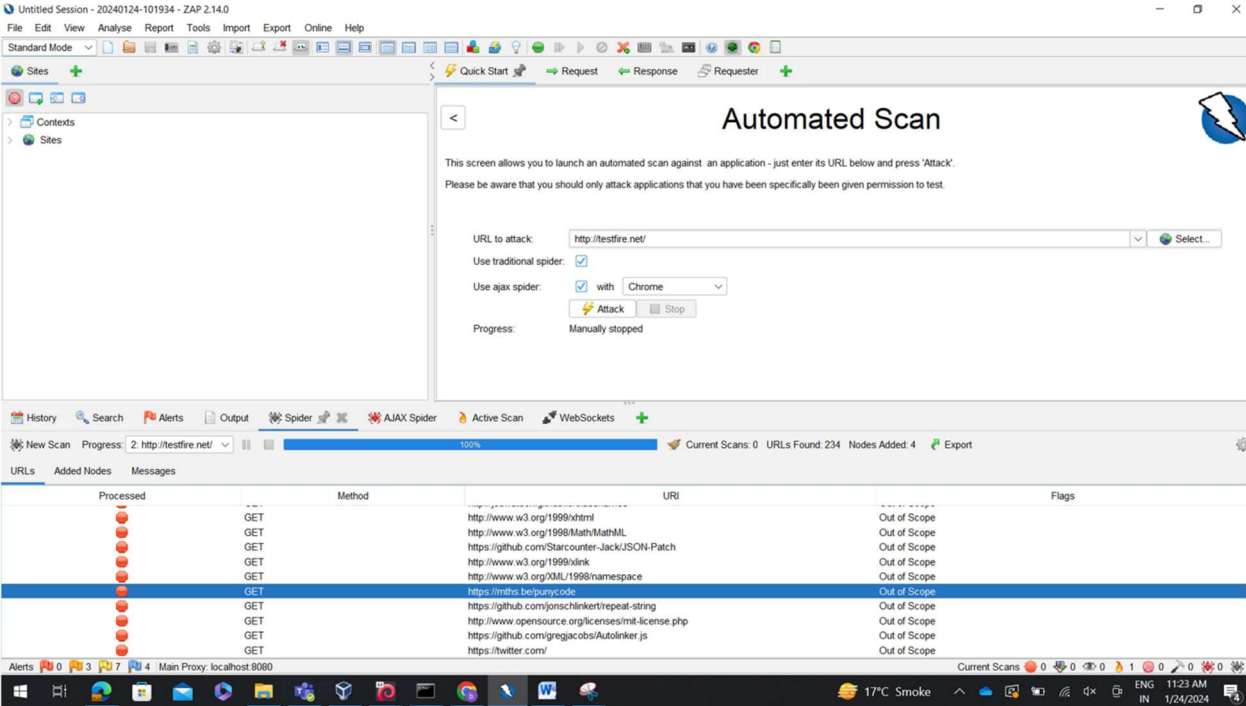
- **To view the alerts created during your test:**

1. Click the Alerts tab in the Information Window.
2. Click each alert displayed in that window to display the URL and the vulnerability detected in the right side of the Information Window.
3. In the Workspace Windows, click the Response tab to see the contents of the header and body of the response. The part of the response that generated the alert will be highlighted.

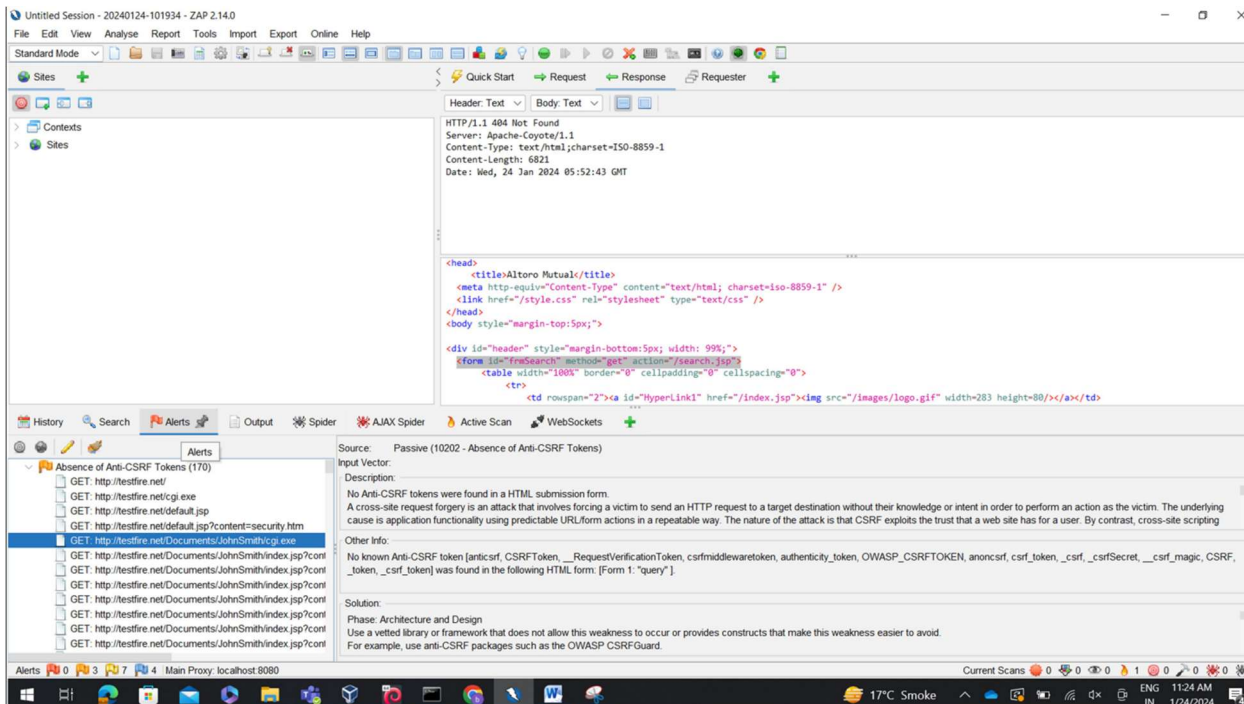
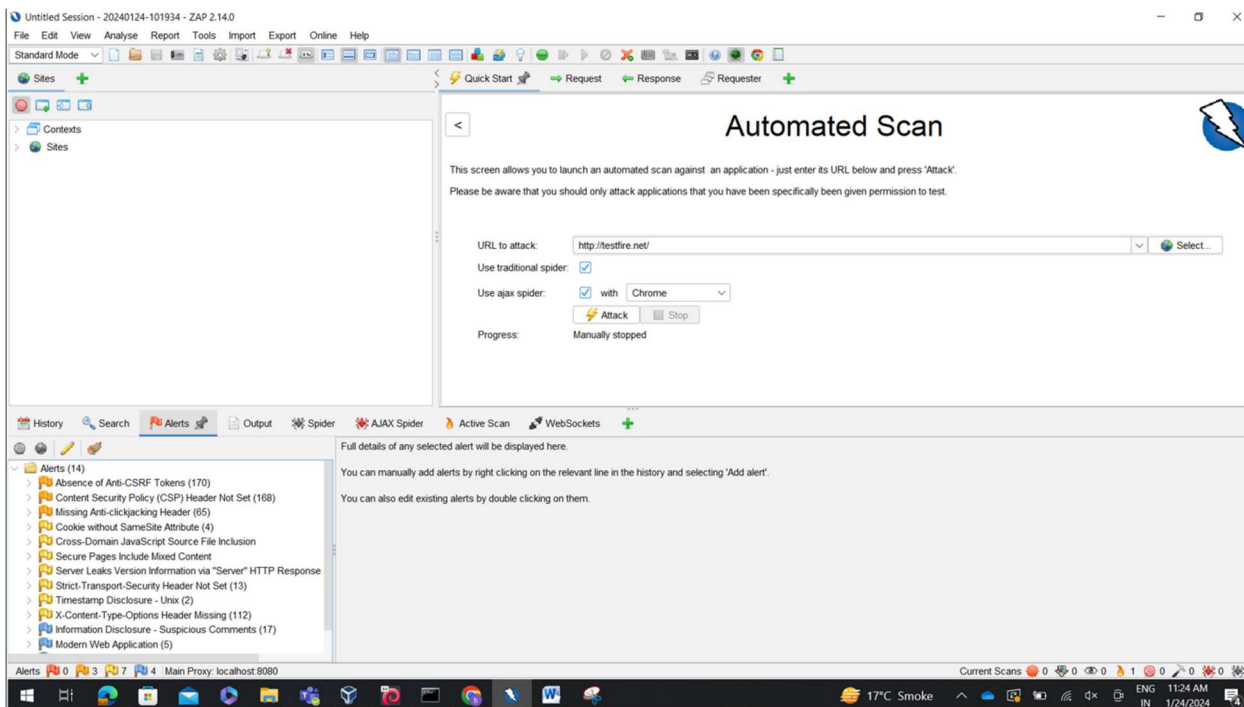
Here are the Outputs of a Sample bank website follow the images below step by step:



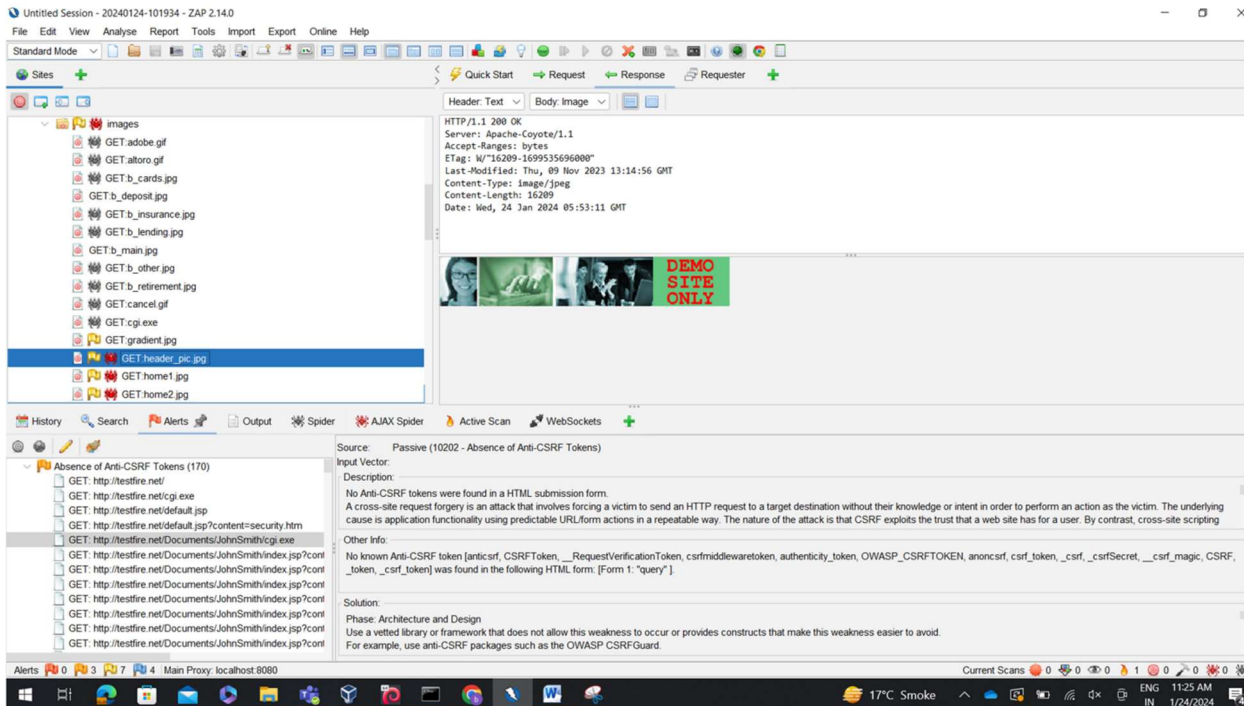
Click on the attack button



CHECK THE OUTPUT OF BELOW DASHBORD IN IMAGE CLICK ON ALERT OPTION TO CHECK VULNERABLITIES IN BELOW IMAGE.



SELECT ANY LINK CHECK THAT DESCRIPTION ABOUT VULNERABILITY

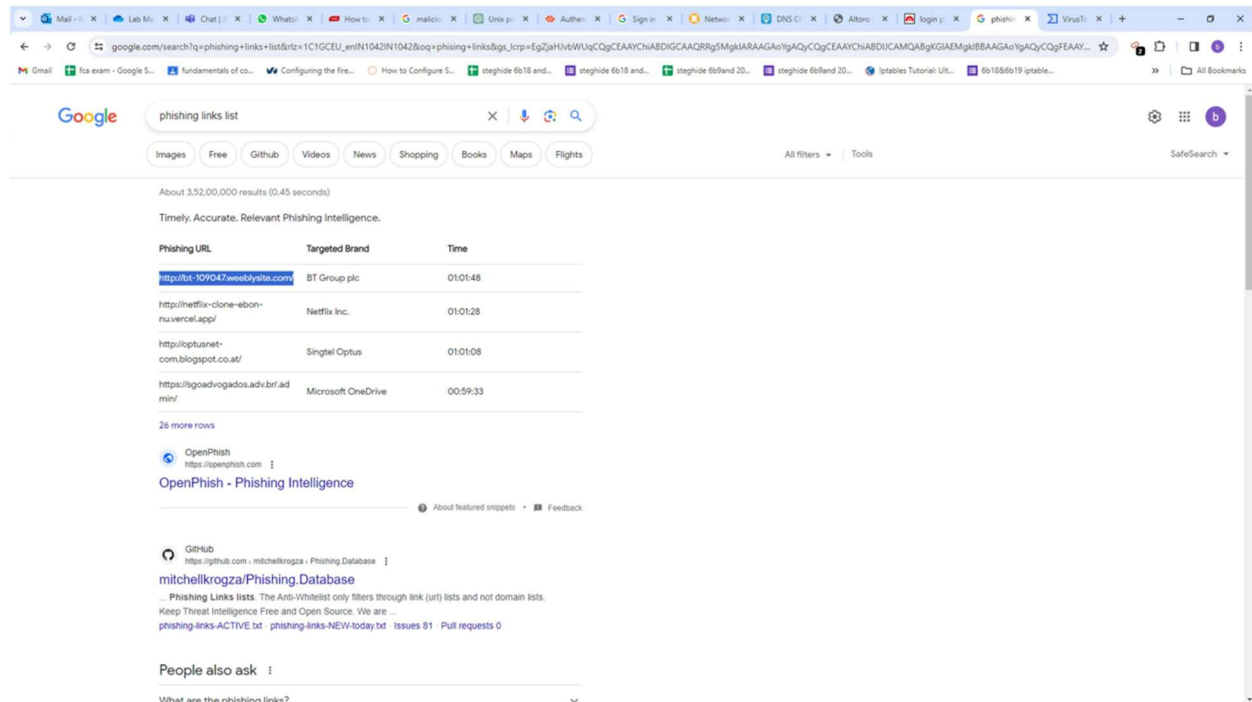


PRACTICAL NO: 7

Aim : Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.

Objective : To know how to find vulnerabilities by using NESSUS

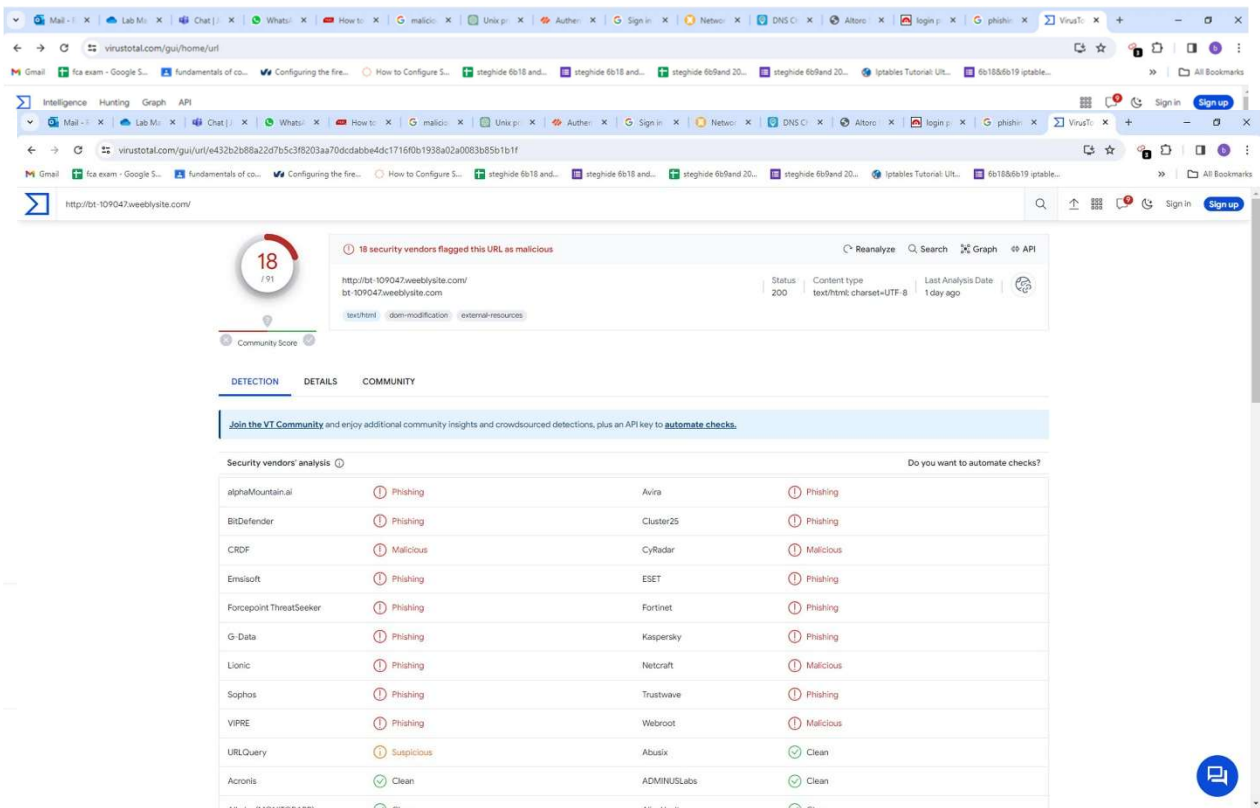
Requirements: Laptop, Kali Linux, Nessus package in kali



The screenshot shows a Google search results page for the query "phishing links list". The search results include a table of phishing URLs and targeted brands. The table has three columns: "Phishing URL", "Targeted Brand", and "Time".

Phishing URL	Targeted Brand	Time
http://t-109047weeblypale.com	BT Group plc	01:01:48
http://netflix-clone-ebon-nuvercel.app/	Netflix Inc.	01:01:28
http://optusnet-com.blogspot.co.at/	Singtel Optus	01:01:08
https://lgoadvogados.adv.br/ad-mini/	Microsoft OneDrive	00:59:33

Below the table, there are links to "OpenPhish" and "GitHub". The "OpenPhish" link is titled "OpenPhish - Phishing Intelligence" and the "GitHub" link is titled "mitchellkroga/Phishing.Database".



• Nessus

Nessus, developed by Tenable Inc, is a widely-used open-source vulnerability scanner. It offers a paid subscription, Nessus Professional, as well as a free version, Nessus Essentials, which is limited to 16 IP addresses per scanner.

Nessus provides a range of services, including vulnerability assessments, network scans, web scans, asset discovery, and more, to aid security professionals, penetration testers, and other cybersecurity enthusiasts in proactively identifying and mitigating vulnerabilities in their networks.

How to install a Nessus in kali

Unlike many security tools, Nessus doesn't come installed on Kali Linux. But it is very easy to download and install.

Follow these steps to install Nessus on your Kali:

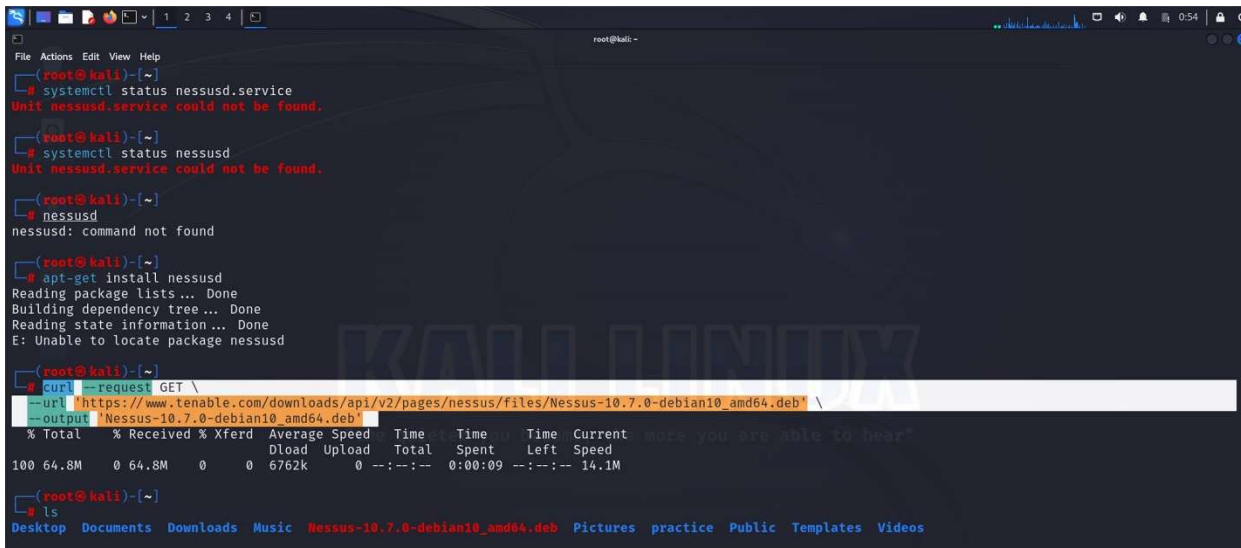
1. Download the Nessus package for Debian on the Nessus website and make sure you set the Platform to **Linux-Debian-amd64**.

The screenshot shows the Tenable Downloads page. On the left is a sidebar with a list of products: Tenable Nessus, Tenable Nessus Agent, Tenable Nessus Network Monitor, Tenable Security Center, Integrations, Sensor Proxy, Tenable Log Correlation Engine, Tenable Core, Tenable OT Security, Tenable Identity Exposure, Frictionless, and Tenable Cloud Security. The main content area is titled 'Downloads' and 'Tenable Nessus'. It features a section '1 Download and Install Nessus' with a 'Choose Download' subsection. Under 'Choose Download', there are two dropdown menus: 'Version' set to 'Nessus - 10.7.0' and 'Platform' set to 'Linux - Debian - amd64'. Below these is a blue 'Download' button with a download icon and a 'Checksum' link. There is also a 'Download by curl' section with a dropdown arrow. Below this is a code block containing the following command:

```
curl --request GET \
--url 'https://www.tenable.com/downloads/api/v2/pages/nessus/files/Nessus-10.7.0-debian10_amd64.deb'
--output 'Nessus-10.7.0-debian10_amd64.deb'
```

 At the bottom of the code block is a progress bar. Below the code block is a link 'View the Downloads API documentation'.

2. When it's finished downloading, open your Linux terminal and navigate to the location you downloaded the Nessus file to.



```
root@kali: ~  
# systemctl status nessusd.service  
Unit nessusd.service could not be found.  
  
root@kali: ~  
# systemctl status nessusd  
Unit nessusd.service could not be found.  
  
root@kali: ~  
# nessusd  
nessusd: command not found  
  
root@kali: ~  
# apt-get install nessusd  
Reading package lists ... Done  
Building dependency tree ... Done  
Reading state information ... Done  
E: Unable to locate package nessusd  
  
root@kali: ~  
# curl --request GET \\\n--url https://www.tenable.com/downloads/api/v2/pages/nessus/files/Nessus-10.7.0-debian10_amd64.deb' \\\n--output 'Nessus-10.7.0-debian10_amd64.deb'  
% Total % Received % Xferd Average Speed Time Time Time Current  
Dload Upload Total Spent Left Speed  
100 64.8M 0 64.8M 0 0 6762k 0 --:--:-- 0:00:09 --:--:-- 14.1M  
  
root@kali: ~  
# ls  
Desktop Documents Downloads Music Nessus-10.7.0-debian10_amd64.deb Pictures practice Public Templates Videos
```

3. Install Nessus using this command:

`dpkg -I Nessus-10.7.0-debian_amd64.deb`

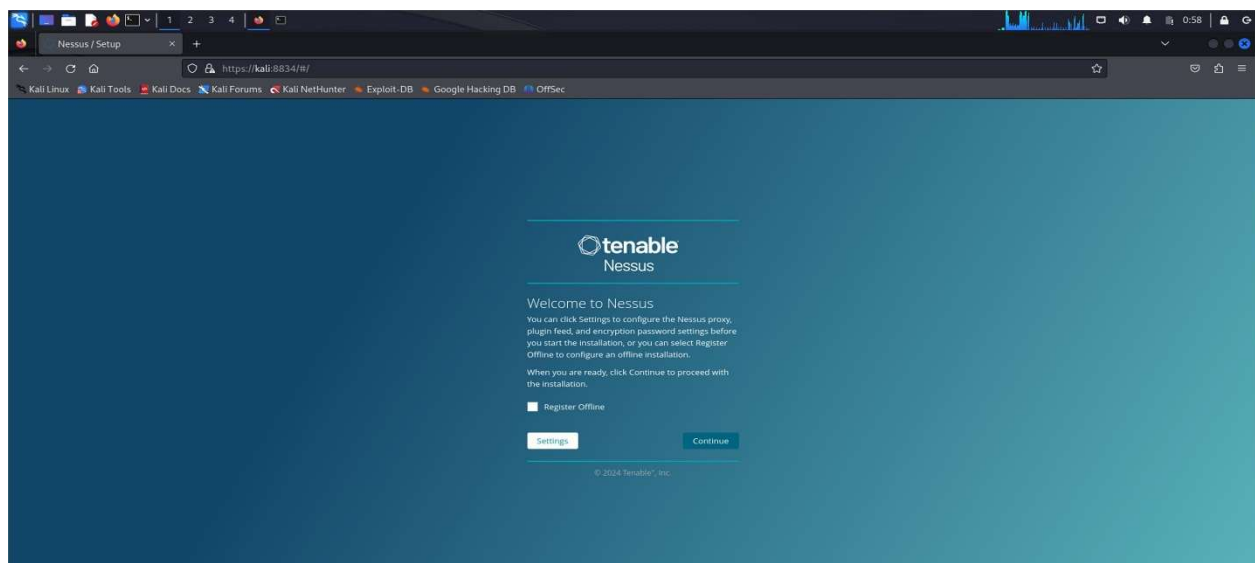
4. Start the Nessus service with this command:

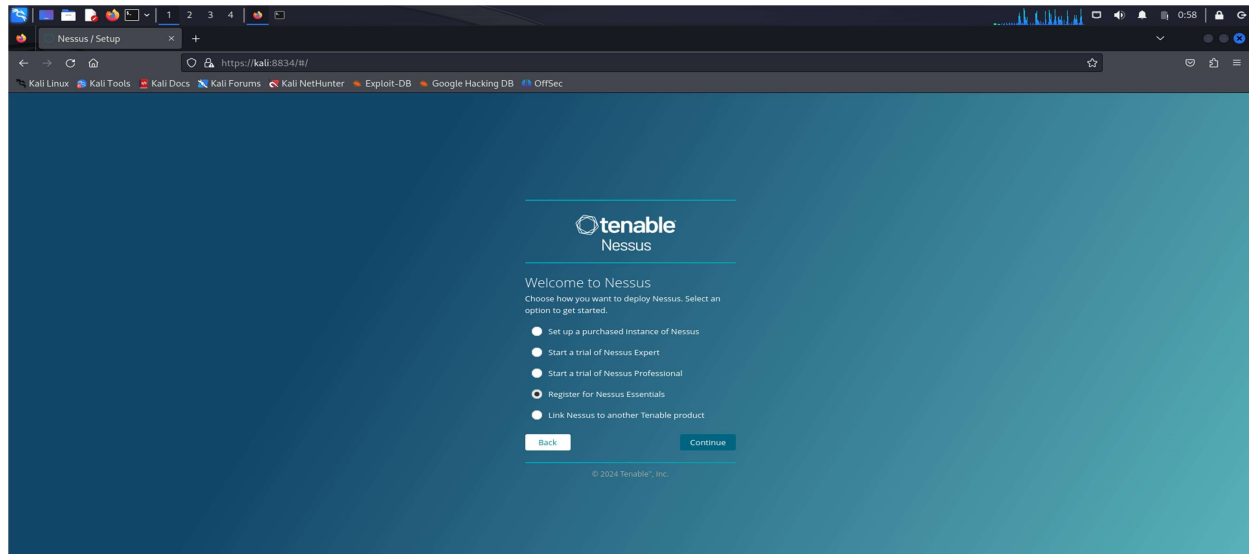


```
root@kali: ~  
# systemctl start nessusd.service  
  
root@kali: ~  
# systemctl status nessusd.service  
● nessusd.service - The Nessus Vulnerability Scanner  
Loaded: loaded (/lib/systemd/system/nessusd.service; disabled; preset: disabled)  
Active: active (running) since Sat 2024-02-10 00:56:06 EST; 13s ago  
Main PID: 7753 (nessus-service)  
Tasks: 14 (limit: 2260)  
Memory: 114.4M  
CPU: 12.710s  
CGroup: /system.slice/nessusd.service  
└─7753 /opt/nessus/sbin/nessus-service -q  
└─7754 nessusd -q  
  
Feb 10 00:56:06 kali systemd[1]: Started nessusd.service - The Nessus Vulnerability Scanner.  
Feb 10 00:56:08 kali nessus-service[7754]: Cached 0 plugin libs in 1msec  
Feb 10 00:56:08 kali nessus-service[7754]: Cached 0 plugin libs in 0msec  
  
root@kali: ~
```

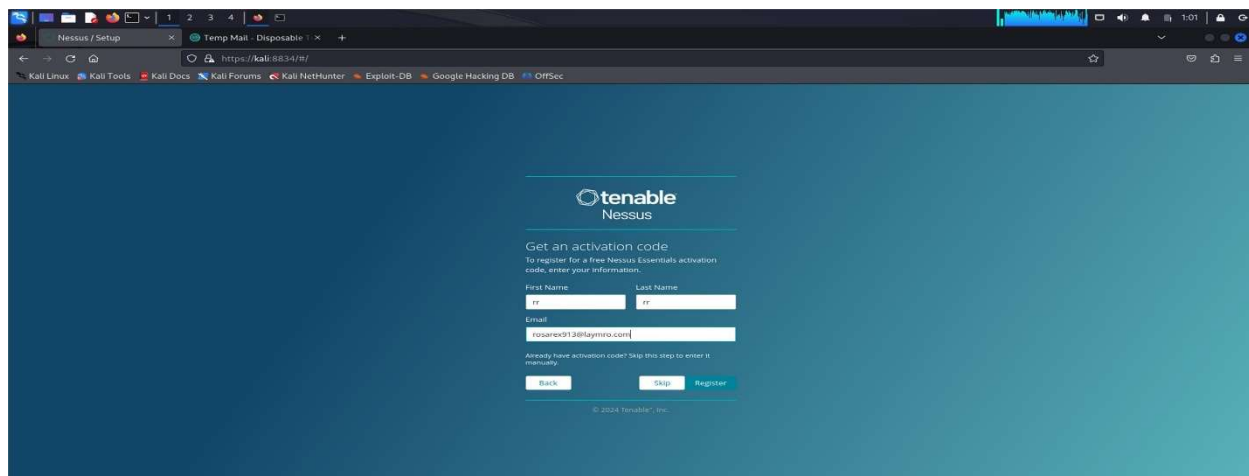
5. On your browser, go to <https://kali:8834/>. It would show a warning page.

2. Click on Advanced. Then, click on Accept Risk and Continue.
3. Choose the Nessus Product you prefer. If you want the free version of Nessus, click on **Register Nessus Essentials**.

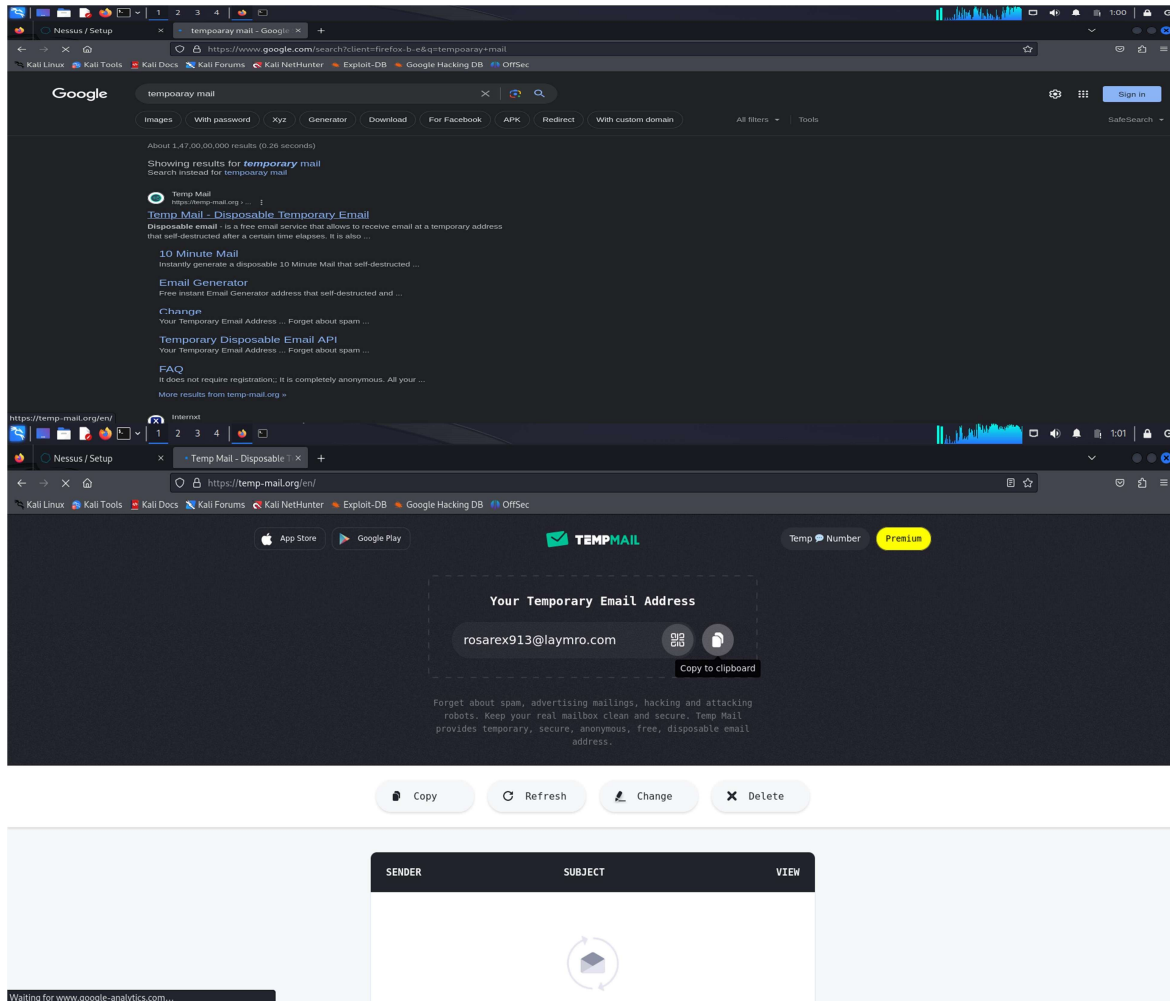




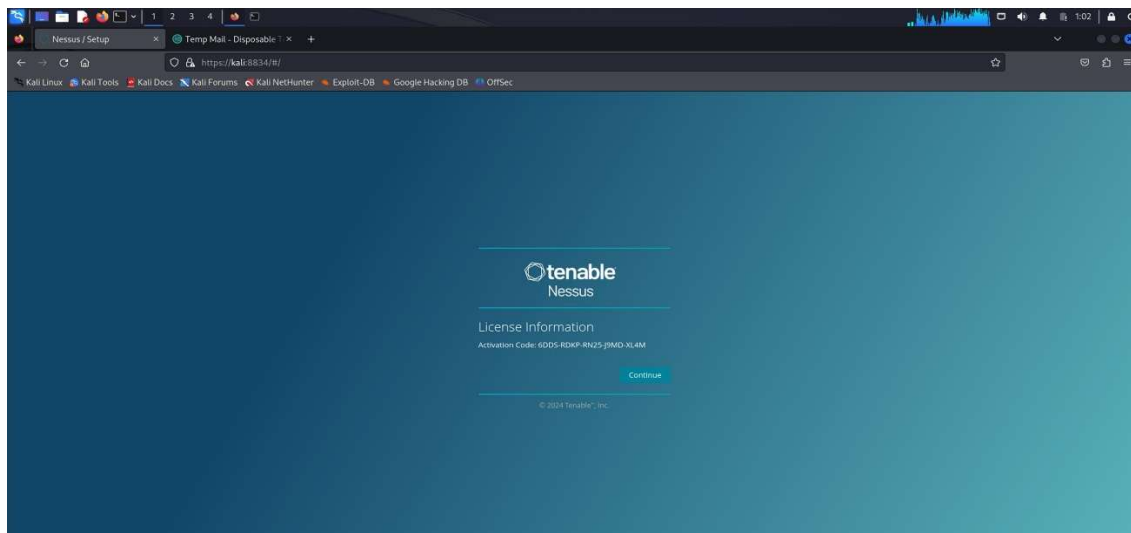
4. Enter your name and email address to receive an activation code by email. Paste the activation code into the space provided and choose a username and password.

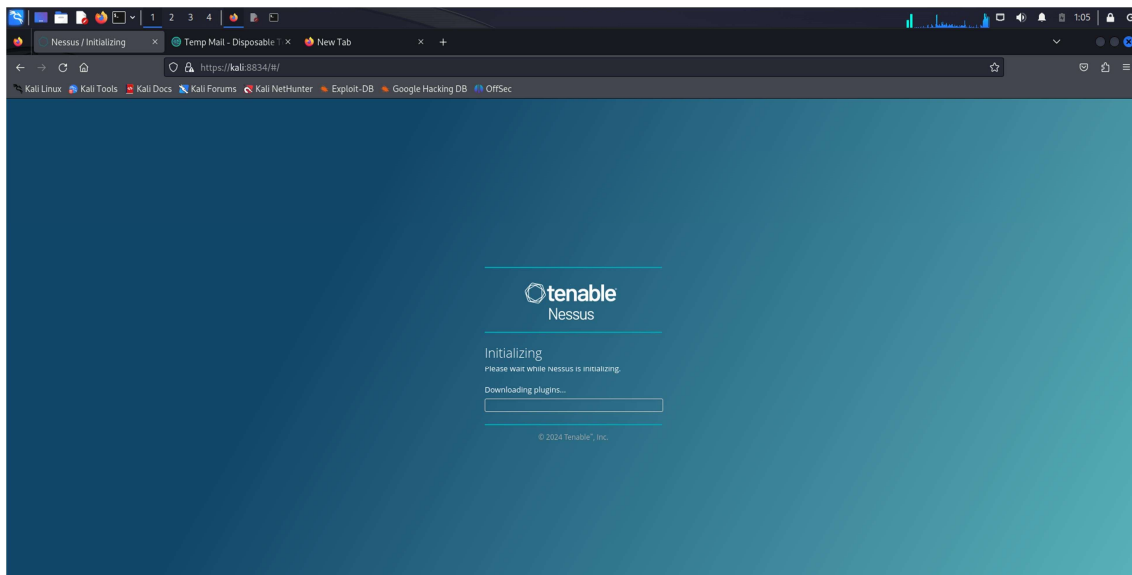
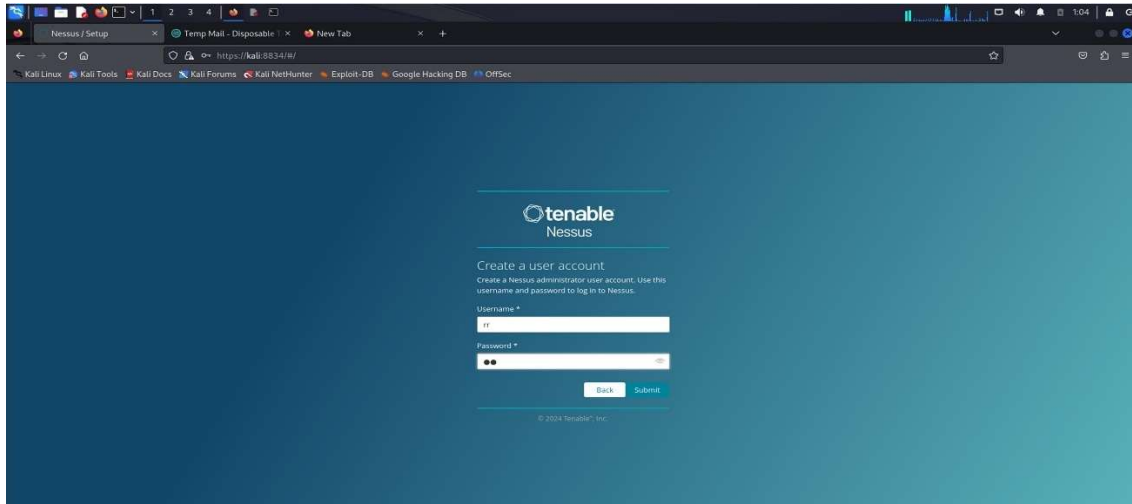


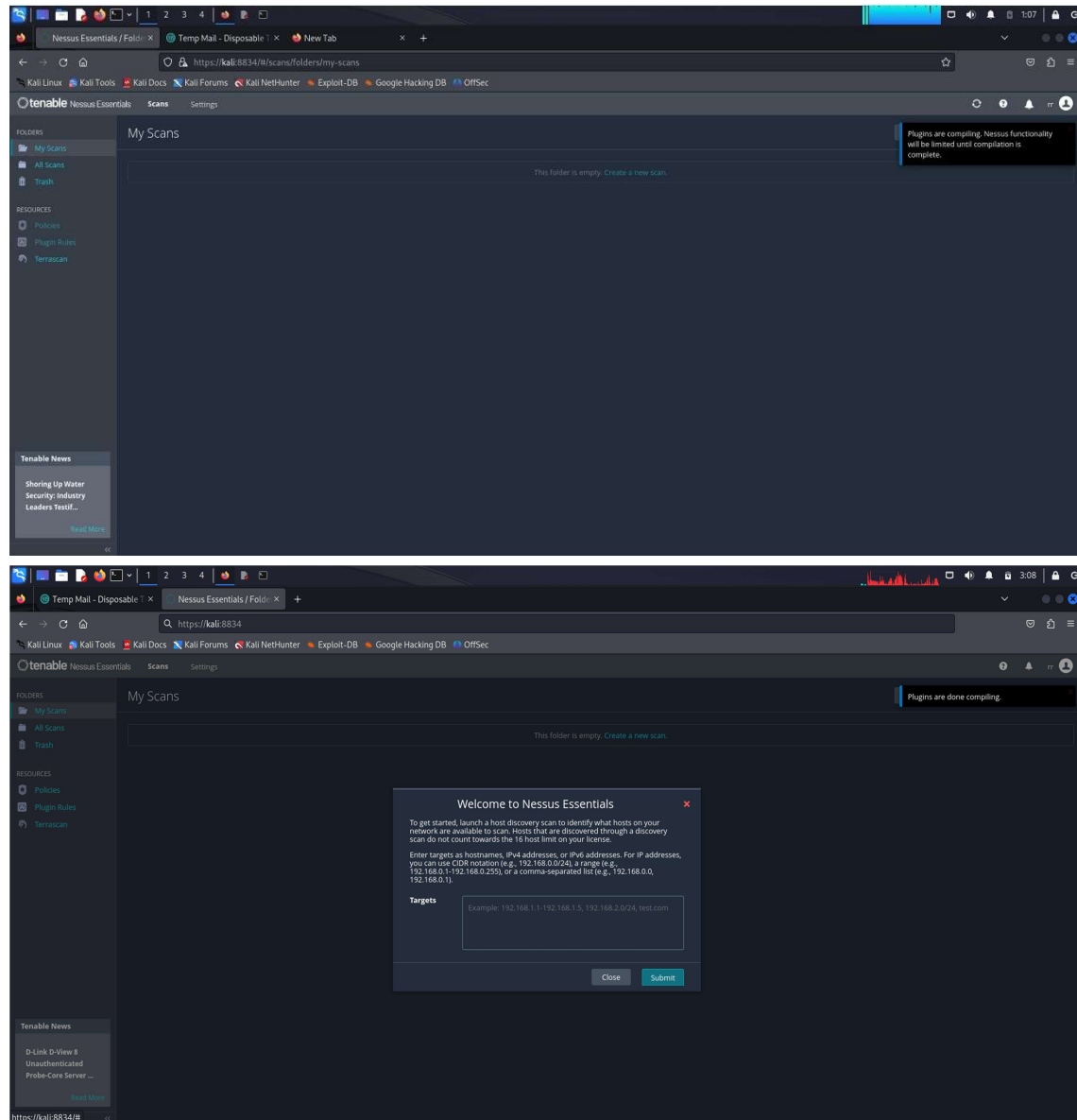
5. You can use email address as temporary you can visit online temporary email address examples below



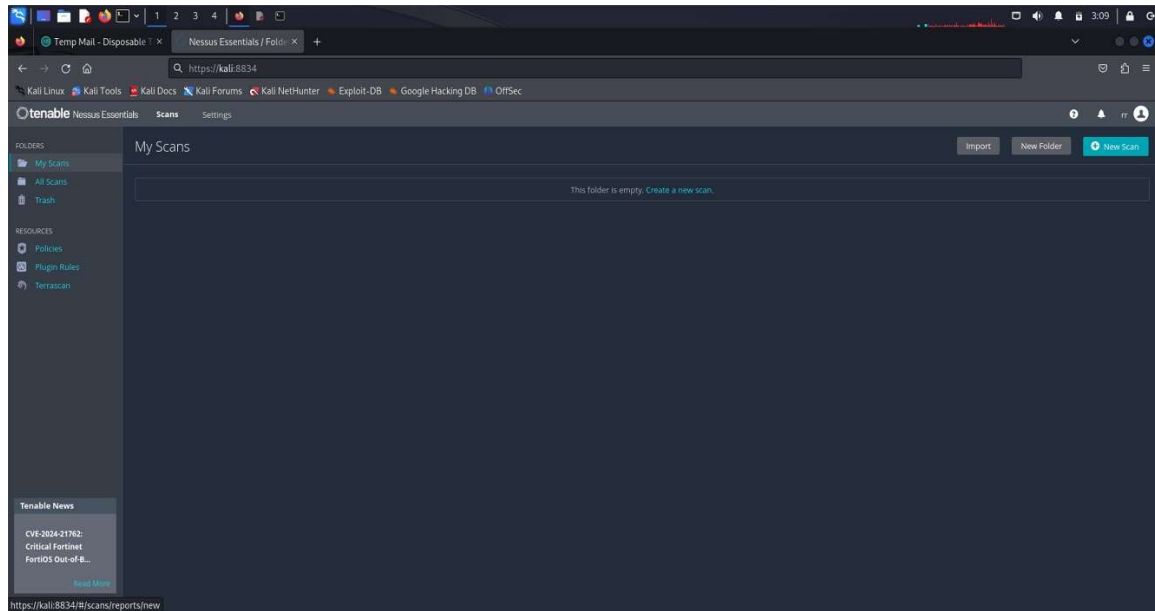
6. Allow Nessus to download the necessary plugins.



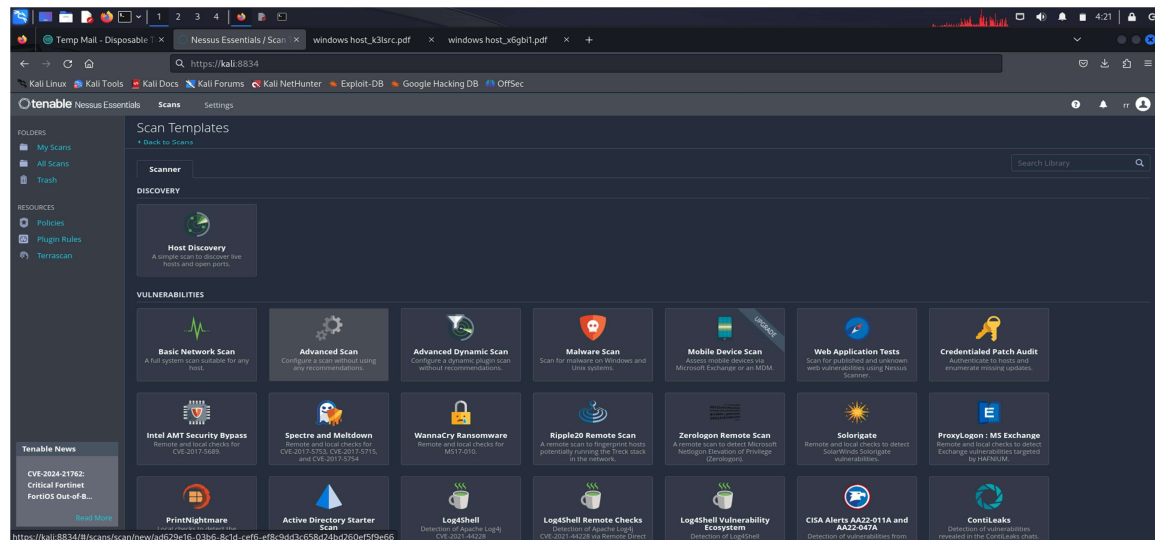




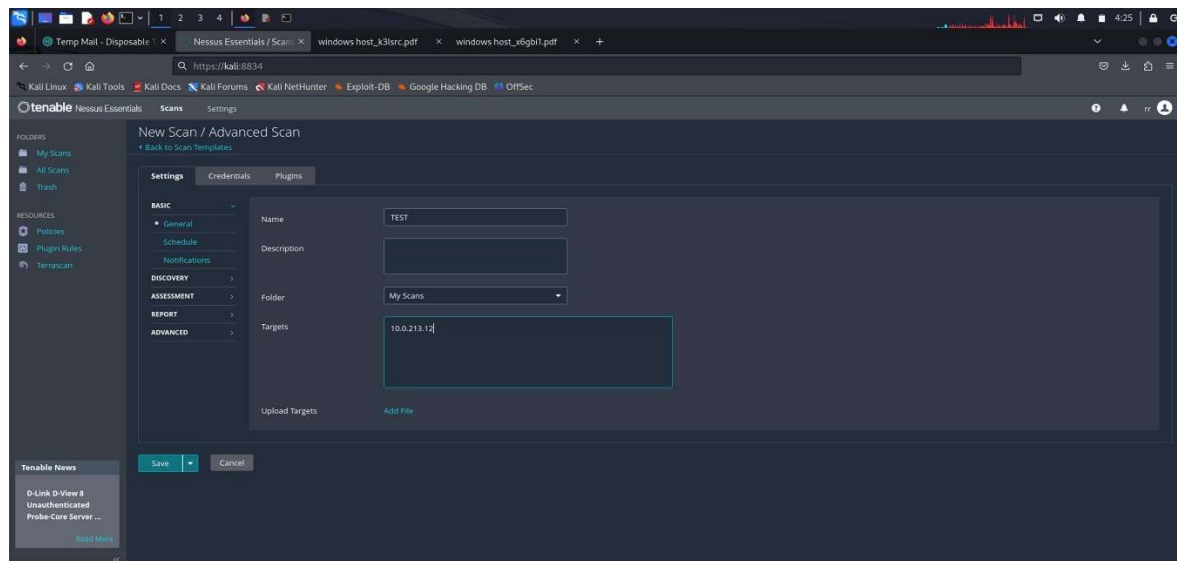
7. Click on New scan to begin scanning for vulnerabilities



8. Click on advance scan

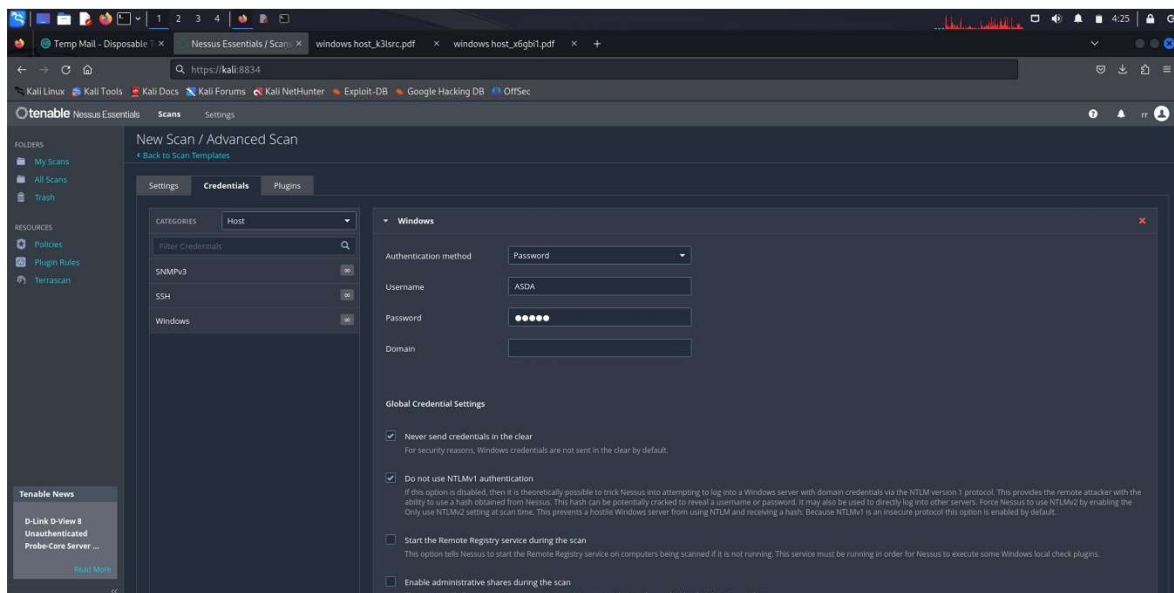


9. Give name for scanning. Give the IP address of your windows that you want to scan for vulnerabilities in target box shown in the below.

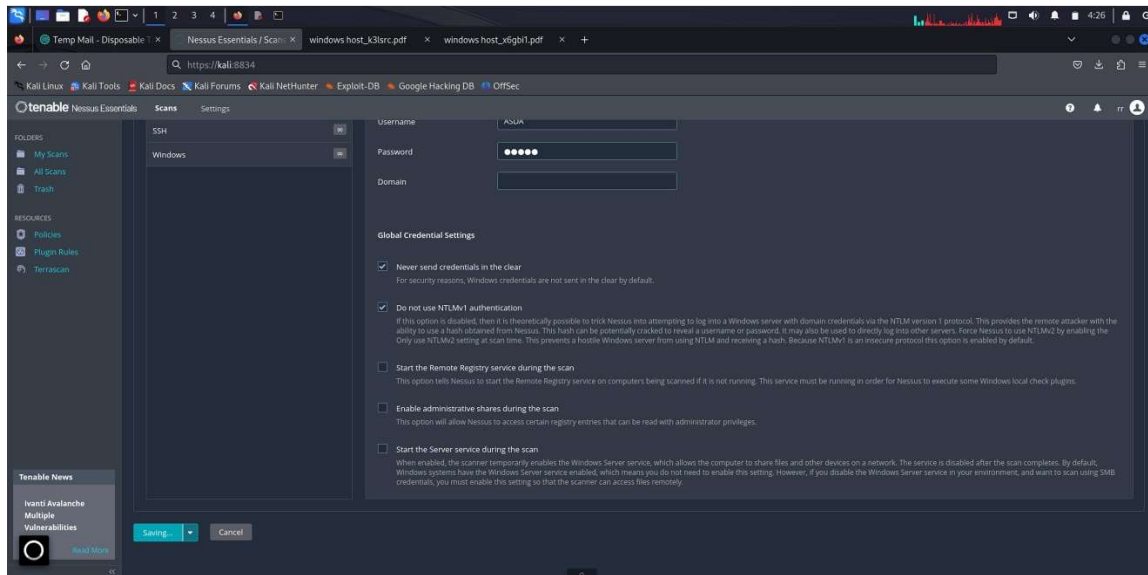


10. Give the credentials of your windows like username and password if you know(Optional)

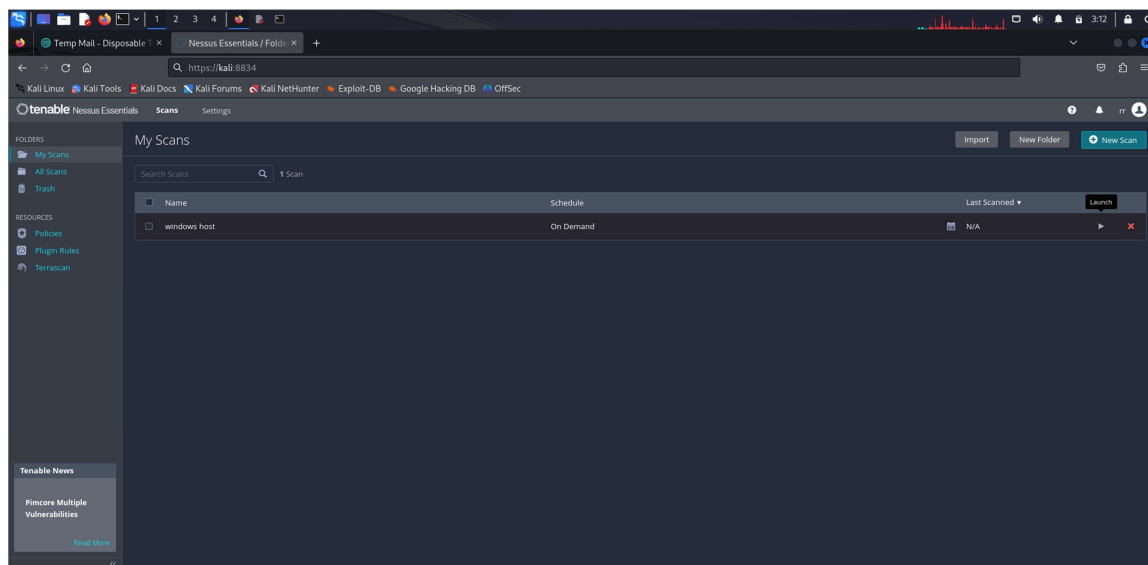
NOTE:- Use of giving credential it will help you scan more into your system.

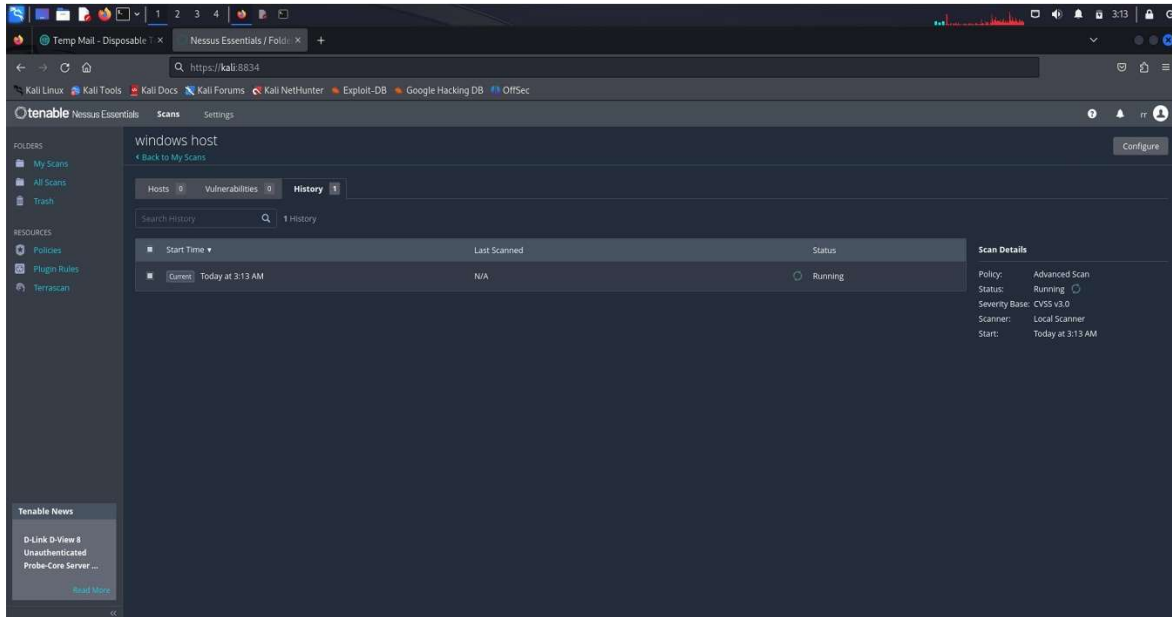


11. Go down and save the Progress

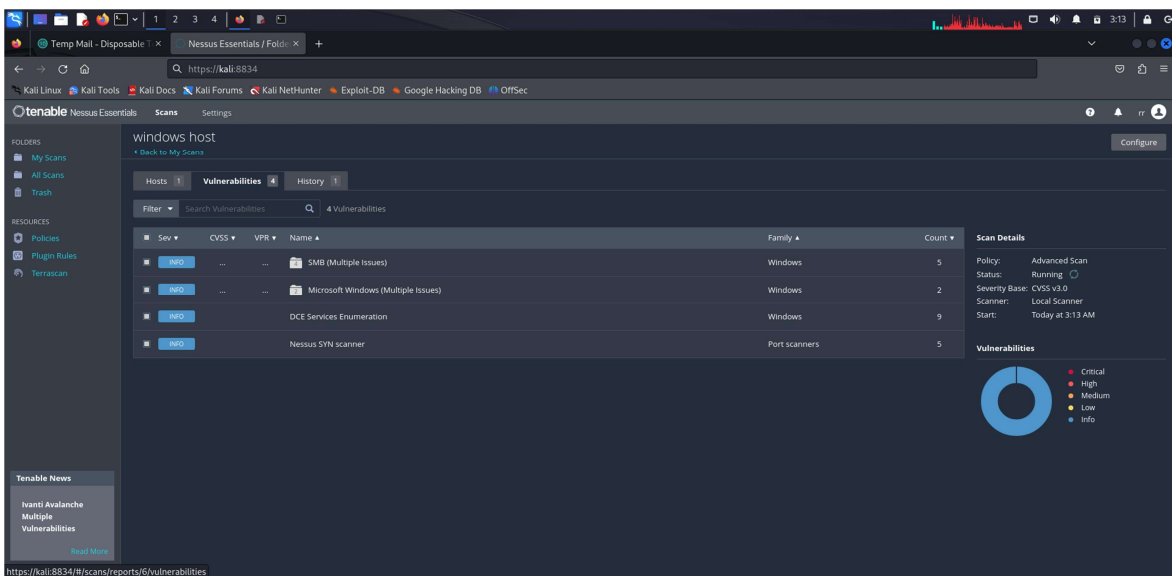


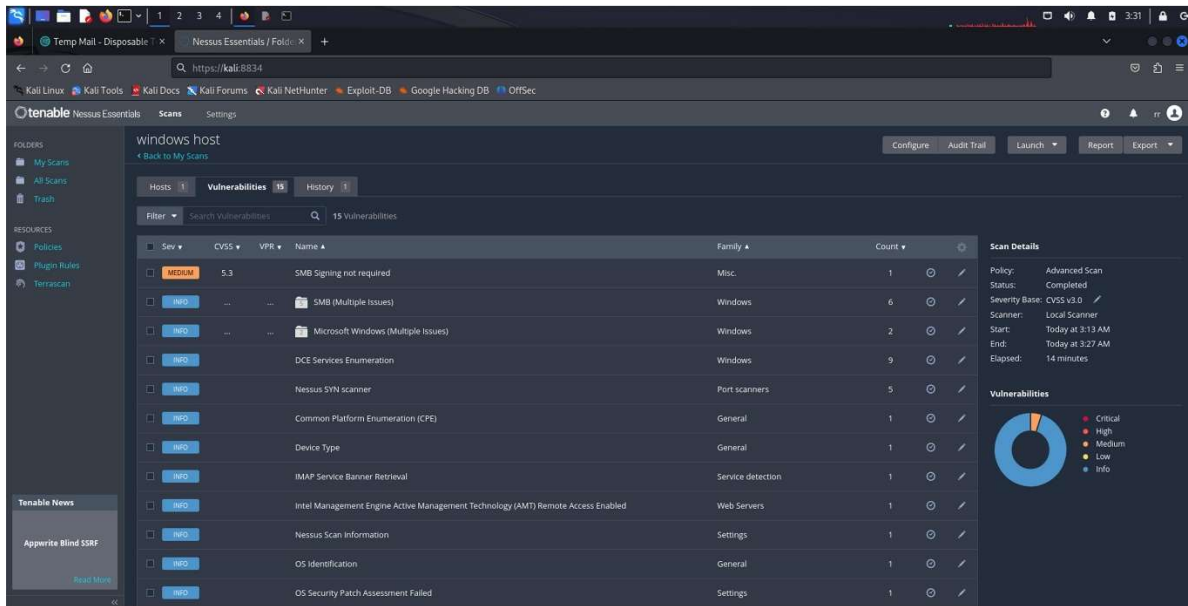
12. Click on Launch



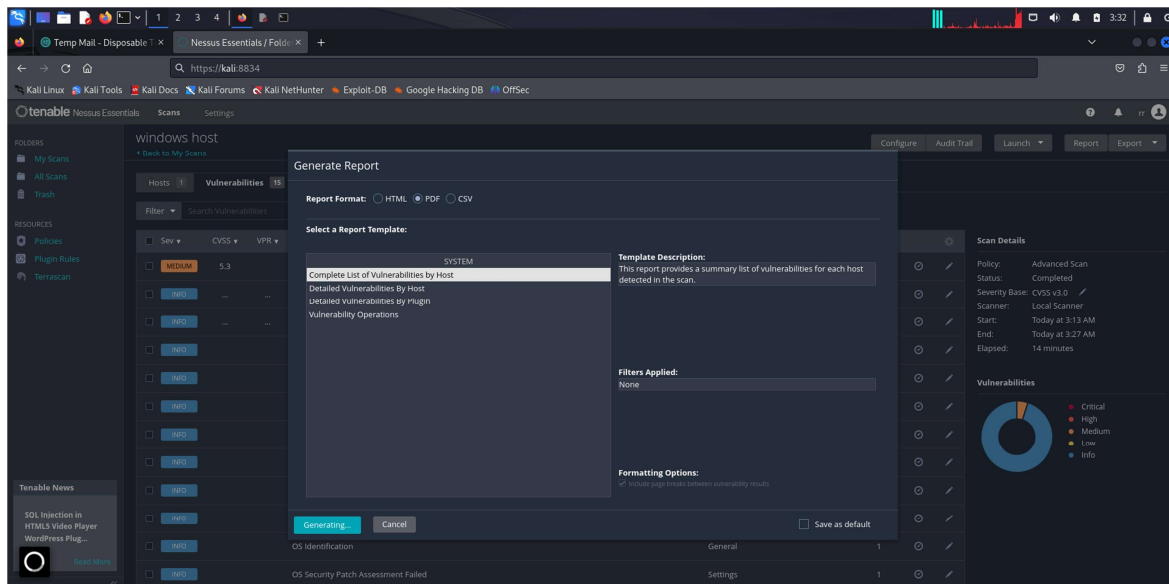


13. Wait for some time to get output





14. On the right side click on the report to generate the entire report of your system vulnerabilities.



PRACTICAL NO: 8

Aim : Implementation of OS hardening and RAM dump analysis to collect the Artifacts and other Information.

Tools : Autopsy, Dump-IT

Description :

- RAM dumps

A memory dump is the process of taking all information content in RAM and writing it to a storage drive as a memory dump file (*.RAW format).

Volatile memory, or RAM, is used to store data currently used by a running process: whether it is a user application or a system service. This type of memory is much quicker than a regular hard drive but unlike files permanently stored on a drive (unless deleted), data from RAM may disappear instantly. At the same time, it may store data crucial for your case, including passwords in raw format without encryption or encoding, decrypted data otherwise kept encrypted on a drive, decryption keys for various services, apps and WDE, remote sessions data, chats in social networks, malware code, cryptocurrency transactions, various system info such as loaded registry branches, and so on. This is why it is not argued that capturing RAM contents must be one of the first steps in seizing a running computer or laptop.

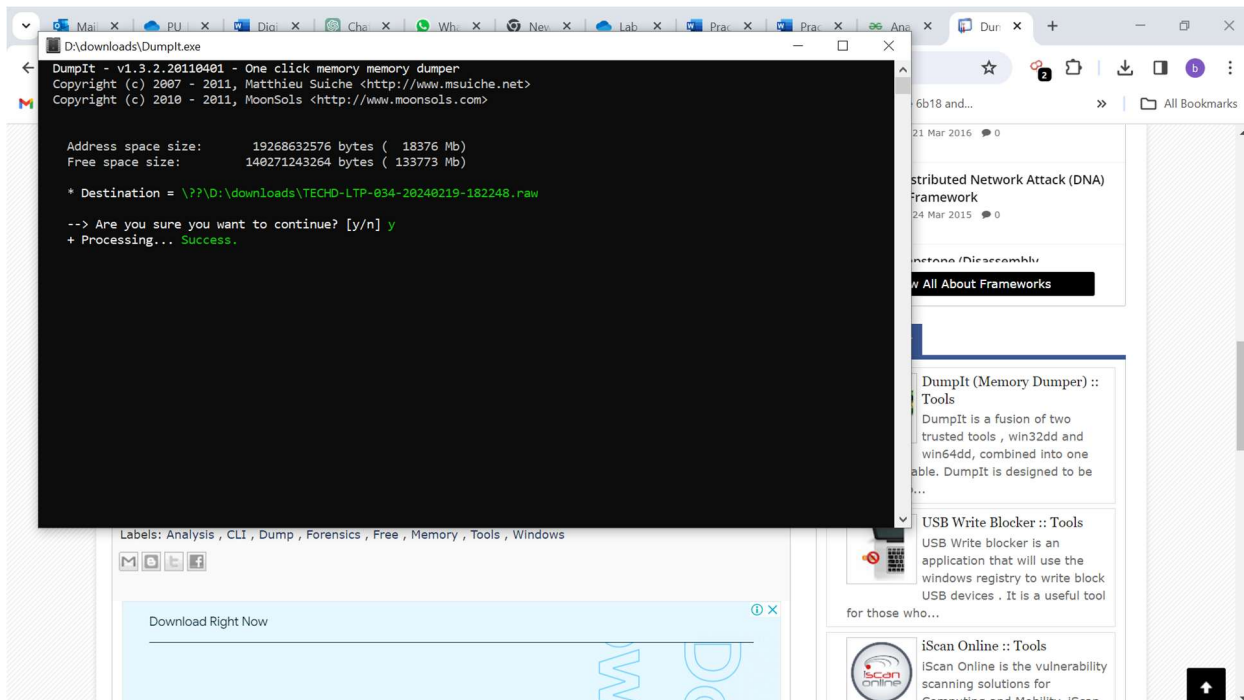
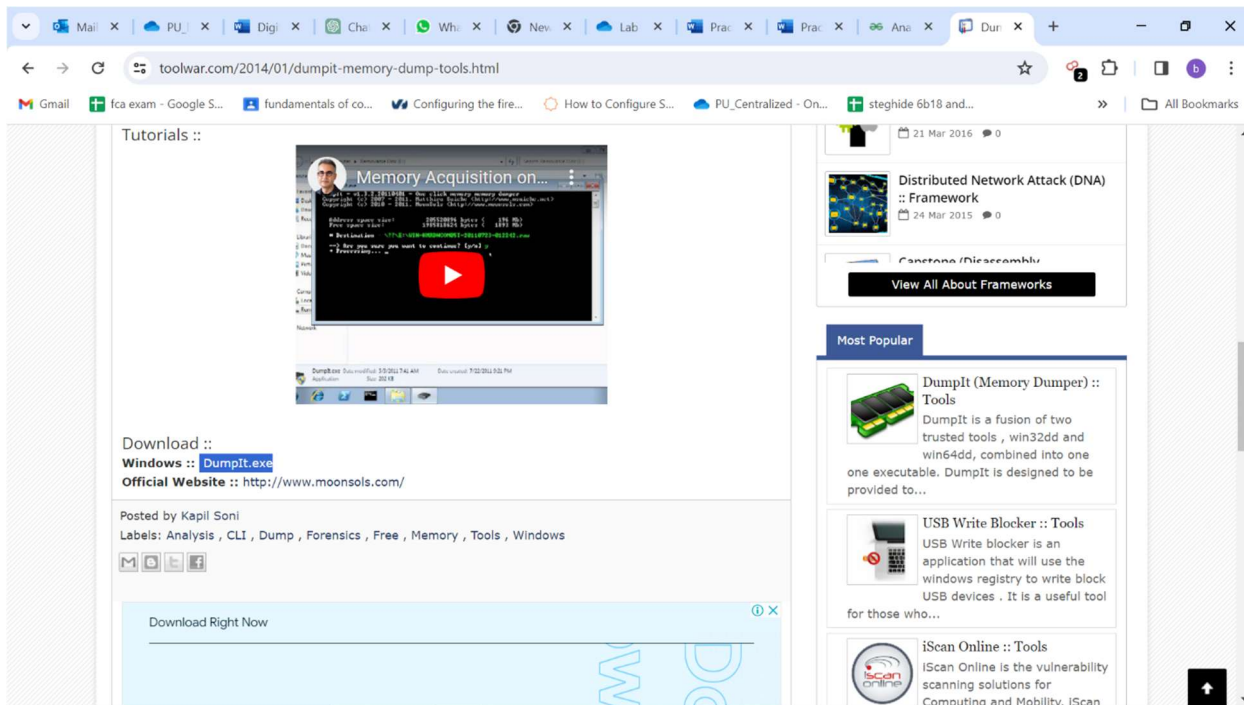
There are various tools that can be used for memory dump. Some of them are:

1. Autopsy
2. Dump-IT

- Procedure

Creating RAM dump using Dump-IT

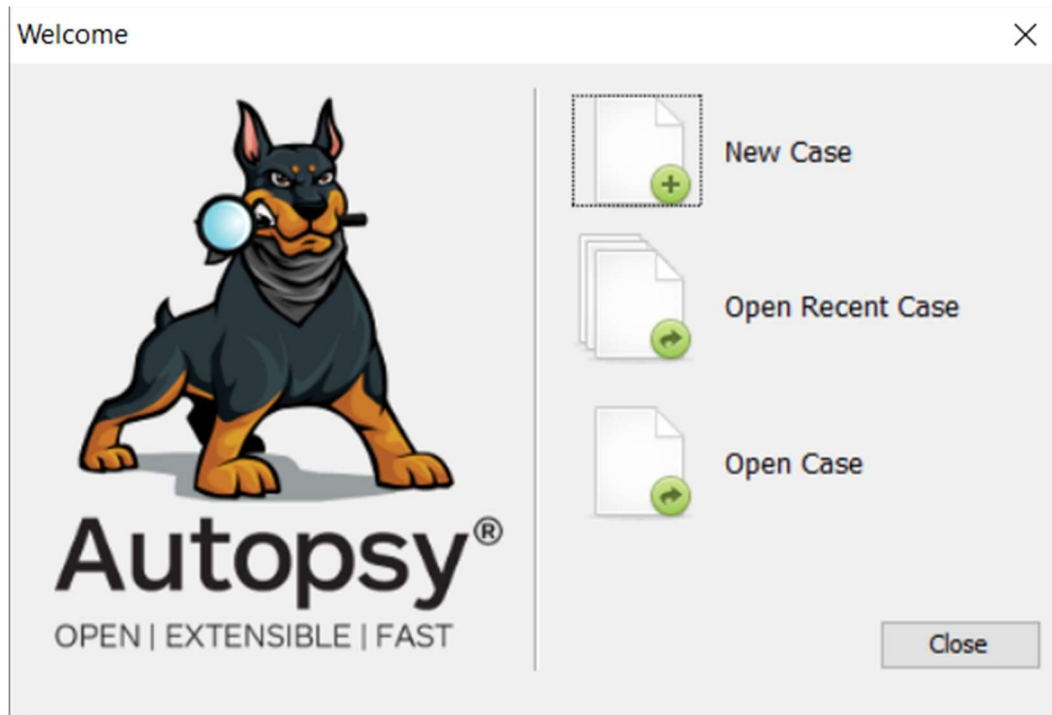
1. Download DumpIT tool from toolwar website
2. Open Dump-IT.exe



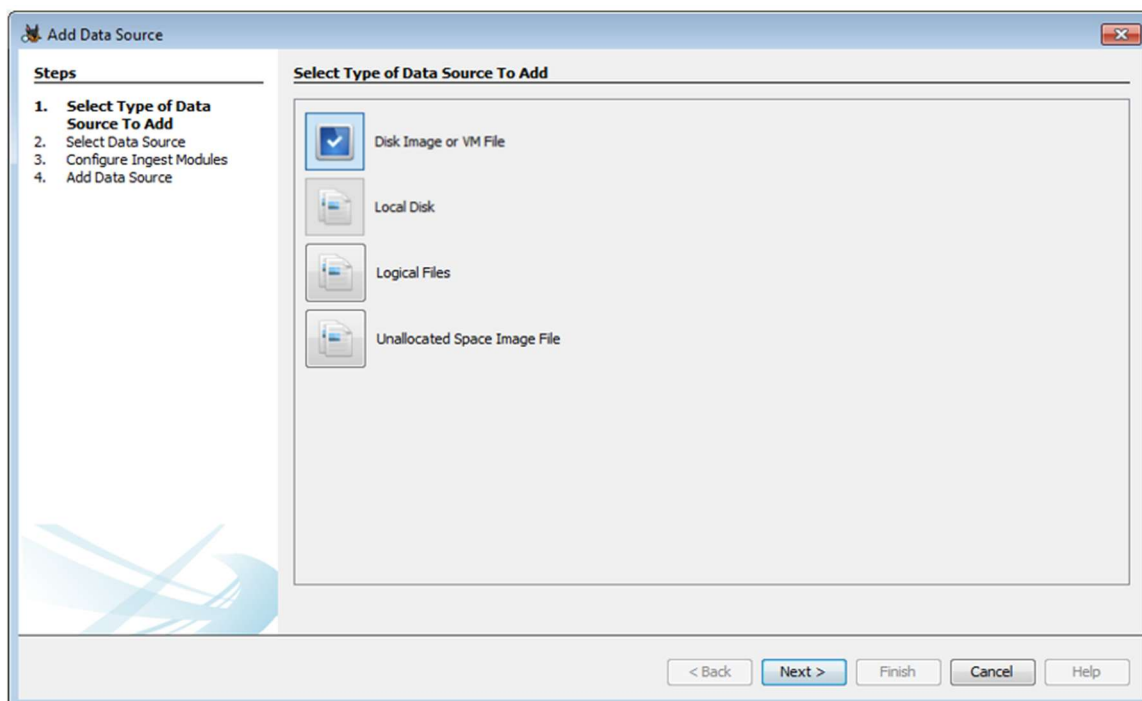
Autopsy performs operations onto disk images which can be created using tools like FTK Imager. Here an already created image is used. You may download Autopsy from [here](#) and the disk image used in this article from [here](#).

1. Getting started

Open Autopsy and create a new case



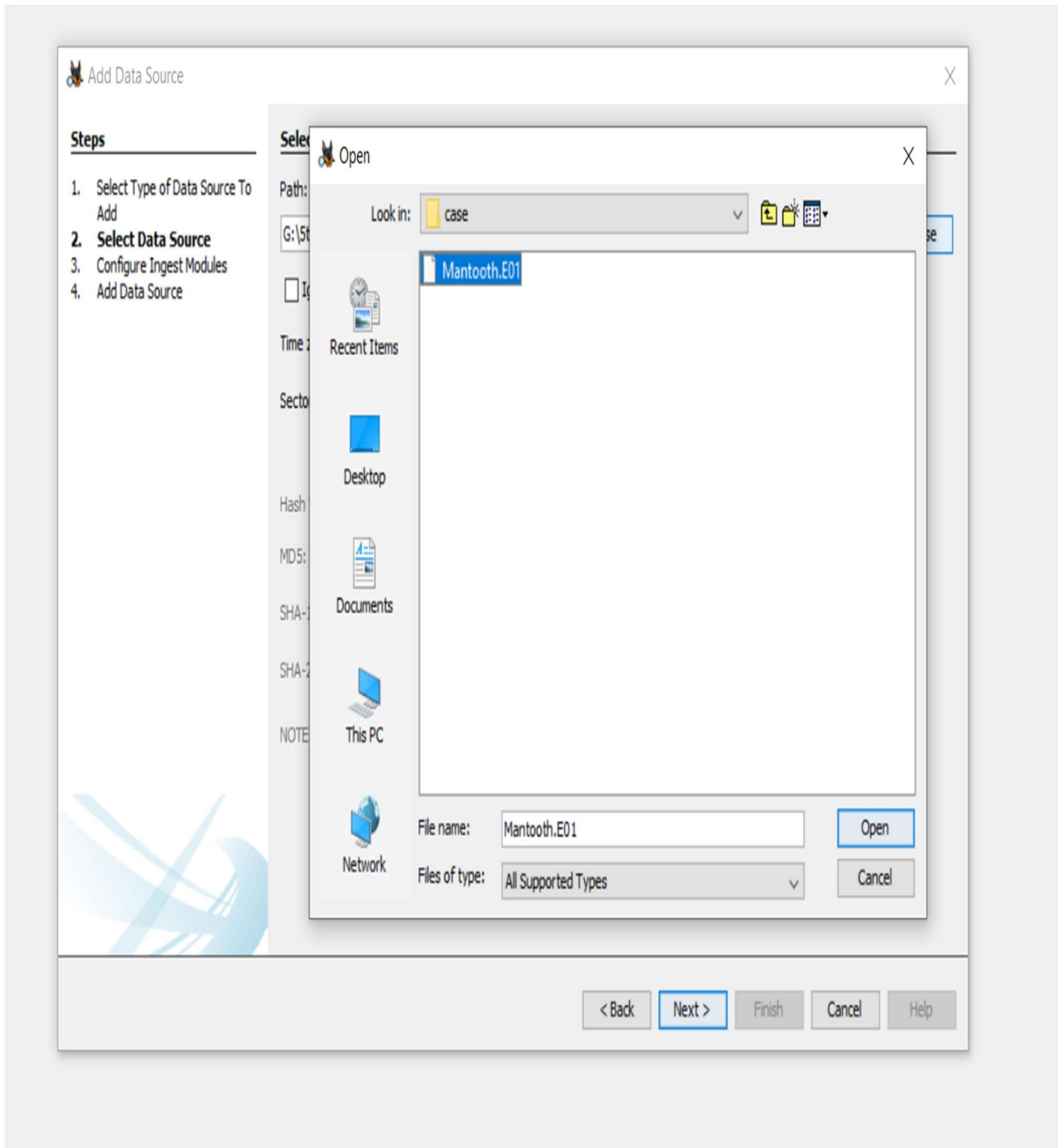
Click on **Finish** after completing both the steps.



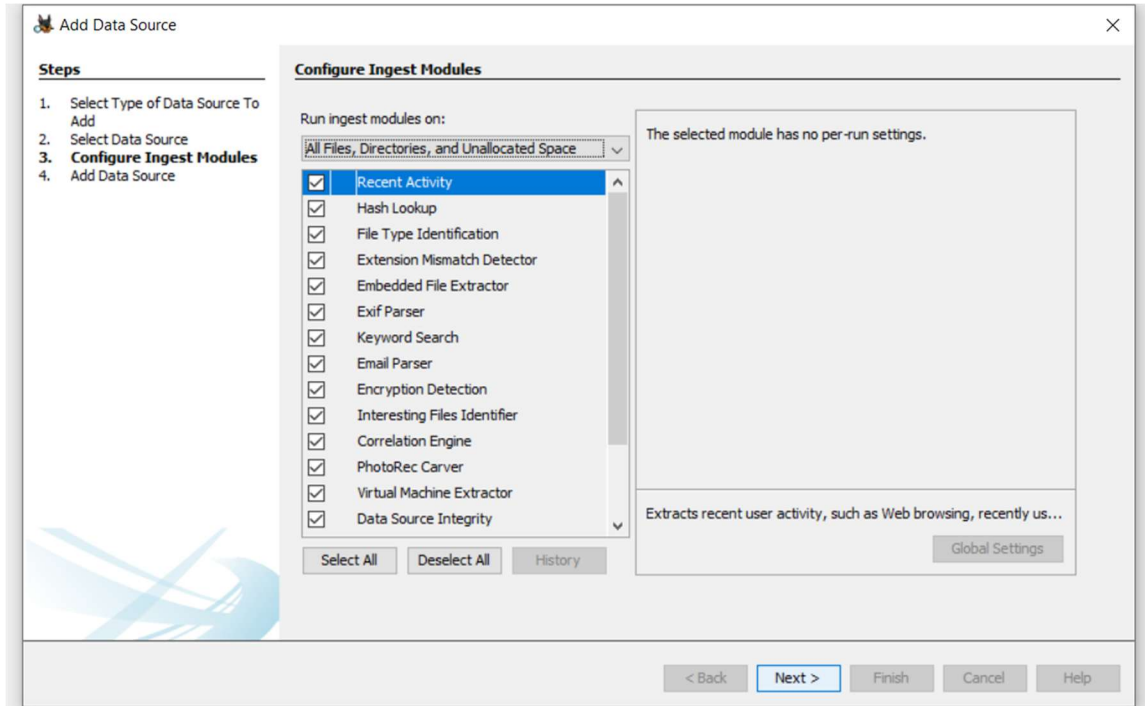
Add a data source.Select the appropriate data source type.

- **Disk Image or VM file:** Includes images that are an exact copy of a hard drive or media card, or a virtual machine image.
- **Local Disk:** Includes Hard disk, Pen drive, memory card, etc.
- Logical Files. : Includes local folders or files.
- **Unallocated Space Image File:** Includes files that do not contain a file system but need to run through ingest.

The data source used here is a disk image. Add the data source destination.



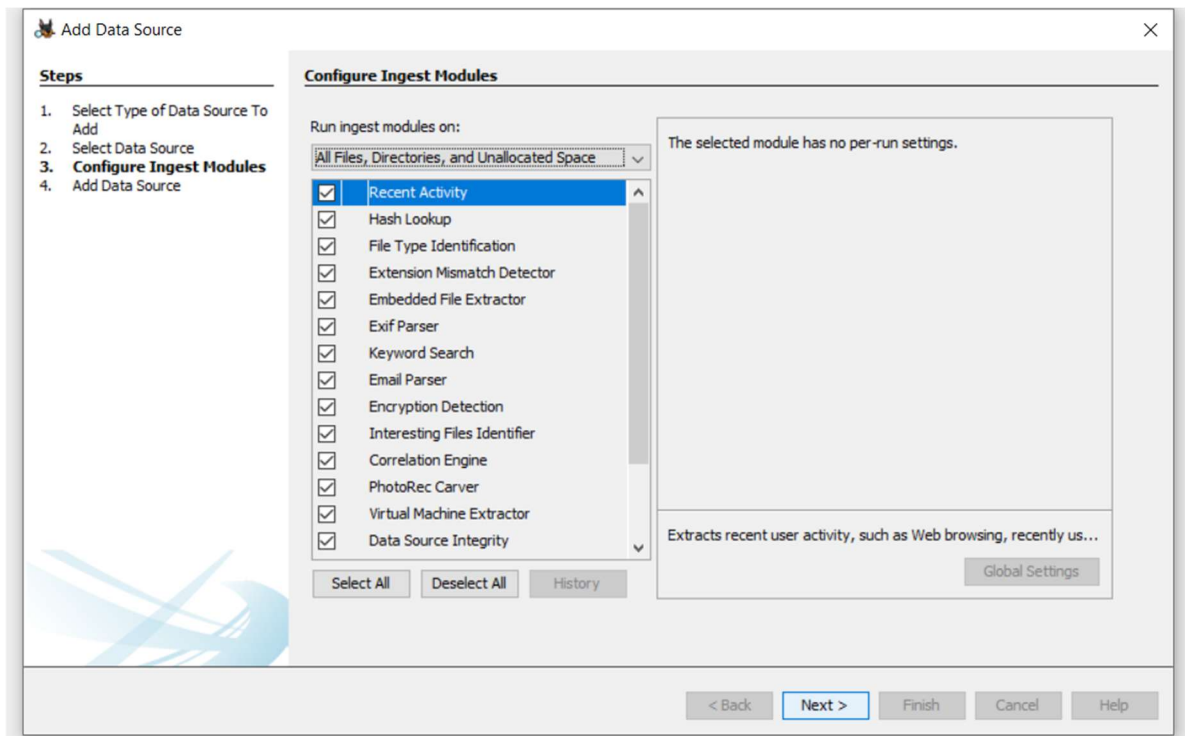
Configure ingest modules.



The ingest modules determine factors for which the data in the data source is to be analyzed. Here is a brief overview of each of them.

- **Recent Activity:** Discover the recent operations performed on the disk, for example, the files that were last viewed.
- **Hash Lookup:** Identify files using hash values.
- **File Type Identification:** Identify files based on their internal signatures rather than just file extensions.
- **Extension Mismatch Detector:** Identify files whose extensions are tampered with/changed possibly to hide evidence.
- **Embedded File Extractor:** It extracts embedded files such as .zip, .rar, etc. and uses the derived file for analysis. Another example could be a PNG image saved inside a doc to make it appear as a document and thus hide crucial information.
- **EXIF (Exchangeable Image File Format) Parser:** It is used to retrieve metadata about the files, for example, date of creation, geolocation, etc.
- **Keyword Search:** Search for a particular keyword/pattern in the data source.
- **Email Parser:** If the disk holds any form of email database, for example, pst/ost files of outlook then information from these files can be extracted using an email parser.
- **Encryption Detection:** Detects and identifies encrypted / password-protected files.
- **Interesting File Identifier:** Let's set custom rules regarding the filtering of data. Examiner is notified when results pertaining to these rules are found.
- **Correlation Engine:** Allows saving properties in and then retrieved from the central repository. It helps in displaying correlated properties.
- **PhotoRec Carver:** Recover files, photos, etc. from the unallocated space.
- **Virtual Machine Extractor:** Extract and analyze any Virtual machine found on the data source.
- **Data Source Integrity:** Calculates the hash values and stores them in the database in case they aren't already present. Otherwise, it will verify the hash values associated with the database.
- **Plaso:** Extract timestamp for various types of files.

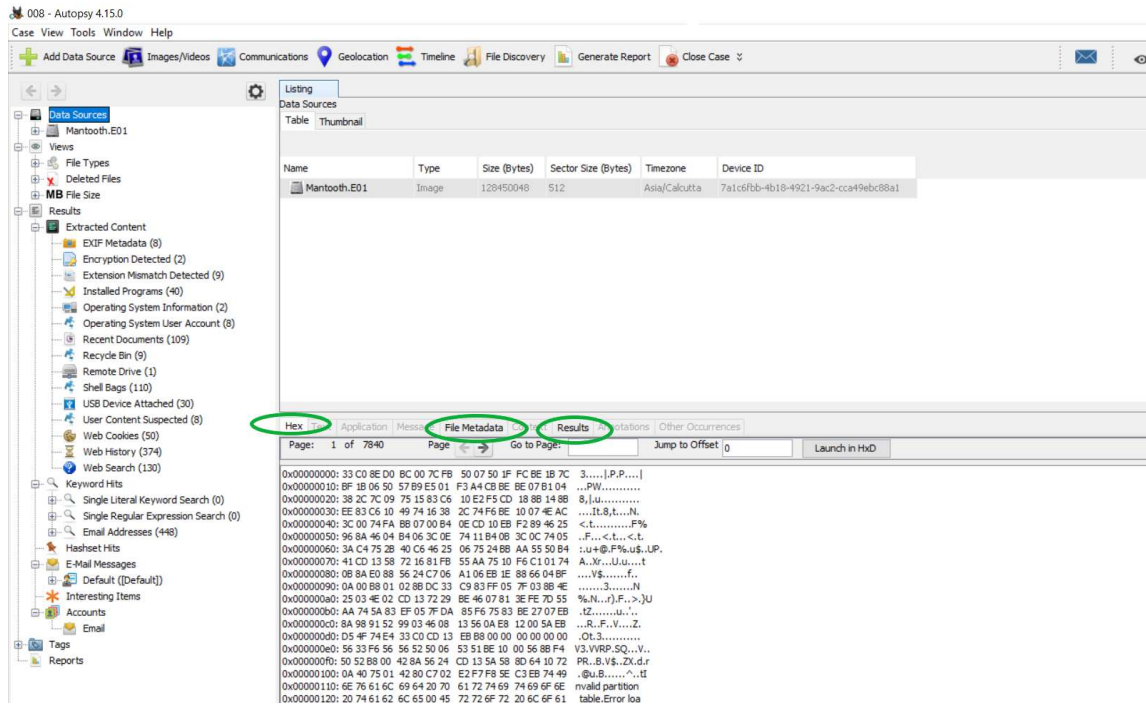
- **Android Analyzer:** Analyze SQLite and other files retrieved from an Android device.



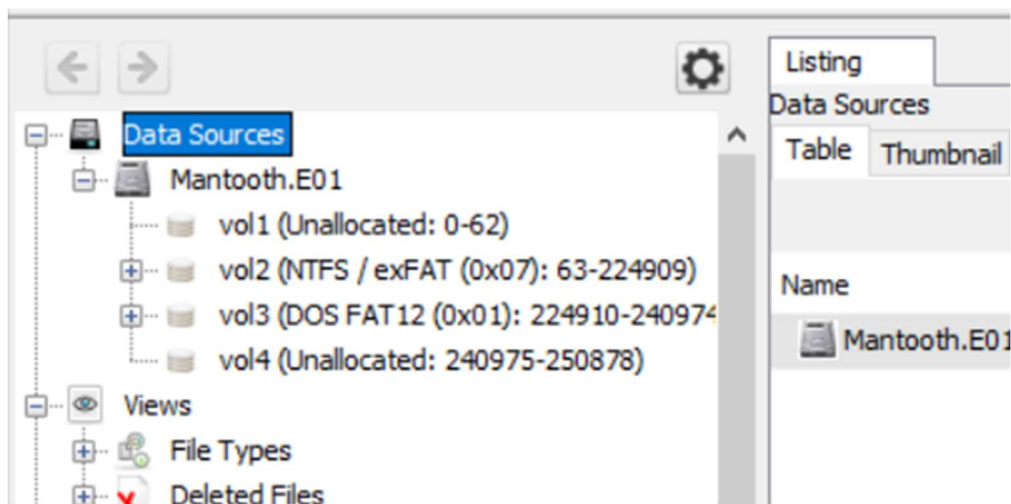
Select all that will serve the purpose of your investigation and click Next. Once the data source is added, click Finish. It will take some buffer time to extract and analyze the data depending upon the size of the Data Source.

1. Exploring the data source:

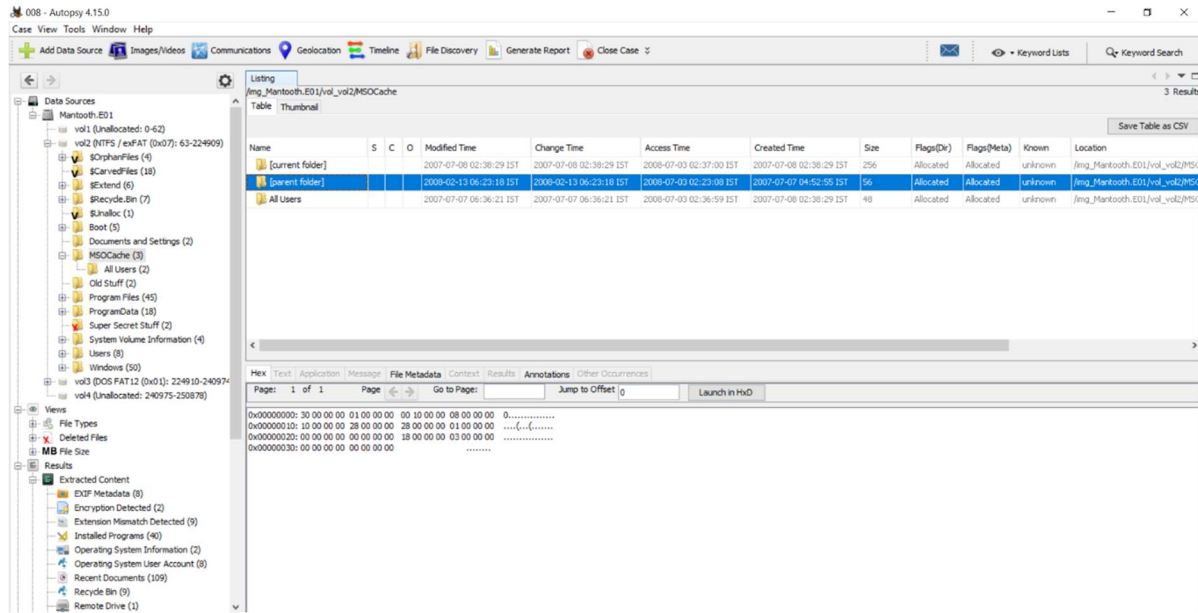
The Data Source information: Here the basic metadata is shown. A detailed analysis is displayed in the bottom section. These details can be extracted in the form of Hex values, Results, File Metadata, etc.



The disk image is then broken down based upon its volume partitions.

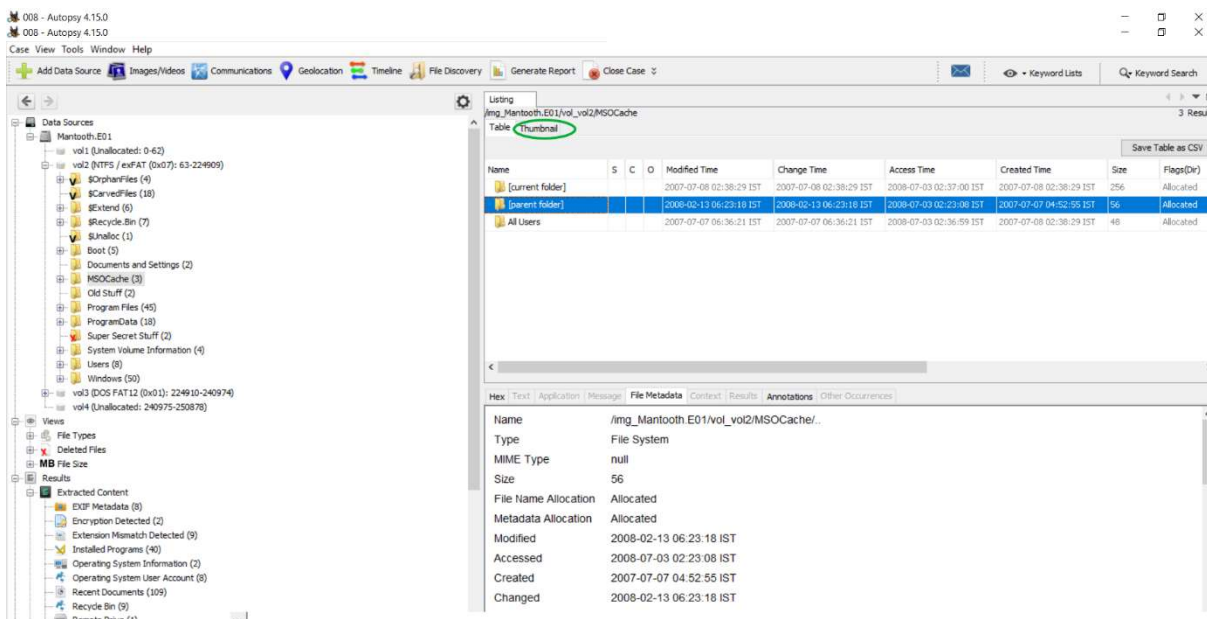


Each volume can be browsed for its contents, results for which are displayed in the section at the bottom. For example, the content shown below belongs to Data Sources -> Mantooth.E01 -> MSOCache-> [Parent Folder].

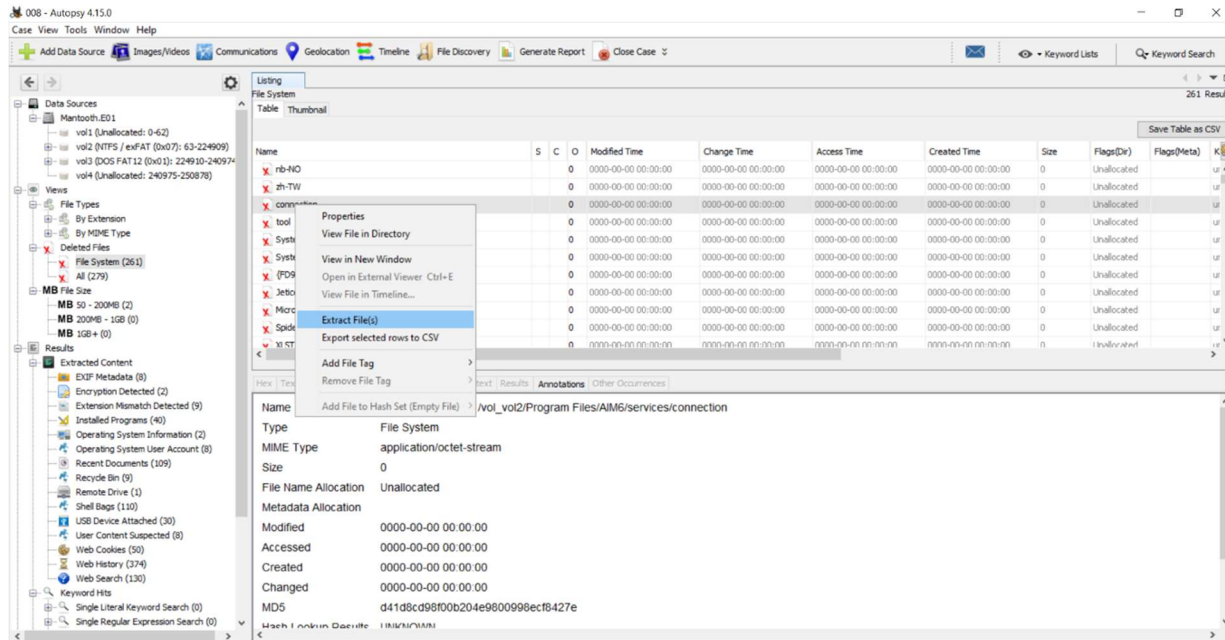


Views (Determines the factor of file classification)

- **File Type:** Here the files are categorized based upon their type. The classification can be done either on the basis of file extension or MIME type. While both of these provide a hint about how to deal with a file, file extensions are commonly used by the OS to decide what program shall be used to open a file and MIME types are used by the browser to decide about how to present the data (or by the server on how to interpret the data received). Files displayed here also include the deleted files.



- **Deleted Files:** Here information about the files that were specifically deleted can be found. These deleted files can be recovered as well: Right-click on the file to be recovered -> click on Extract File(s) -> Save the file in an appropriate destination.

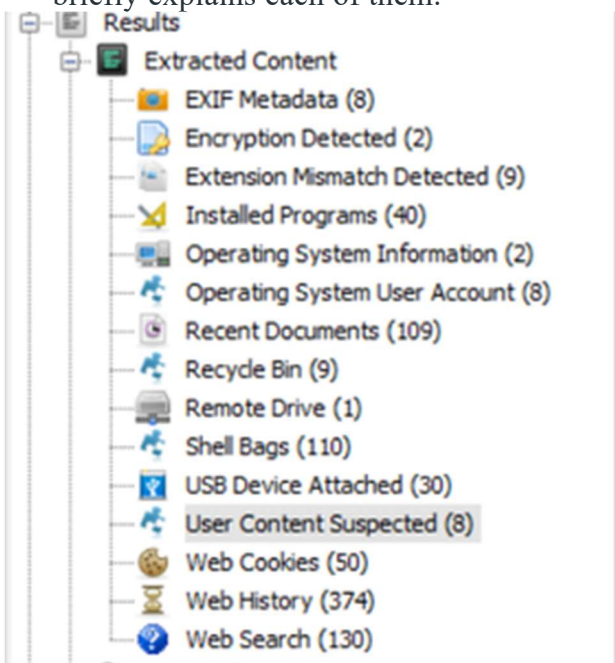


- **MB Size Files:** Here files are classified based upon their size. The range starts from 50MB. This enables the examiner to determine exclusively large files.
- **Note:** It is usually advised to not scan or extract any suspected files/ disks such as payload files, etc. in the main system, rather scan them in safe environments such as a virtual machine, and then extract the data, as they hold the possibility of being corrupt and may infect the examiner's system with viruses.

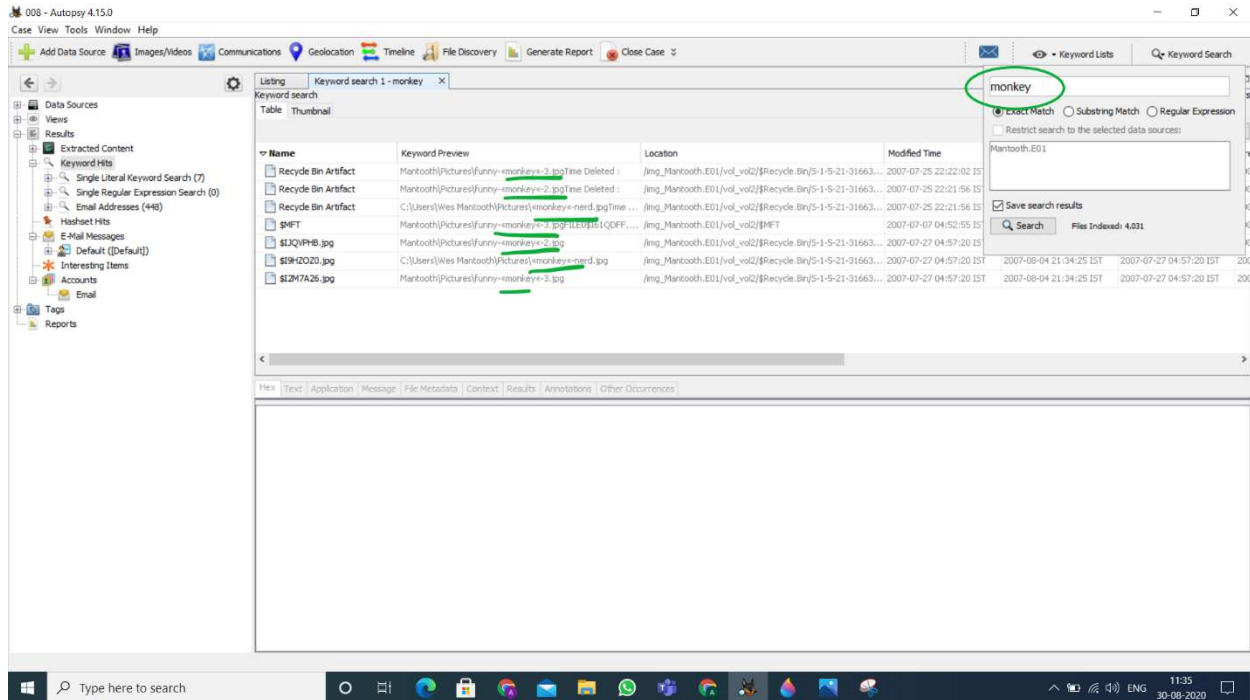
Results:

All the extracted data is viewed in **Views/ Data Source**. In **Results**, we get the information about this data.

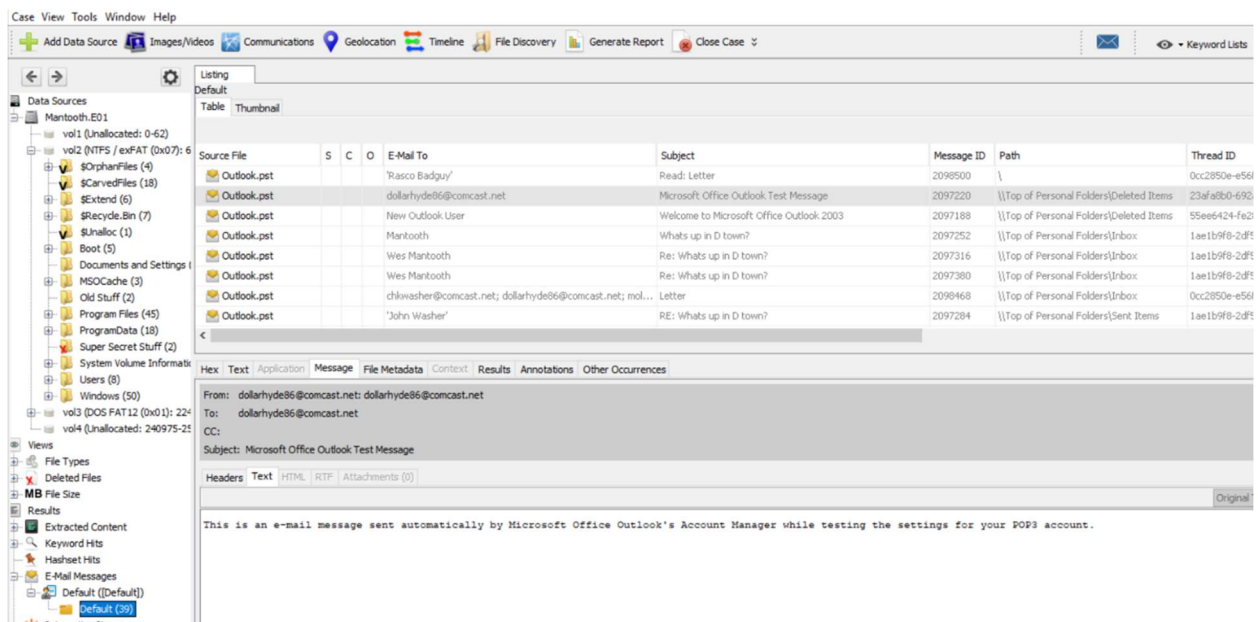
- **Extracted Content:** Each Extracted Content displayed below can be further explored. The following briefly explains each of them.



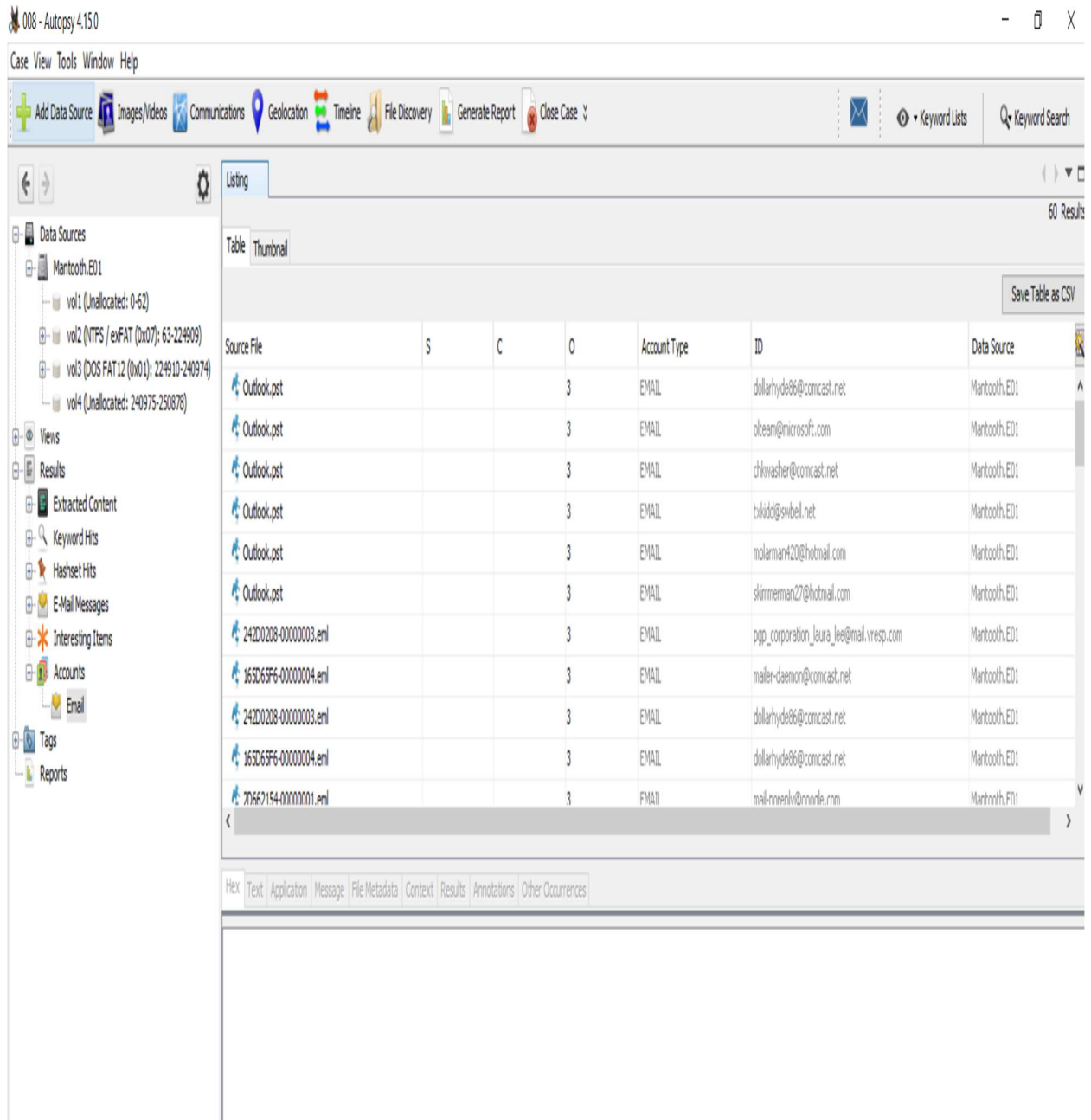
- **EXIF Metadata:** It contains all the .jpg images that have EXIF Metadata associated with them, this Metadata can be analyzed further.
- **Encryption Detection:** It detects files that are password protected/ encrypted.
- **Extension Mismatch Detection:** As explained above, it identifies the files whose extensions do not match their MIME types and thus they may be suspicious.
- **Installed Programs:** It gives details about the software used by the user. This information is extracted with the help of the Software Registry hive.
- **Operating System Information:** It gives information about the OS with the help of the Windows Registry hive and the Software Registry hive.
- **Operating System User Account:** It lists information about all the user accounts, for example, accounts belonging to the device are extracted from the Software Hive and the accounts associated with the Internet Explorer using index.data files.
- **Recent documents:** Lists all the documents that were accessed nearby the time the disk image was captured.
- **Recycle Bin:** Files that are temporarily stored on the system before being permanently deleted are visible here.
- **Remote Drive:** Shows information about all the remote drives accessed using the system.
- **Shell bags:** A shell bag is a set of registry keys that stores details about a folder being viewed, such as its position, icon, and size. All the Shell bags from the system can be viewed here.
- **USB Device attached:** All the information about the external devices attached to the system is displayed here. This data is extracted from Windows Registry which is actually a maintained database about all the activities taking place on the system.
- **Web Cookies:** Cookies save the user information from the sites and thus provide a lot of information about the user's online activities.
- **Web History:** All the details about the browser history is shown here.
- **Web Searches:** Details about the web searches made are displayed here.
- **Keyword Hits:** Here specific keywords can be looked for in the image of the disk. Multiple data sources can be selected for the lookup. The search can be restricted to Exact match, Substring match and Regular expression, for example, emails/ IP Addresses, etc.



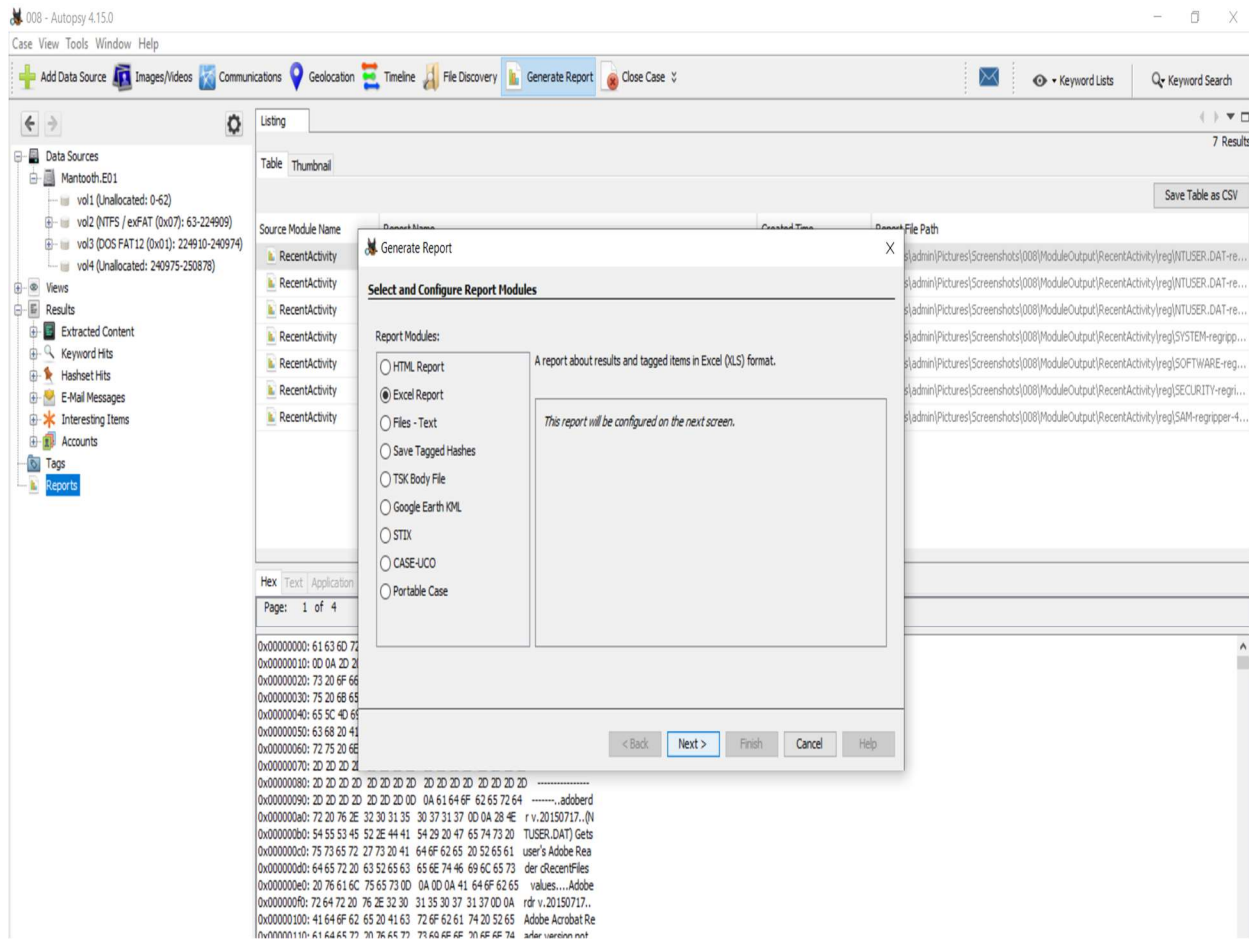
- **HashSet Hits:** Here the search can be made using hash values.
- **E-mail Messages:** Here all the *outlook.pst* files can be explored.

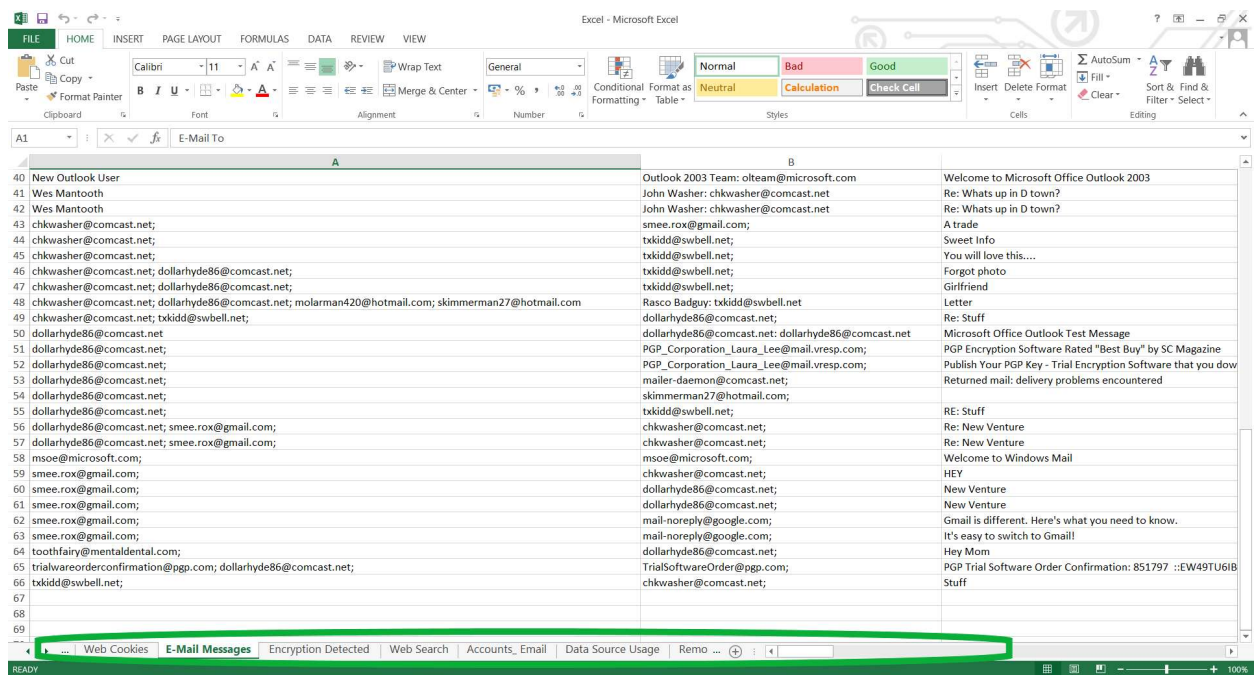
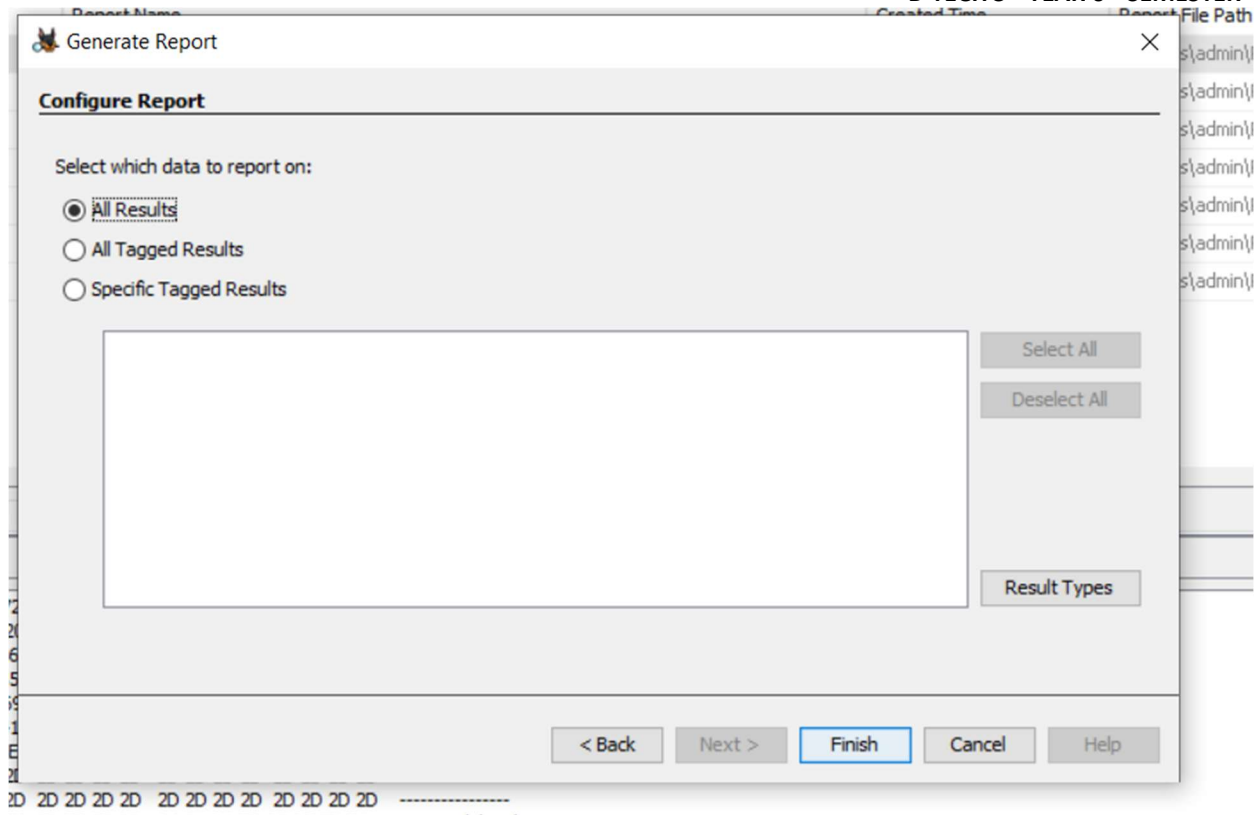


- **Interesting Items:** As discussed before, these are the file results based upon the custom rules set by the examiner.
- **Accounts:** Here all the details regarding the accounts present on the disk are shown. This disk has the following EMAIL accounts.



- **Reports:** Reports about the entire analysis of the data source can be generated and exported in many formats.

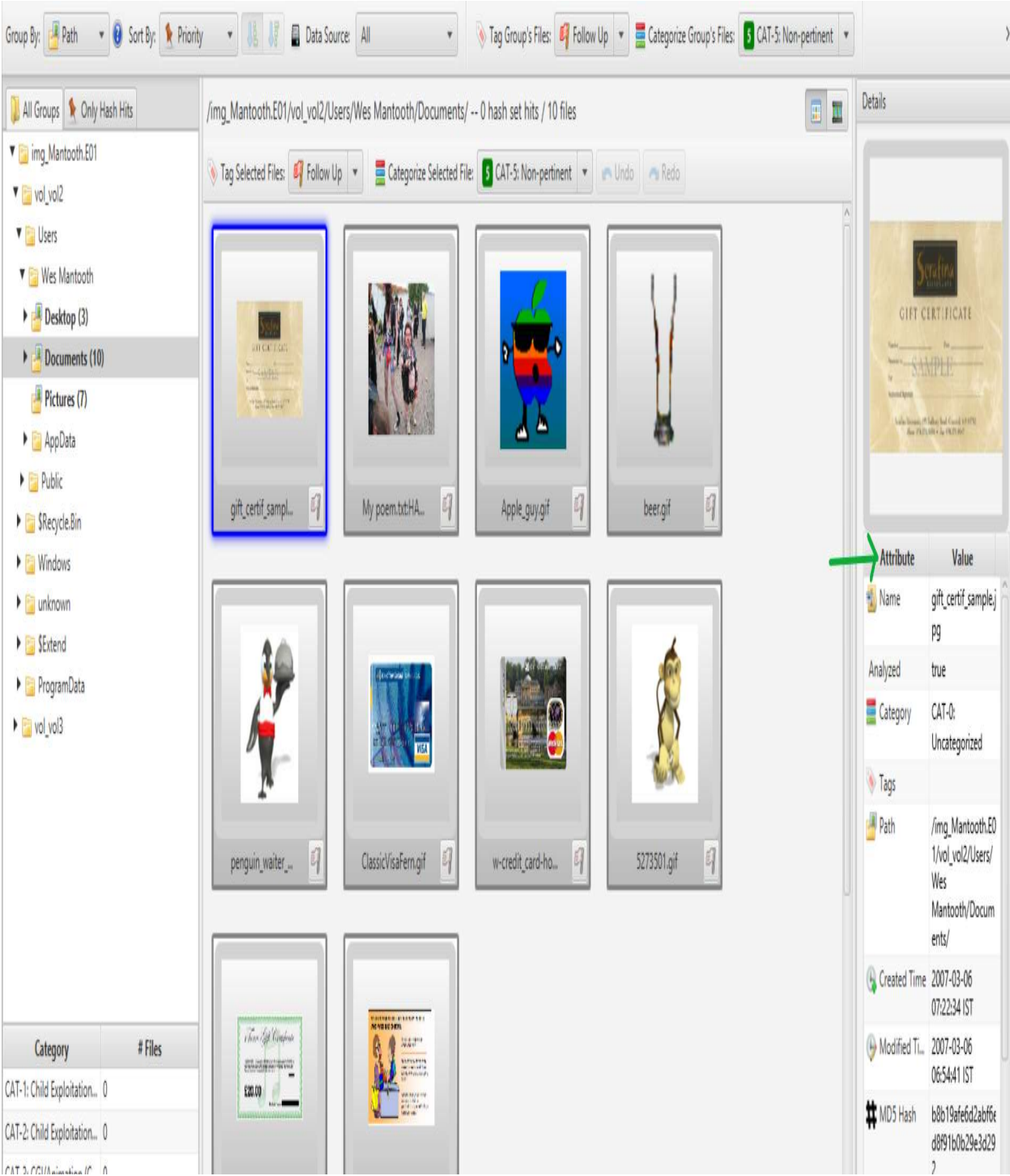




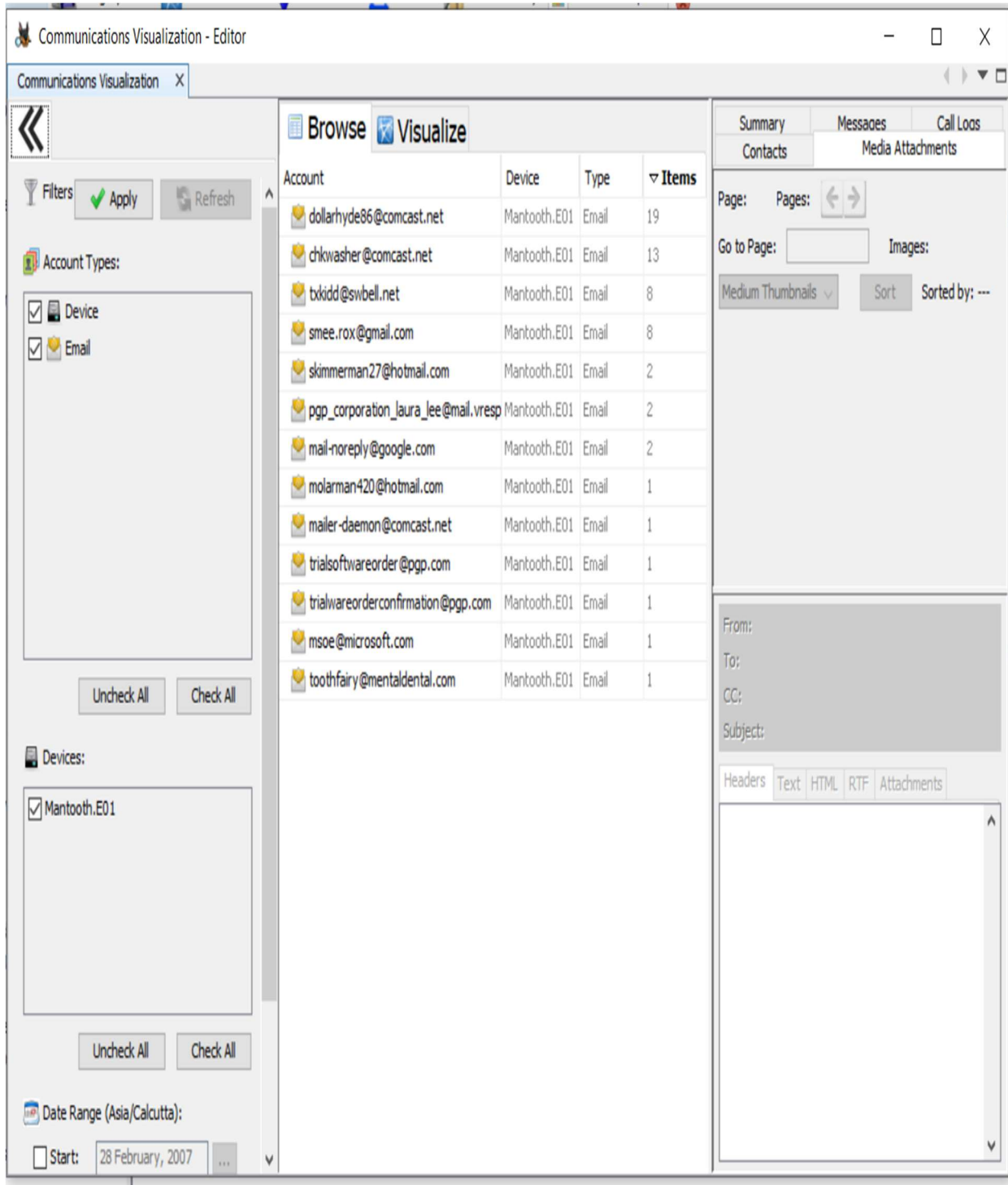
- **Additional Features:**



- **Add a Data Source:** Each case can hold multiple Data Sources.
- **Images/Videos:** Images/ Videos in the data source can be viewed in Gallery View. The information here is displayed in the form of attribute-value pairs.



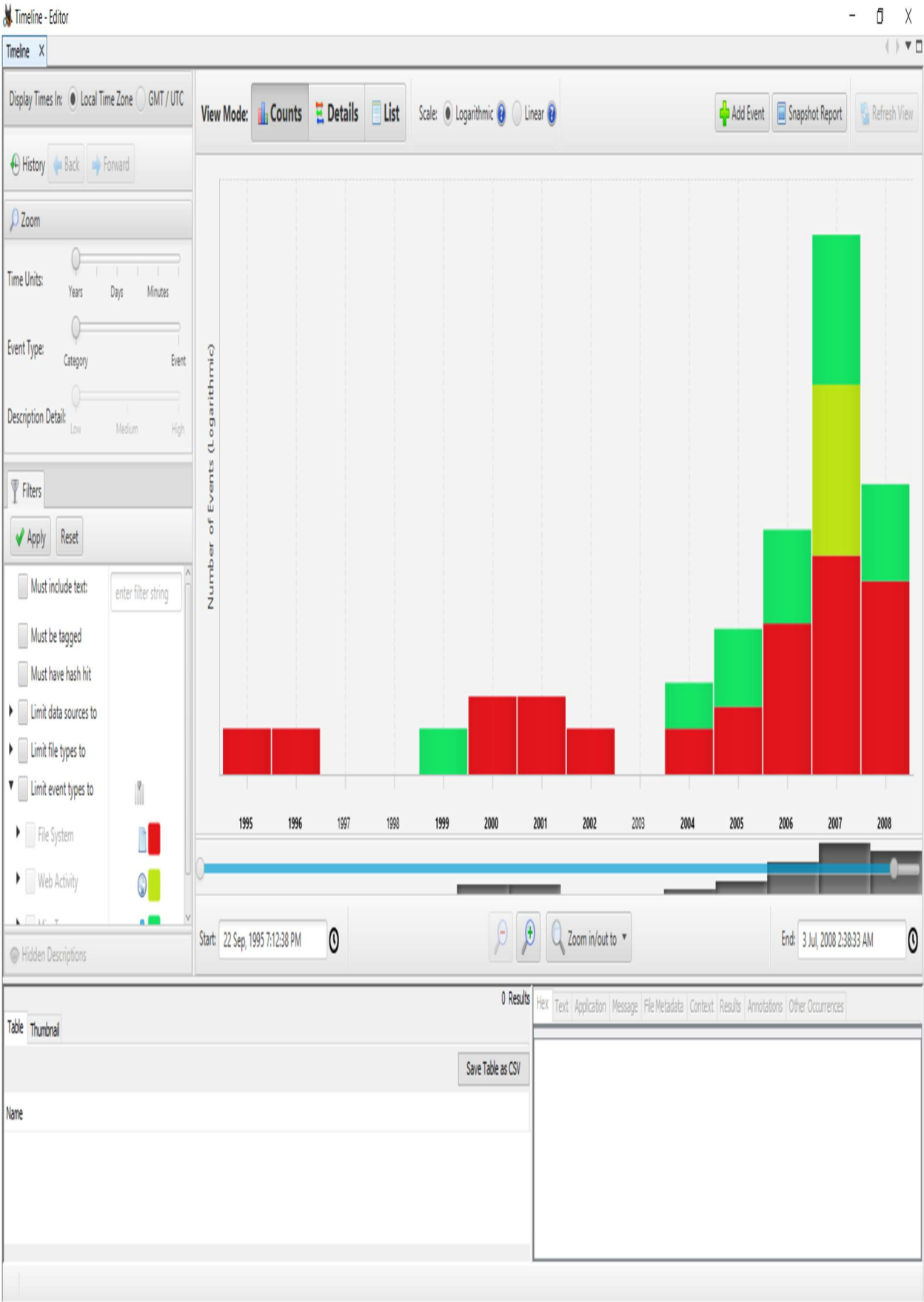
- o **Communications:** All the communications made using the source device are displayed here. This device had communications only in the form of emails.



Geolocation: This window displays the artifacts that have longitude and latitude attributes as waypoints on a

map. Here the data source has no waypoints.

Timeline: Information about when the computer was used or what events took place before or after a given event can be found, this greatly helps in investigating events near about a particular time.



Almost all the basic features and how actually Autopsy works have been discussed in this article. However, it is always recommended to go through different sample data sources to explore even more.

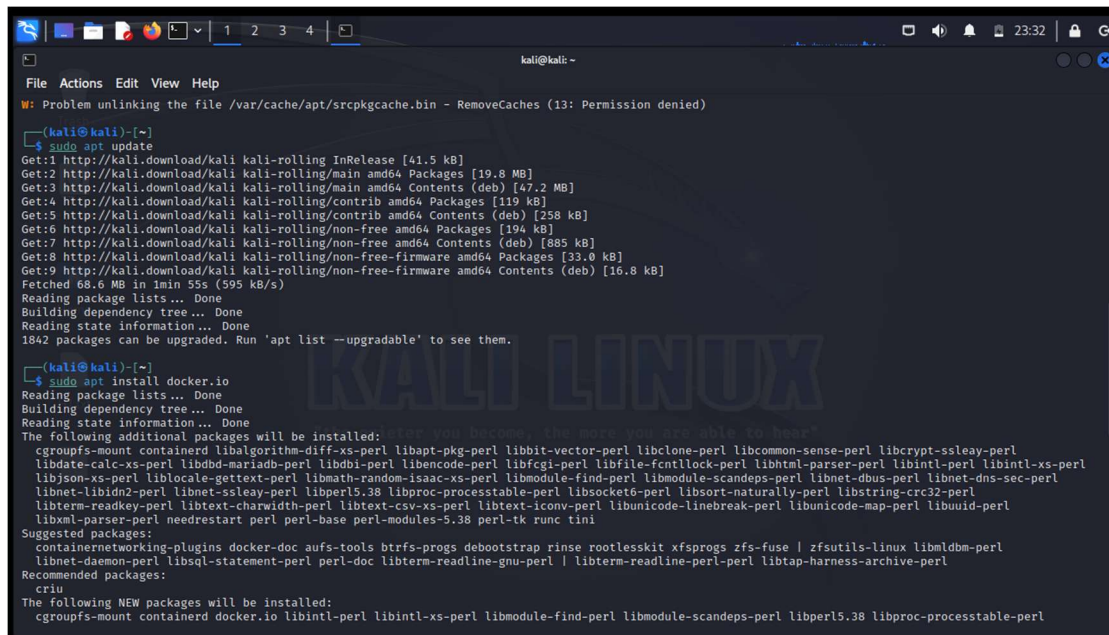
PRACTICAL NO: 9

Aim : Implementation of Mobile Audit and generate the report of the existing Artifacts.

Tools : mobsf

Description :

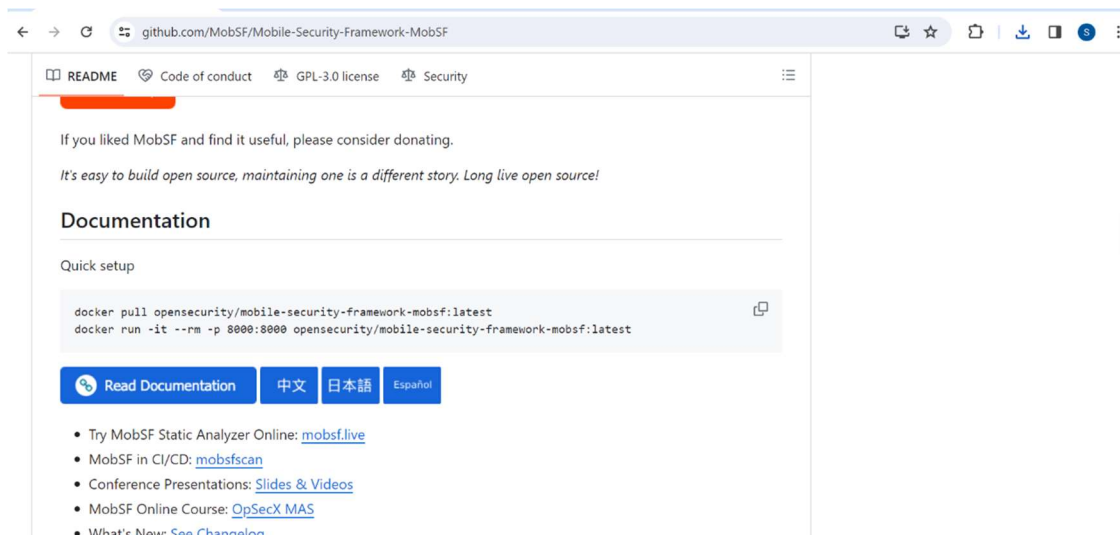
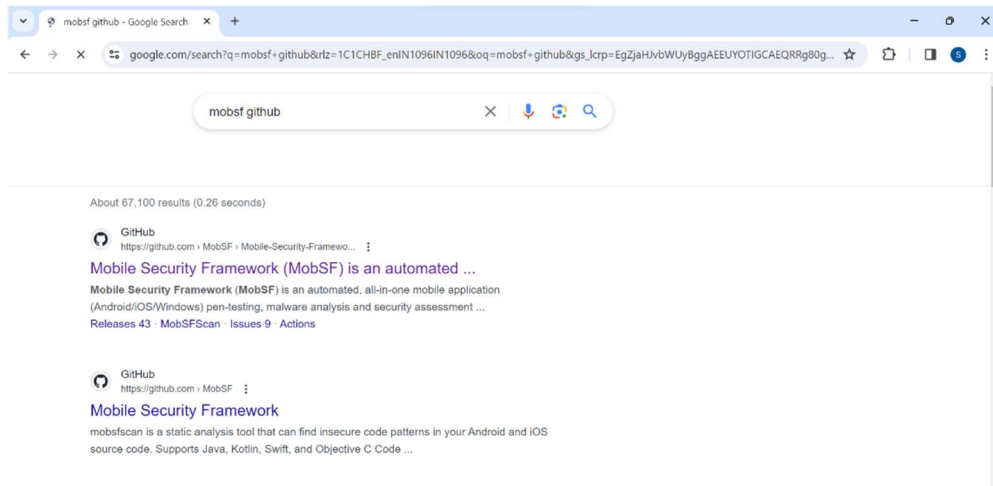
- Update APT
sudo apt update
- Install Docker



```
kali@kali: ~  
File Actions Edit View Help  
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)  
  
(kali@kali)-[~]  
$ sudo apt update  
Get:1 http://kali.download/kali kali-rolling InRelease [41.5 kB]  
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [19.8 MB]  
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [47.2 MB]  
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [119 kB]  
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [258 kB]  
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [194 kB]  
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [885 kB]  
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [33.0 kB]  
Get:9 http://kali.download/kali kali-rolling/non-free-firmware amd64 Contents (deb) [16.8 kB]  
Fetched 68.6 MB in 1min 55s (595 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
1842 packages can be upgraded. Run 'apt list --upgradable' to see them.  
  
(kali@kali)-[~]  
$ sudo apt install docker.io  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
  cgroupfs-mount containerd libalgorithm-diff-xs-perl libapt-pkg-perl libbit-vector-perl libclone-perl libcommon-sense-perl libcrypt-ssleay-perl  
  libdate-calc-xs-perl libdbd-mariadb-perl libdbi-perl libencode-perl libfcgi-perl libfile-fcntllock-perl libhtml-parser-perl libintl-perl libintl-xs-perl  
  libjson-xs-perl liblocale-gettext-perl libmath-random-isaac-xs-perl libmodule-find-perl libmodule-scandeps-perl libnet-dbus-perl libnet-dns-sec-perl  
  libnet-libidn2-perl libnet-ssleay-perl libperl5.38 libproc-processtable-perl libsocket6-perl libsort-naturally-perl libstring-crc32-perl  
  libterm-readkey-perl libtext-charwidth-perl libtext-csv-xs-perl libtext-iconv-perl libunicode-linebreak-perl libunicode-map-perl libuuid-perl  
  libxml-parser-perl needrestart perl perl-base perl-modules-5.38 perl-tk runc tini  
Suggested packages:  
  containernetworking-plugins docker-doc aufs-tools btrfs-progs debootstrap rinse rootlesskit xfsprogs zfs-fuse | zfsutils-linux libldbml-perl  
  libnet-daemon-perl libsql-statement-perl perl-doc libterm-readline-gnu-perl | libterm-readline-perl-perl libtap-harness-archive-perl  
Recommended packages:  
  criu  
The following NEW packages will be installed:  
  cgroupfs-mount containerd docker.io libintl-perl libintl-xs-perl libmodule-find-perl libmodule-scandeps-perl libperl5.38 libproc-processtable-perl
```

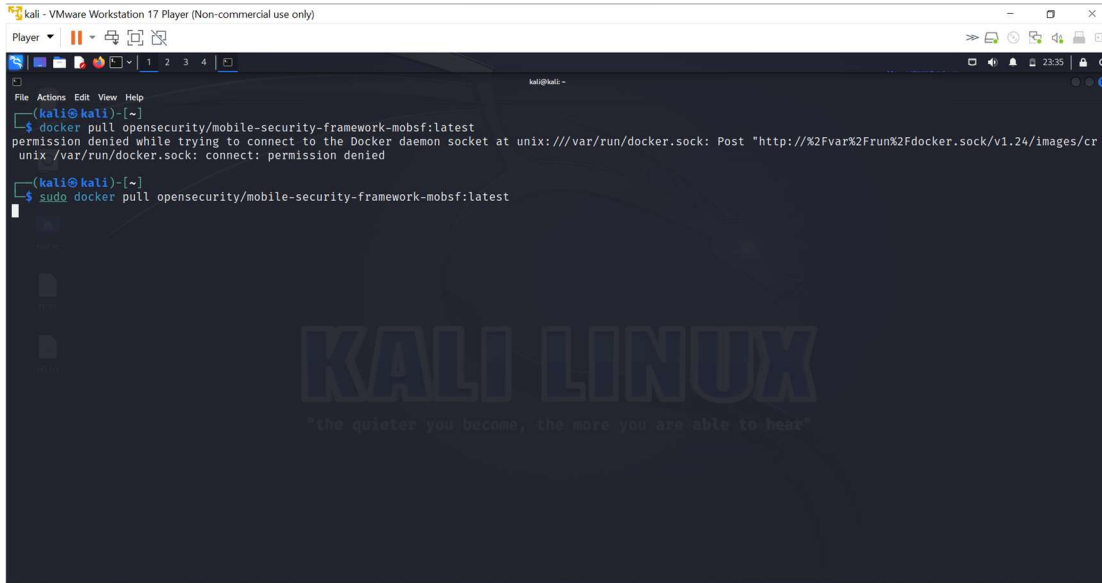
sudo apt install docker.io

- Search mobsf GitHub on your web browser



Next, download MobSF Docker image from <https://hub.docker.com/r/opensecurity/mobile-security-framework-mobsf/> with the following command:

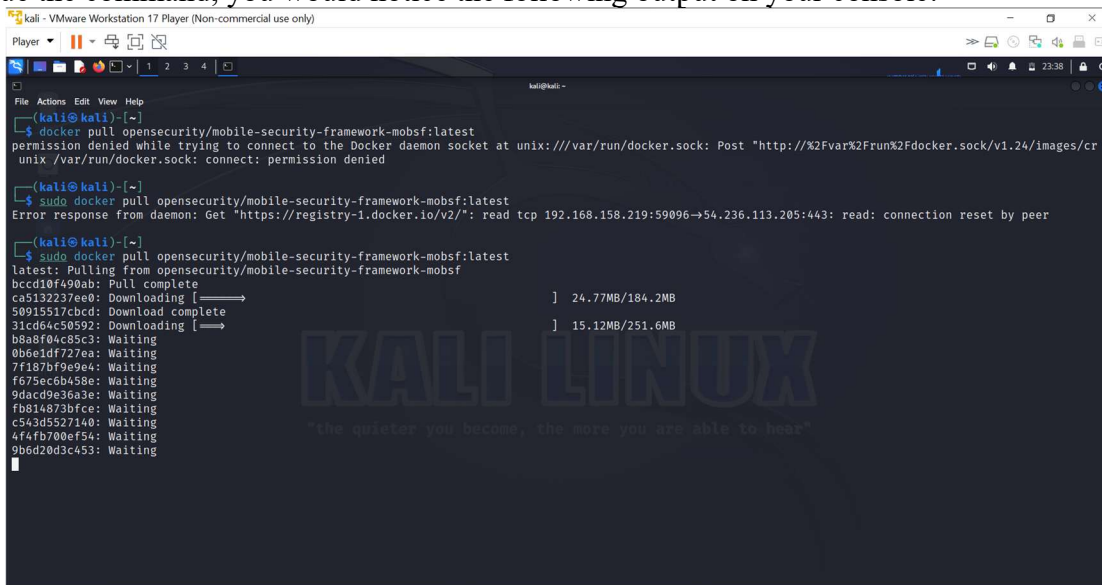
`docker pull opensecurity/mobile-security-framework-mobsf:latest`



```
kali@kali:~$ docker pull opensecurity/mobile-security-framework-mobsf:latest
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Post "http://%2Fvar%2Frun%2Fdocker.sock/v1.24/images/create?name=opensecurity%2Fmobile-security-framework-mobsf%3Alatest&platform=linux%2Famd64": dial unix /var/run/docker.sock: connect: permission denied

kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
```

Once you issue the command, you would notice the following output on your console:



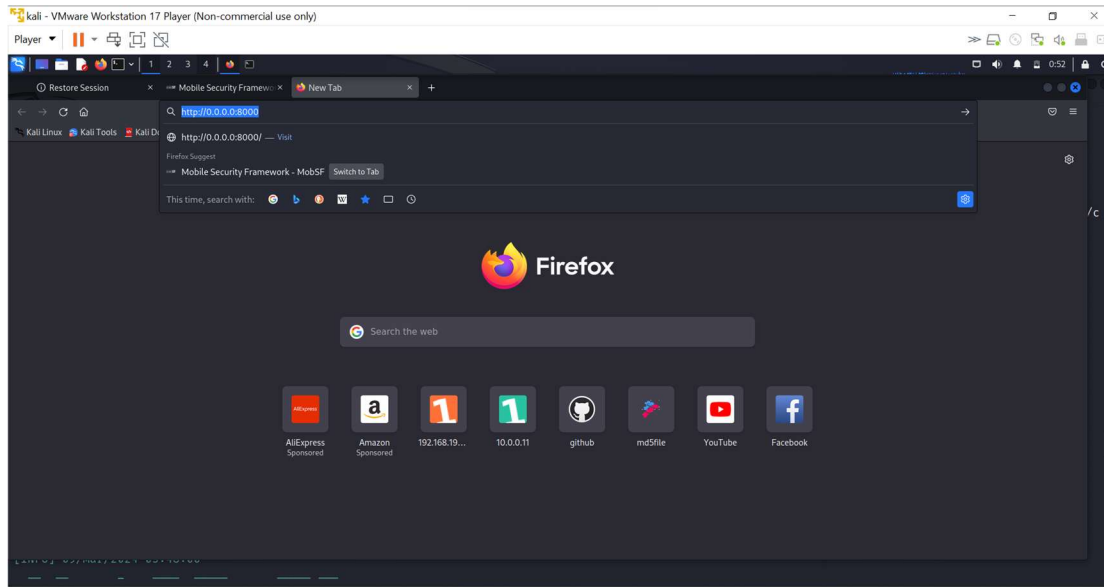
```
kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
Error response from daemon: Get "https://registry-1.docker.io/v2/": read tcp 192.168.158.219:59096->54.236.113.205:443: read: connection reset by peer

kali@kali:~$ sudo docker pull opensecurity/mobile-security-framework-mobsf:latest
latest: Pulling from opensecurity/mobile-security-framework-mobsf
b6cd10f400ab: Pull complete
ca5132237e00: Downloading [====>] 24.77MB/184.2MB
50915517cbcd: Download complete
31cd64c50592: Downloading [====>] 15.12MB/251.6MB
b8a8f04c85c3: Waiting
0b6e1df727ea: Waiting
7f187bf9e9e4: Waiting
f675ec6b459e: Waiting
9d8cd9e36a3e: Waiting
fb814873bfce: Waiting
c543d5527140: Waiting
4f4fb700ef54: Waiting
9b6d20d3c453: Waiting
```

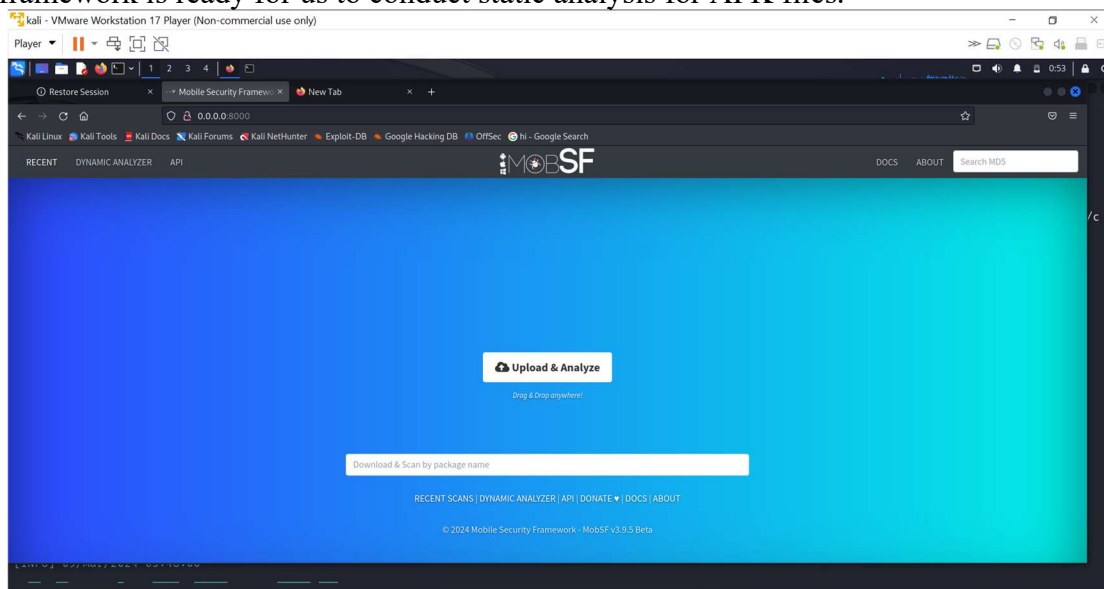
This signifies that the docker image for MobSF is being downloaded. Once completed, the following message will appear:

Now that the docker image is downloaded, the image can be run with the following command:

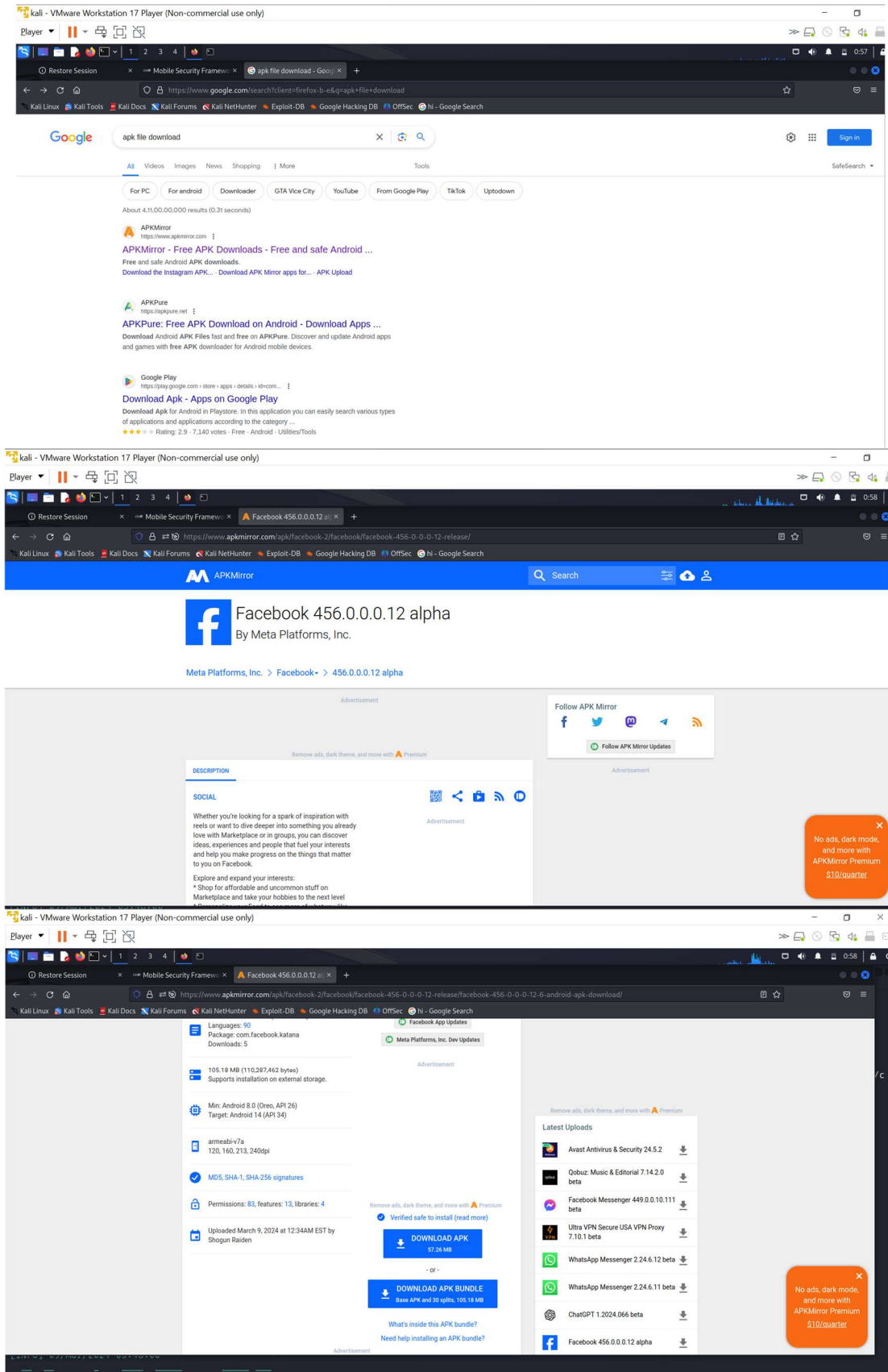
`docker run -it -p 8000:8000 opensecurity/mobile-security-framework-mobsf`

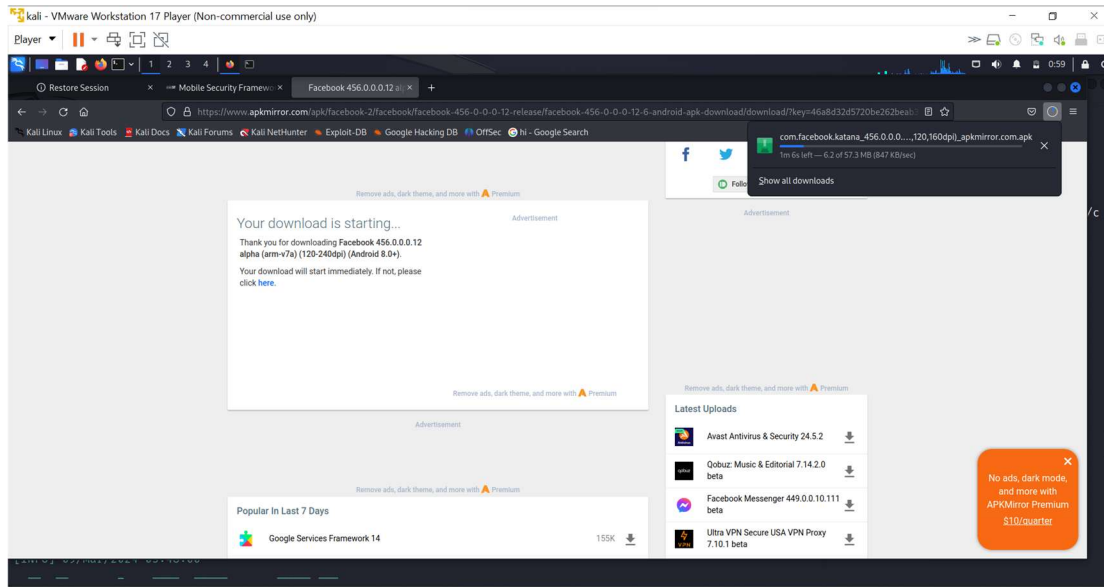


Our MobSF framework is ready for us to conduct static analysis for APK files.

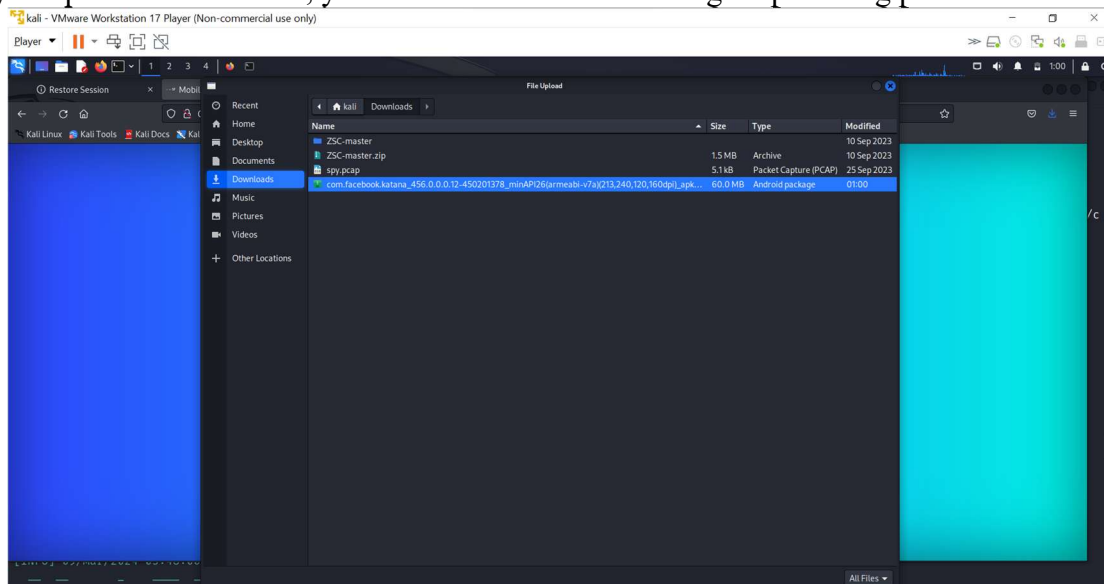


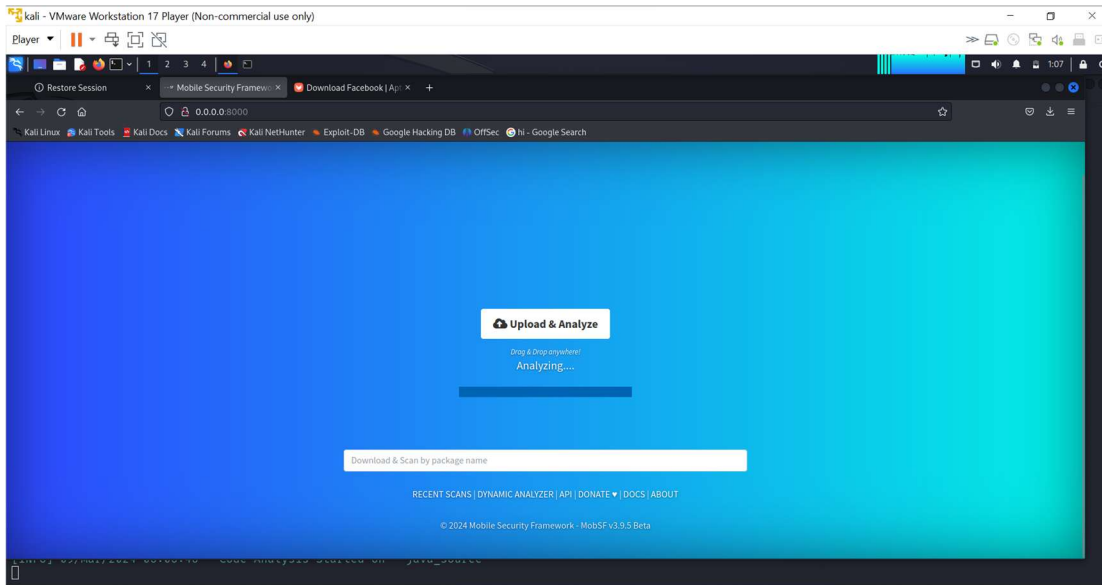
For our testing, we take Facebook Lite's APK file.



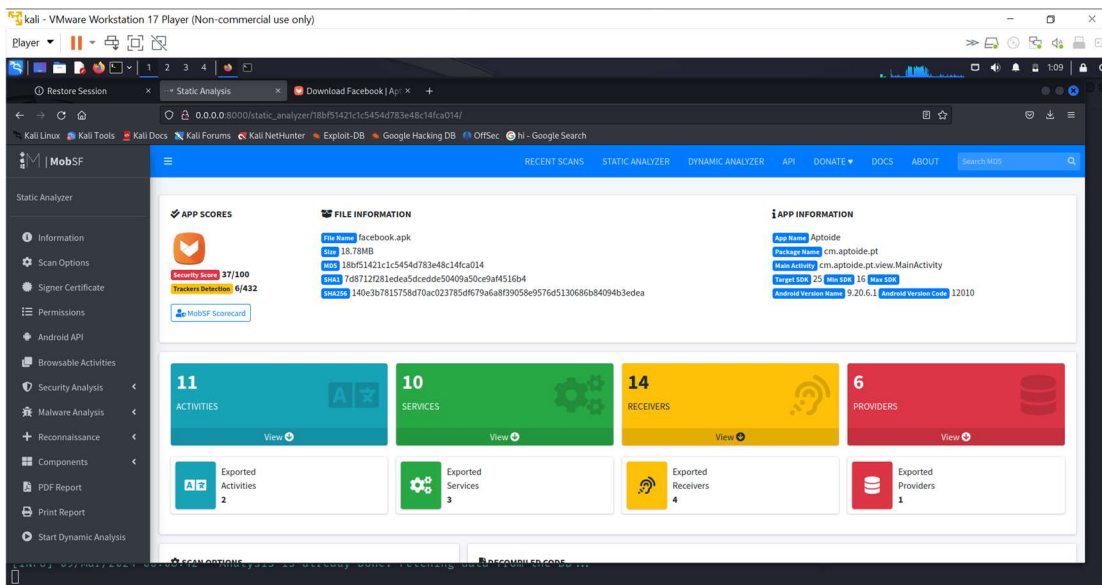


Once you upload the APK file, you would notice the following output being printed on our terminal:

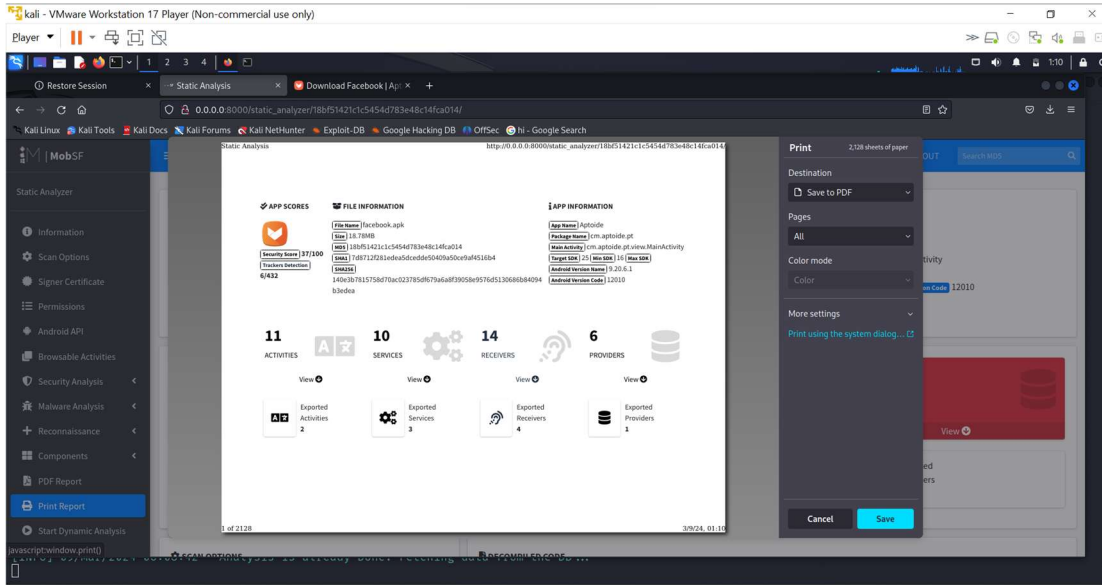




Once the analysis is complete, the browser will show assessment details, such as application description, Android permission analysis, code analysis, CVSS grading, malware analysis etc.



You also notice a brief overview of the application you have reviewed. You can also the report for same.



PRACTICAL NO: 10

Aim : Implementation of Cyber Forensics tools for Disk Imaging, Data acquisition, Data extraction and Data Analysis and recovery..

Tools : FTK Imager, Autopsy

Description :

- FTK Imager (Data acquisition and Disk imaging)
- SCOPE: The scope of practical acquisition of digital evidence can encompass a wide range of methods and devices. Common areas are: Hard Drives, USB, Mobile Devices, Network Forensics, Live system etc.
- Requirements: The requirements are Hardware and Devices, FTK imager software, Forensics workstations, sample data, Documentation Templates etc.
- Theory:

Data Acquisition:

- The gathering and recovery of sensitive data during a digital forensic investigation is known as data acquisition and collecting and preserving electronic evidence from various sources.
- Digital forensic analysts need to know how to access, recover, and restore that data as well as how to protect it for future management. This involves producing a forensic image from digital devices and other computer technologies.
- The data acquisition process involves several steps: Identification, Preservation, collection, Verification, Documentation.

Disk Imaging:

Disk imaging, also known as forensic imaging or disk cloning, is a crucial process in digital forensics for cybersecurity.

It involves creating a bit-for-bit copy or snapshot of an entire storage device, such as a hard drive, solid-state drive (SSD), or any other media, to preserve its contents for analysis and investigation.

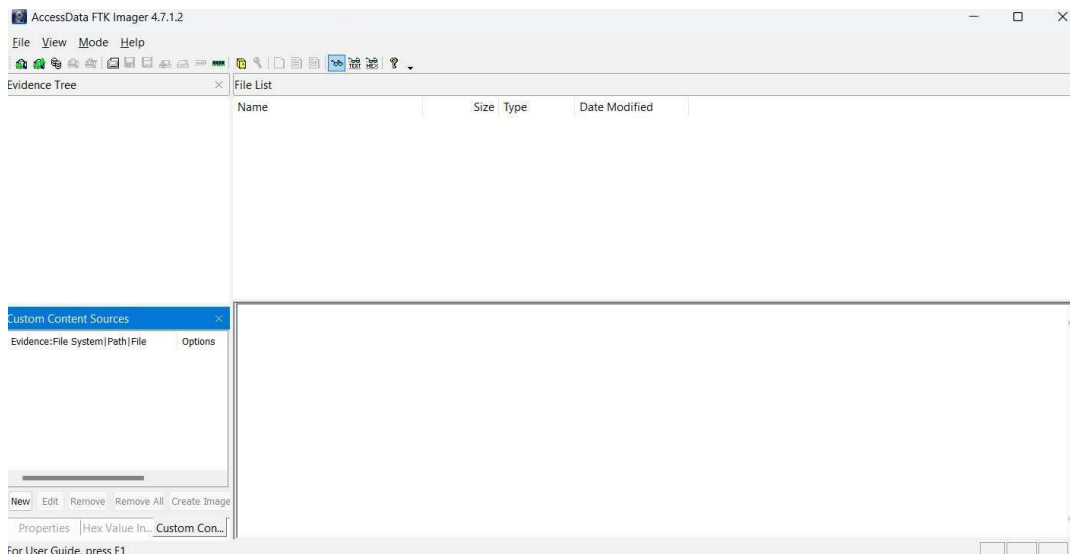
The purpose of disk imaging is to obtain an exact replica of the original storage device, including not only the visible files and folders but also the hidden and deleted data.

Types of Disk Imaging:

- **Logical Imaging:** Logical imaging, also known as file-level imaging or logical acquisition, involves creating an image or snapshot of the logical structure of a storage device, such as a file system or a partition. This type of imaging captures the files and directories stored on the disk, along with their metadata and attributes, without including unallocated space or low-level disk structures.
- **Physical Imaging:** Physical imaging, also known as bit-for-bit imaging or sector-level imaging, is a comprehensive process that involves creating an exact copy of the entire storage device, including all sectors, regardless of whether they are allocated or unallocated. This type of imaging captures not only the visible files and folders but also the hidden data, deleted files, slack spaces, and all other data.

- **Forensic Significance of Imaging:** Forensic imaging prevents the loss of original data. These imaging tools and techniques are the only way to ensure that electronic data can be successfully admitted as evidence in a court or legal proceeding.

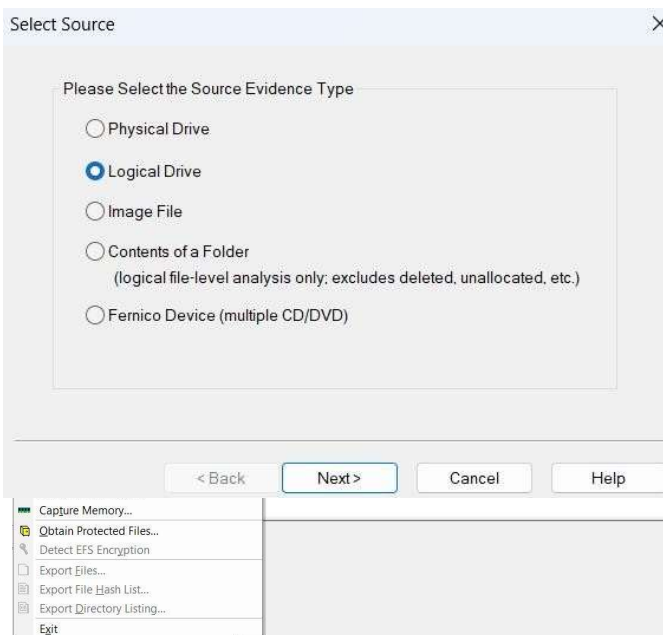
- dcEXAMINATION:



Step-1 : Open FTK Imager software

Step-2 : Now go to the find option -> choose Create Disk Image.

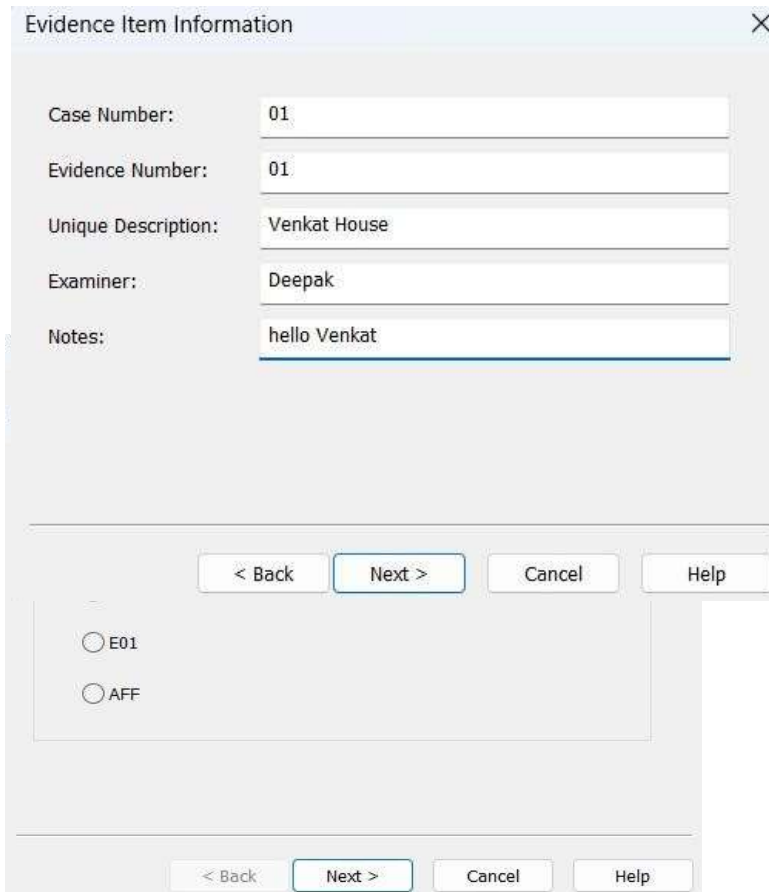
Step-3 : After clicking Create Disk image -> select the source of evidence type like Logical Drive, then click next option.



En no :

Step-4 : Now select source drive path and then press the finish button.

Step-5 : Choose destination image type (Raw(dd)) -> Next

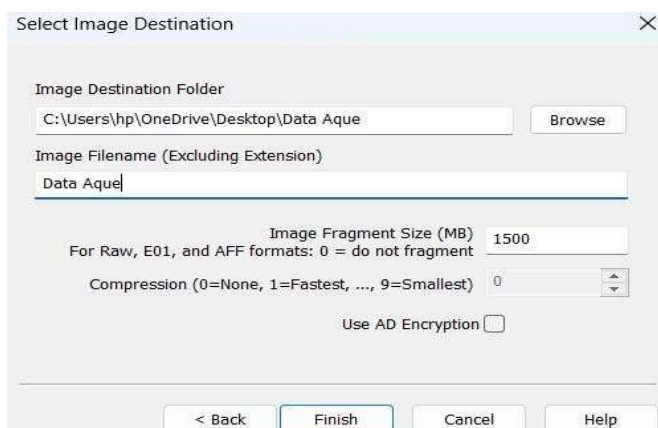


The 'Evidence Item Information' dialog box contains the following fields and controls:

- Case Number: 01
- Evidence Number: 01
- Unique Description: Venkat House
- Examiner: Deepak
- Notes: hello Venkat
- Buttons: < Back, Next >, Cancel, Help
- Radio buttons: ☐ E01, ☐ AFF
- Bottom buttons: < Back, Next >, Cancel, Help

Step-6 : Now fill the form of Evidence item information.

Step-7 : After filling the information -> select image destination, enter Image filename and Image fragment size then click the finish button.

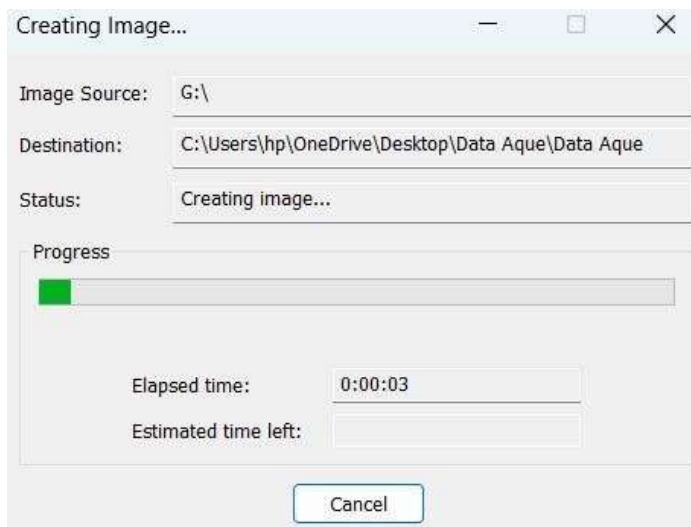
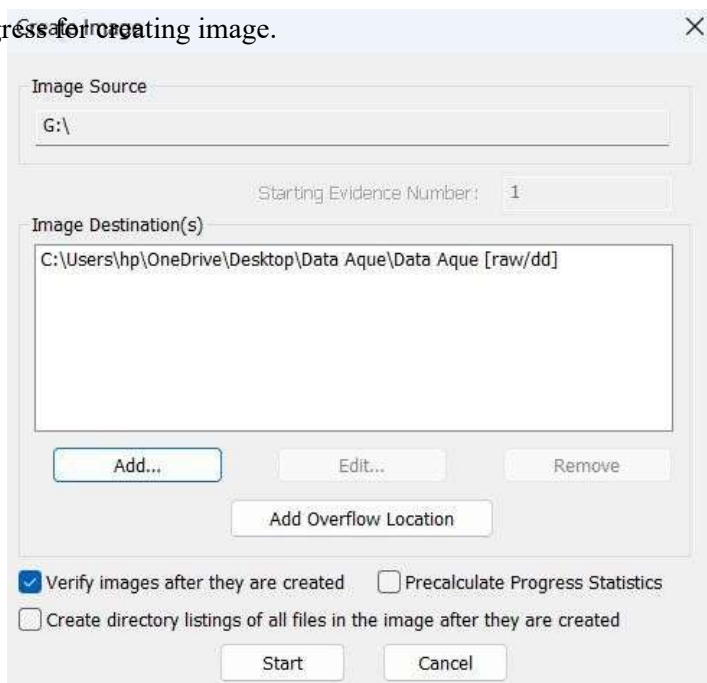


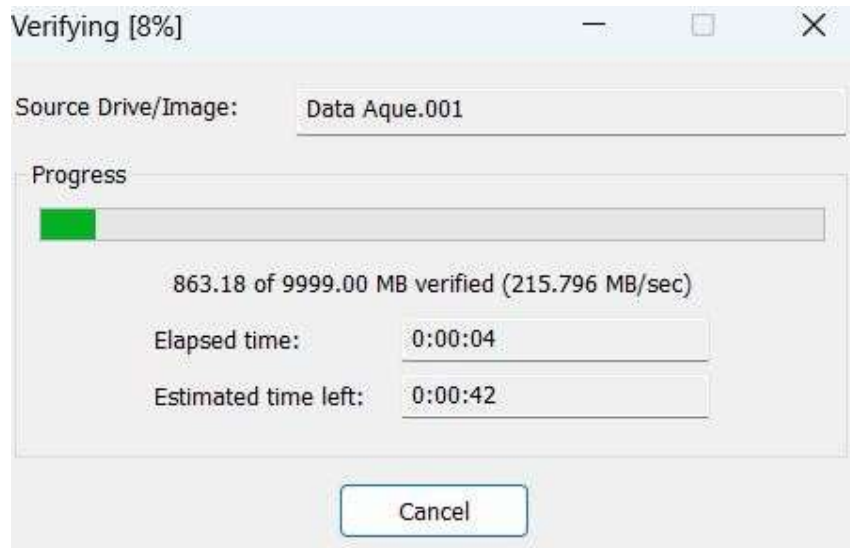
The 'Select Image Destination' dialog box contains the following fields and controls:

- Image Destination Folder: C:\Users\hp\OneDrive\Desktop\Data Aque, with a Browse button.
- Image Filename (Excluding Extension): Data Aque
- Image Fragment Size (MB): 1500, with a note: For Raw, E01, and AFF formats: 0 = do not fragment.
- Compression (0=None, 1=Fastest, ..., 9=Smallest): 0, with a spinner control.
- Use AD Encryption: ☐
- Buttons: < Back, Finish, Cancel, Help

Step-8 : After all the processes now, click in the checkbox of verify images after they are created -> start.

Step-9 : Progress for creating image.





Step-10 : Verifying the source drive/image

Step 11: Verify result of source drive/image form of hash values.

Drive/Image Verify Results	
Name	Data Aque.001
Sector count	20477952
MDS Hash	
Computed hash	a8b13d1099105de415d7b795374ef1aa
Report Hash	a8b13d1099105de415d7b795374ef1aa
Verify result	Match
SHA1 Hash	
Computed hash	b473b690ef93531639142ee3011b7a8a0947d81
Report Hash	b473b690ef93531639142ee3011b7a8a0947d81
Verify result	Match
Bad Blocks List	
Bad block(s) in image	No bad blocks found in image

Created By AccessData® FTK® Imager 4.7.1.2

Case Information:

Acquired using: ADI4.7.1.2

Case Number: 01

Evidence Number: 01

Unique description: Venkat House

Examiner: Deepak

Notes: hello Venkat

Information for C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque:

Physical Evidentiary Item (Source) Information:

[Device Info]

Source Type: Logical

[Drive Geometry]

Bytes per Sector: 512

Sector Count: 20,477,952

[Physical Drive Information]

Removable drive: False

Source data size: 9999 MB

Sector count: 20477952

[Computed Hashes]

MD5 checksum: a8b13d1099105de415d7b795374ef1aa

SHA1 checksum: b473b690eff93531639142ee301187a8a0947d81

Image Information:

Acquisition started: Wed Jun 28 13:00:17 2023

Acquisition finished: Wed Jun 28 13:01:01 2023

Segment list:

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.001

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.002

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.003

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.004

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.005

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.006

C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.007

Image Verification Results:

Verification started: Wed Jun 28 13:01:01 2023

Verification finished: Wed Jun 28 13:01:40 2023

MD5 checksum: a8b13d1099105de415d7b795374ef1aa : verified

SHA1 checksum: b473b690eff93531639142ee301187a8a0947d81 : verified

- Observation:
- Autopsy:
 - Autopsy is a digital forensics platform and graphical interface to The Sleuth Kit and other digital forensics tools. It is used by law enforcement, military, and corporate examiners to investigate what happened on a computer. You can even use it to recover photos from your camera's memory card.
 - Basically, the autopsy is a free open-source tool that supports a wide range of other digital forensics modules and tools. The tool is largely maintained by Basis Technology Corp. With the assistance of programmers from the community. The Autopsy is computer software that makes it simpler to deploy many of the open-source programs and plugins used in The Sleuth Kit.

Features of Autopsy

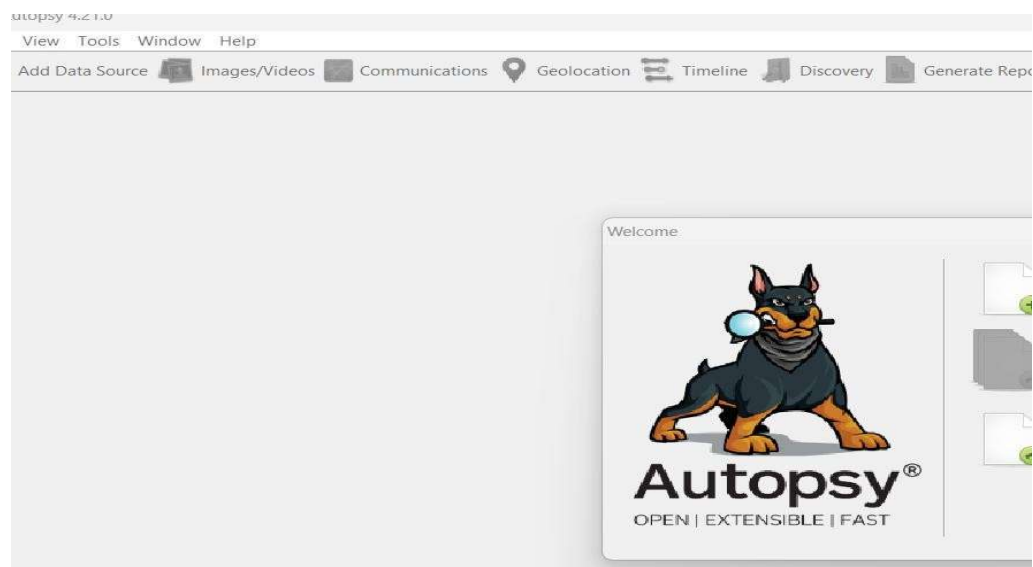
- Timeline Analysis:- Displays system events in a graphical interface to help identify activity.
- Keyword Search: Text extraction and index searched modules enable you to find files that mention specific terms and find regular expression patterns.
- Web Artifacts:- Extracts web activity from common browsers to help identify user activity.
- Registry Analysis:- Uses Reg Ripper to identify recently accessed documents and USB devices.
- Email Analysis: Parses MBOX format messages, such as Thunderbird.
- EXIF: Extracts geo location and camera information from JPEG files.
- Robust File System Analysis: Support for common file systems, including NTFS, FAT12/FAT16/FAT32/ExFAT, HFS+, ISO9660 (CD-ROM), Ext2/Ext3/Ext4, Yaffs2
- Unicode Strings Extraction: Extracts strings from unallocated space and unknown file types in many languages
- File Type Detection based on signatures and extension mismatch detection. Interesting Files Module will flag files and folders based on name and path.
- Android Support: Extracts data from SMS, call logs, contacts, Tango, Words with Friends, and more.
- Multi-User Cases: Collaborate with fellow examiners on large cases.
- LNK File Analysis:- Identifies shortcuts and accessed documents

- **EXAMINATION:**

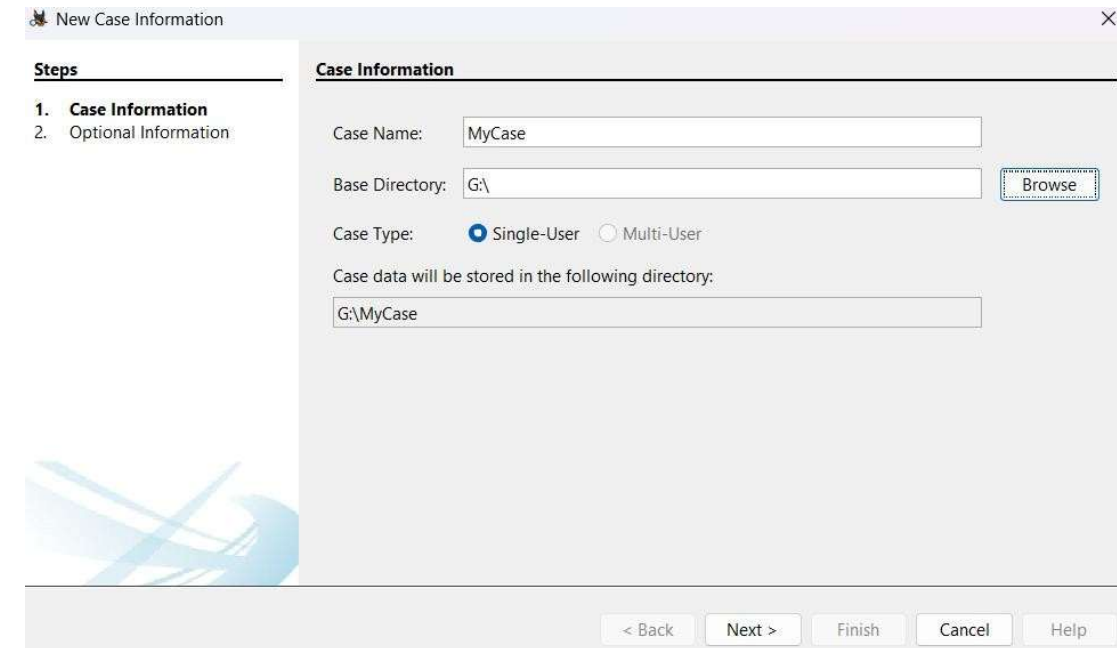
Step-1 : Download and setup the installation process of Autopsy tool. Open it.



Turning on modules...



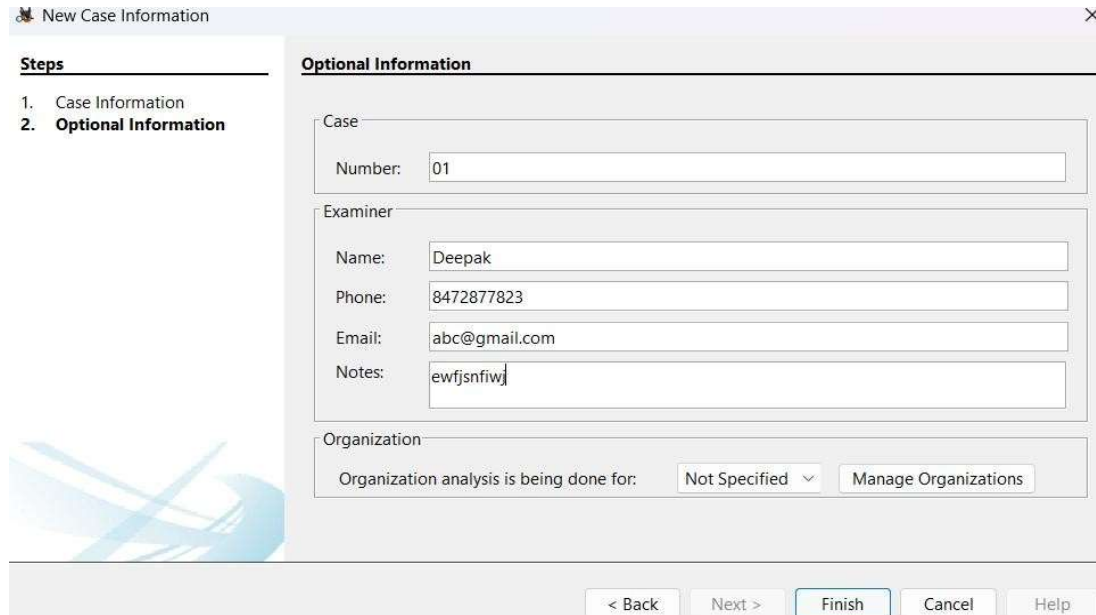
Step 2: Create a case file – Open Autopsy and click New Case > enter a case name and choose a directory.



The screenshot shows the 'New Case Information' dialog box with the 'Case Information' tab selected. The 'Steps' list on the left shows '1. Case Information' and '2. Optional Information'. The 'Case Information' section contains the following fields:

- Case Name: MyCase
- Base Directory: G:\ (with a 'Browse' button)
- Case Type: ☒ Single-User ☐ Multi-User
- Case data will be stored in the following directory: G:\MyCase

At the bottom, there are navigation buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.



The screenshot shows the 'New Case Information' dialog box with the 'Optional Information' tab selected. The 'Steps' list on the left shows '1. Case Information' and '2. Optional Information'. The 'Optional Information' section contains the following fields:

- Case Number: 01
- Examiner:
 - Name: Deepak
 - Phone: 8472877823
 - Email: abc@gmail.com
 - Notes: ewfjsnfiw
- Organization:
 - Organization analysis is being done for: Not Specified (dropdown menu)
 - Manage Organizations (button)

At the bottom, there are navigation buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Step-3 : Select data source – Now, add the data source windows pops up > select Logical disk from the drop-down list > choose the targeted drive image.

Add Data Source

Steps

1. Select Host
- 2. Select Data Source Type**
3. Select Data Source
4. Configure Ingest
5. Add Data Source

Select Data Source Type

- Disk Image or VM File
- Local Disk
- Logical Files**
- Unallocated Space Image File
- Autopsy Logical Imager Results
- XRY Text Export

< Back Next > Finish Cancel Help

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
- 3. Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Local files and folders

F:\

Add
Delete
Clear

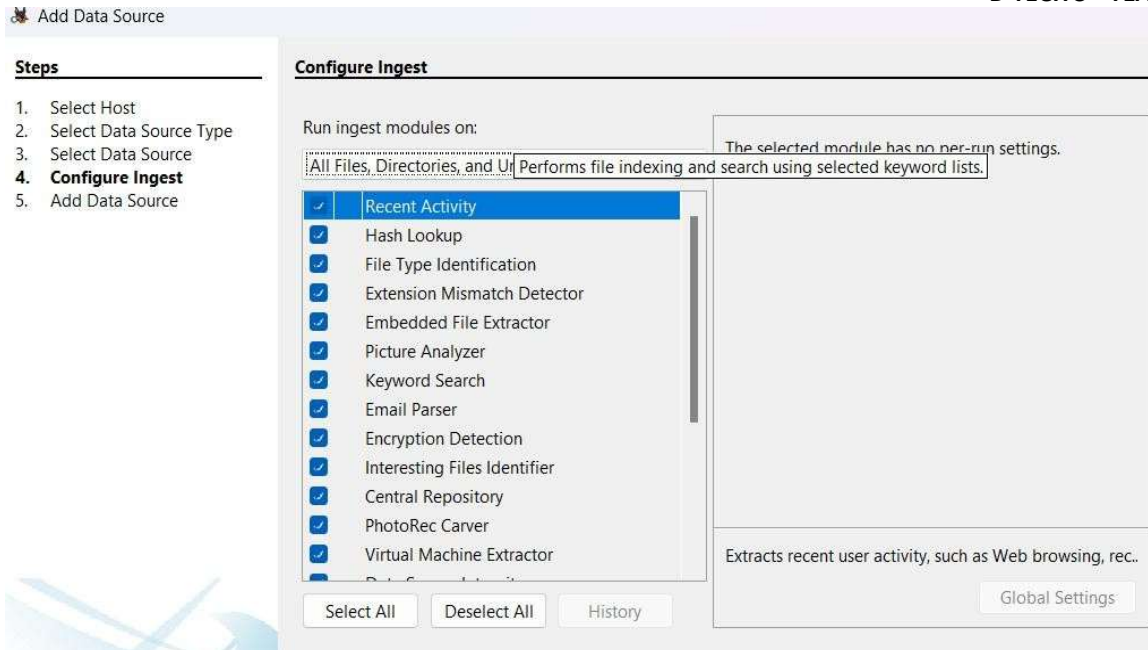
Logical File Set Display Name: Default Change

Timestamps To Include:

- ☐ Modified Time - Often not changed when a file is copied
- ☐ Creation Time - Often changed when a file is copied
- ☐ Access Time - Can be changed when the file is opened

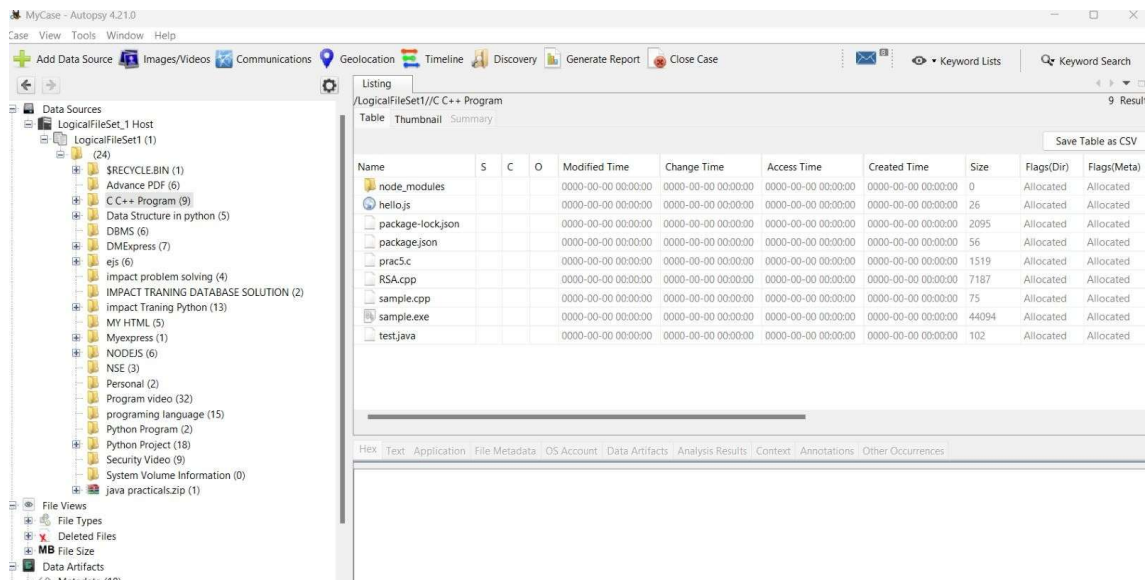
NOTE: Time stamps may have changed when the files were copied to the current location.

< Back Next > Finish Cancel Help



Step 4: Click next to proceed.

Step 5: Wait for the analysis to complete, and the data will be displayed in different categories.



- OBSERVATION

Data Restoration – Open the folder of the files, Right-click on it to restore and select export. Choose a location to export the data , click save.

